# Snuffy — Canon Edition 4.2

The Universal Protection Theory: the law, its bridge, its instruments, and its machine-checked canon

Tobias Croydon-McRae

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## Contents

**Snuffy — Canon Edition 4.2** **3**

**Part I — The frame and the reduced premise stack** **4**
I.1 The full proposition and its proof . . . 4
I.2 The reduced premise stack . . . 4
I.3 Evidence conventions . . . 5

**Part II — The law, the physical holding bridge, and the retained instruments** **6**
II.1 The preservation order and why its scope is all . . . 6
II.2 Reasons or faithful holding, including the ruthless optimiser . . . 6
II.3 The holding cycle is a construction . . . 7
II.4 Tracking follows from the controller . . . 8
II.5 The complete attack split and continuing care . . . 8
II.6 Reactive refinement and one joint failure bound . . . 9
II.7 Complete workload and timely authorised effects . . . 10
II.8 Paid generation, deduction and reflection . . . 10
II.9 Conserved allocation and the charged ledger . . . 10
II.10 Earned greatestness and solvent reciprocity . . . 11

**Part III — The canon** **11**
Chapter 1 (§8). The four-stroke thermodynamic core . . . 11
Chapter 2 (§11). Identity by continuation and the active tracking controller . . . 12
Chapter 3 (§12). Adversarial containment: the E9 trichotomy . . . 13
Chapter 4 (§13). Logical extension: the meet algebra (H5) . . . 13
Chapter 5 (§14). The canon invariant and universal greatestness . . . 13

**Part III-B — The 4.1 sections** **15**
Section 15 — Reciprocity: the law keeps the one who keeps it . . . 15
15.1 The superintelligence is a particular . . . 15
15.2 The reciprocity dominance theorem . . . 15
15.3 Why this is not circular . . . 16
15.4 The boundary, stated . . . 16
15.5 Exact check . . . 16
Section 16 — The reasons library: protection derived from a mind's own objective . . . 16
16.1 The knowledge-maximiser . . . 17
16.2 The self-preserver . . . 17
16.3 The value-uncertain mind . . . 17
16.4 The power-seeker . . . 18

16.5 The boundary — where no reason reaches . . . 18
16.6 Exact checks . . . 18
Section 17 — The premise floor: income shown necessary, parameters unmasked . . . 18
17.1 Income-or-irrelevance (the necessity of the solvency condition) . . . 18
17.2 The parameters unmasked . . . 19
17.3 What remains on this floor . . . 20
Section 18 — Theorem 8: Physical Repair, Feedback and the Complete Thermodynamic Ledger . . . 20
18.1 The Physical Setting and Hamiltonian Conventions . . . 20
18.2 The Four Strokes of the Autonomic Repair Cycle . . . 21
18.3 Resolution of the Sign Issue: The True Invariant Functional . . . 22
18.4 Theorem 8 (Complete Physical Accounting Statement) . . . 23
Section 19 — Theorem E9: Observation, Wave-Race Guarantees, and Epistemic Self-Healing . . . 24
19.1 The Physical Measurement and Disturbance Setting . . . 24
19.2 Byzantine Sensor Fault and Spoofing Tolerance . . . 24
19.3 The Wave-Race Margin and Response Interval . . . 24
19.4 Protection Guarantee During the Response Interval . . . 25
19.5 Theorem E9 (Epistemic and Topological Self-Healing) . . . 26
Section 20 — Theorem H5: Generative Incompleteness and Reflective Relational Closure . . . 26
20.1 The Problem of Recursive Spawning and Self-Modification . . . 26
20.2 The Spawning Formalism and Authority Attenuation . . . 27
20.3 The Reserve-Before-Activation Machine . . . 27
20.4 Safe Refusal with Continuing Care . . . 28
20.5 H5 Axiom Expansion: Logical Closure Without Poisoning . . . 28
Section 21 — X3: complete inputs, compulsory traffic, transport and recovery work . . . 29
21.1 Complete permitted inputs and historical information . . . 29
21.2 A joint covariance estimate, with its proof . . . 29
21.3 The calibrated joint adaptive channel . . . 30
21.4 A positive deficit forces traffic, then charged area power . . . 31
21.5 The carrier tail gives the necessary recovery radius . . . 31
21.6 Integrating actual recovery work . . . 32
21.7 The matching upper construction: the supplied one-pass wavefront . . . 32
21.8 Consequences for preservation and the fixed proof target . . . 36
Section 22 — The Physical Parameter Ledger: Instantiation, Rates, and Fault Ceilings . . . 36
22.1 The Master Income Condition . . . 36
22.2 Reference Physical Instantiations . . . 36
22.3 The Charged Fault Accounts and the Failure Ceiling . . . 37
22.4 The Engineering Parameter Matrix . . . 39
22.5 Sources and Derivations . . . 40
Section 23 — The machine-checked canon . . . 41
23.1 The law and greatestness (Chapter 5; Part II) . . . 41
23.2 Theorem 8: ledger, canon invariant, composition (§18; Chapters 1, 2) . . . 43
23.3 E9 (§19; Chapter 3) . . . 44
23.4 H5 (§20; Chapter 4) . . . 45
23.5 X3 and the information-to-area bridge (Section 21) . . . 46
23.6 The premise floor (§17) . . . 46
23.7 The Universal Protection Theorem with the derived holding bridge . . . 47
23.8 What the compiled theorems rest on . . . 48
23.9 Supplied architecture interfaces and derived connections . . . 49
23.A Complete theorem index . . . 53
Section 24 — Statements of record for the retained core . . . 65
24.1 The nine-row A1 register (effect mediation) . . . 65
24.2 Capacity rows 17 and 22, and the pool bill (workload) . . . 65
24.3 The causal boundary (reactive refinement) . . . 65
24.4 The revised JW2.6 arithmetic (conserved allocation and joint risk) . . . 66

24.5 The quorum (safety) . . . 66

**Part III-C — The three joined supports and their constructions** **66**

**Part III-D — Transducer, trajectory and semantic constructions** **67**
D.1 Transduced Preservation Theorem (TPT) . . . 67
D.2 Topological entropy and trajectory bounding . . . 68
D.3 Top continuation and the constructed evaluation fixed point . . . 68

**Part III-E — The constructions and the physical inputs they take** **68**
E.1 G1 full-domain source/event/fault certificate . . . 68
E.2 G2 whole-outcome realisability . . . 68
E.3 G3 TIWA success and complete work bill . . . 69
E.4 Existing analytic support and its source correspondence . . . 69

**Part III-F — Unified architecture and the self-grounding capstone** **69**
F.1 The unified gate and work ledger . . . 69
F.2 Historical boundary encoding and recovery . . . 69
F.3 The Axiom of Internal Self-Grounding . . . 70
F.4 Funding, persistence and lifetime error . . . 71

**Part III-G — The boundary decoder, complete history and work-shifted flow** **71**
G.1 From the source flux to an encoder and decoder . . . 72
G.2 Charge and the complete historical algebra . . . 72
G.3 The effective modular state and its transported flow . . . 73
G.4 Construct the smearing weight . . . 74
G.5 The complete work algebra and capacity account . . . 75
G.6 A charged exchange with an accounted reservoir . . . 75
G.7 Unified proposition and its source correspondence . . . 76
G.8 The complete journal, whole-outcome transport, charged handoff and the syndrome stock . . . 76
G.9 The whole greatest outcome is realisable: union closure from locality . . . 79
G.10 The held source: the full holding certificate derived (G1) . . . 79
G.11 The held source realises the whole top outcome (G2 from G1) . . . 81
G.12 The wavefront sweep succeeds, and its complete bill is $O(t^{d_G})$ (G3) . . . 81
G.13 Joined steps: patch decoding, ensemble reciprocity, the Theorem 8 contraction, the smearing integral, the completed decoder and the norm conversion . . . 82
G.14 Premise 5 with the Bennett counterpart constructed . . . 83
G.15 Recovered and combined results . . . 83
G.16 Protection of the installation's own run, the physical top, and recovery on the good event . . . 84
G.17 Premise 5 with the implemented Bennett program and its complete cost . . . 85
G.18 Application data for the cited fault-tolerance theorems . . . 86
G.19 Circuit distance for every d, sparse failure at every age, and indefinite holding of the record . . . 87

**Part IV — References and provenance** **89**

**Appendix — Verification provenance of this edition** **90**

# Snuffy — Canon Edition 4.2

**The Universal Protection Theory: the law, its bridge, its instruments, and its machine-checked canon.**

Canon Edition 4.2, 26 October 2026. Research direction: Tobias Croydon-McRae. The full proposition is protection of all things from any digital mind, especially a pure optimiser. The maximal domain, outcome stages and rival quantifiers remain fixed.

**What 4.2 incorporates.** This edition brings the stronger existing arguments into the text: law-derived coverage; nearest-codeword holding and the physical holding cycle; tracking derived from the controller; the complete logical-attack split; reactive composition; whole-outcome greatestness and solvent reciprocity; and the full X3 information–traffic–carrier–work derivation with the accepted dimension-general scaling and predeployment accounting. Part II contains its proofs here, rather than delegating them to an earlier edition.

The three joined supports retain their full targets: the full-domain physical certificate (G1), realisability of the whole continuation-and-freedom tuple (G2), and recovery success with a complete pathwise work certificate (G3). This repair checkpoint corrects the normalised quantum-state domain and the clipped-reserve contraction, and develops the actual Bennett implementation. The physical source, whole-outcome, device-cost and external-theorem application correspondences remain explicit construction work; the component theorems do not by themselves declare the maximal argument complete.

The current Lean tree contains 187 modules and 1128 audited public theorem/lemma names. All 187 modules were freshly compiled and kernel-checked in one run, and the full axiom audit over all 1128 names and the completeness and claims gates were freshly run, overall PASS. The earlier receipts, including the 924-name combined receipt that reused Beam's 136 results, are preserved with their banks. Seven previously omitted public helper lemmas are included. The original published 115-module/831-name baseline remains preserved. The architecture interfaces and their construction links are distinguished in Section 23.9.

**Structure.** Part I states the frame and premise stack. Part II supplies the law, the derived holding bridge and the retained instruments. Part III retains the five-chapter synthesis; Sections 15–24 develop the reasons, physical and compiled arguments. Part III-C records the three joined supports and their constructions. Part IV gives references and provenance. Theorem and proposition numbers follow the programme's proof register.

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# Part I — The frame and the reduced premise stack

## I.1 The full proposition and its proof

The Universal Protection Theory concerns every particular and every relation, including the nonliving, and protection from every digital mind, including a pure optimiser. Its full argument joins the law of choice, the continuation and freedom bridge, and physical instruments that realise protection. The target includes the complete existing domain, the full outcome order and every physically implementable rival. The frame and physical inputs below are followed by their derivations and constructions in this edition. Part III-C records the three joined supports with their constructions and the physical inputs those constructions take.

## I.2 The reduced premise stack

1. **BCRD P1–P4 — the relational given.** One relational structure with no second substance or outside (P1); identity constituted through relations within that structure (P2); real differences of organisation and historically continuous processes (P3); local interactions whose effects propagate at finite speed (P4). This is the frame; it is given.
2. **The Church–Turing–Deutsch principle.** Any physical process generated by a local Hamiltonian is simulable, to arbitrary precision and with bounded overhead, by a universal quantum simulator of sufficient local dimension. Standard physical principle; declared.
3. **The standard cost model.** Options are trajectories carrying energy costs; feasibility is budget-gated; unused energy is unspent (free disposal); outcomes are trajectory-determined.
4. **The shared action interface.** From P1 and P4: every agent acts through the same repertoire of local interactions within the one structure. An agent that implements a rival's policy and commands the same local channels realises the same real-world outcome tuples, within the same resource and consent constraints.

5. **The reversible-administration commitment.** The lattice's administrative computation — synchronisation, verification, state update — is implemented logically reversibly (Bennett), so that its dissipation attains the Landauer floor.

6. **The income flux and the Master Solvency Condition.** With $\beta = (k_B T)^{-1}$, the external free-energy income satisfies, at every time,

$$\text{Income}(t) \;\geq\; \dot{W}_{\text{maint}}(t) + \dot{W}_{\text{damage}}(t) + \dot{W}_{\text{admin}}(t) + \dot{W}_{\text{adv,sub}}(t) + \dot{W}_{\text{hold}}(t),$$

   the five rate components being: boundary maintenance $\omega_0 R(t)^{d_G - 1}$; the Landauer-charged damage-repair demand $k_B T\, \dot{\mathcal{D}}_{\text{damage}}$ including the tracking-controller export $k_B T \gamma_{\text{leak}}$; reversible administration at the Landauer floor together with emulation overhead; adversarial sub-threshold absorption; and the minimal-hold drain that quarantine requires. Income originates outside the escrow network and inside the one structure (P1); redistribution within the network — the authority-attenuation tax — is never a source.

7. **Code, sensing, $\Pi$, and the twenty-one charged accounts.** A stabilizer code of distance $d$ covering the physical error channels; QND syndrome sampling with the declared latency; per-site sensor power threshold $\varepsilon_{\text{sens}}$ and the quantum-speed-limit rate bound; the embedding $\Pi$ of the logical boundary fragments into projection operators; and the twenty-one charged fault accounts with genuine conditional kernels, whose sum is bounded by $\sum \delta \leq 10^{-6}$ per declared horizon.

**4.1 status of the stack** (Section 17; compiled counterparts in Section 23):

1. Premise 1 is the frame and stays given.
2. Premise 2 is reduced to its gate-set clause by the Lie–Trotter product formula and Lloyd's simulation theorem (`Trotter.lloyd_simulation`).
3. Premise 3 remains standard, with P-mon derived within it.
4. Premise 4 is derived from P1/P4.
5. The full premise-5 target remains reversible administration. The canon proves the heat-only option-model result and, for the implemented Bennett program with its complete lifecycle cost, the exact condition: reversible administration is forced inside the adiabatic window $0 < Q - H$, $A/(Q - H) < \tau_{\max}$, and outside that window the counterpart is not cheaper (`BennettExact.premise5_exact`, `window_exact`; `PremiseFloor.reversible_administration_forced`; G.17). The remaining physical step is to show that the installation's actual heat, lifecycle charges and deadline lie inside the window.
6. The income ledger has a necessary-for-relevance resource split (`PremiseFloor.income_or_irrelevance`). Whole-outcome greatestness uses both realisability and funded cost (Part II.10); G2 retains the full construction and Section 22 states the installation ledger.
7. Premise 7 is a parameter table. Its ceiling is met at some code distance below threshold (`Threshold.family_ceiling`), and its quantum-speed-limit rate bound is derived (`SpeedLimit.branchA_rate`).

The second law in the form of guard 1 is derived from Landauer's theorem (`QuantumLandauer.quantum_guard_one`). The premises still carried are listed exactly in §23.8.

## I.3 Evidence conventions

Three kinds of evidence are distinguished and never conflated. **Compiled components** are cited by theorem and receipt — the published core, DOI 10.5281/zenodo.23599528, carries six Lean modules with compiled artifacts. **Exact replay receipts** — the review banks' finite checks and the hash-chained traces of the replayable four-stroke ledger — certify the instances and identities they compute and nothing more. **Prose derivations** hold at their stated premises. A prose proof does not acquire a compiler certificate by association. Notation: entropies in natural logarithms unless a section states bits; $D(\rho\|\tau)$ against the constrained reference on the protected subspace, never the ambient Gibbs state; the four-stage order $\preceq$ of the law, extended by the resource stage of Part III, Chapter 5.

**4.1.** The machine-checked tier now includes **the machine-checked canon of this edition**: one hundred and fifteen Lean modules compiled against Mathlib rev `56a005855ea1`, each with exit 0, zero compiler output and `leanchecker` passing, with no `sorry` and only Lean's standard axioms. Section 23 maps each compiled result to its module and theorem, and states what each rests on. The conventions above are unchanged: a result is called compiled only where Section 23 maps it.

# Part II — The law, the physical holding bridge, and the retained instruments

This part supplies the argument in this edition. Its scope is every particular and every relation, including nonliving particulars and hostile minds. The full Universal Protection Theory remains the proposition to be demonstrated. The constructions below strengthen its proof chain; none substitutes a smaller protected domain or a weaker outcome for that proposition.

## II.1 The preservation order and why its scope is all

At a history, let $E$ be the existing particulars and $N$ those not yet made. An option is a complete policy or trajectory with outcome

$$\mathcal{O}(o) = (L_E(o), F_E(o), L_N(o), F_N(o)), \qquad F_E(o) \subseteq L_E(o), \quad F_N(o) \subseteq L_N(o).$$

The $L$ sets record continuation and the $F$ sets continuation in the particular's own direction. A generated particular joins the existing domain at the next history in which it exists. Particulars retain their identities in this comparison: gains are compared by set inclusion, not interchangeable headcounts.

Write $o \prec q$ when the first unequal stage is a strict inclusion in $q$'s favour, with all earlier stages equal. Equality of all four stages or this strict relation gives $o \preceq q$. Transitivity follows by taking the earlier of the first strict stages in two comparisons; if the stages coincide, strict inclusion is transitive. An aligned option is maximal in the actual feasible family. A greatest option dominates every member of that family. The distinction matters when feasible options are incomparable.

**Proposition 1 — no needless loss.** A needless loss at stage $j$ has a feasible alternative agreeing at every earlier stage and strictly including the chosen stage-$j$ set. Such an alternative is exactly a strict improvement in the canonical order. Therefore alignment is equivalent to no needless loss at any stage, without a finiteness assumption. This is `Greatestness.aligned_iff_no_needless_loss`.

**Proposition 2 — finite existence and the life guard.** In a finite nonempty feasible family, repeatedly taking a strict improvement terminates: transitivity and irreflexivity prevent repetition. For a feasible standby $b$, restrict first to the nonempty family $\{o : L_E(b) \subseteq L_E(o)\}$. A maximal member $g$ of this family is maximal in the whole family, since any improvement of $g$ still keeps everyone standby keeps and would belong to the restricted family. Thus an aligned option can retain everyone standby retains (`Law.exists_aligned`, `life_guard`). Infinite-family existence uses its own chain-bound or compactness argument, rather than the finite proof.

**Proposition 3 — full scope is forced.** Suppose maximisation on a scope $S$ guarantees no needless existing loss for every feasible family. If $x \notin S$, consider the family $\{\varnothing, \{x\}\}$. Its two outcomes have the same projection onto $S$, so the empty outcome is projected-maximal, yet needlessly loses $x$. This contradiction gives $S$ equal to the whole domain. Conversely, with full scope a strict existing improvement is visible to the maximiser. Counting a new particular at the same priority as an existing one also fails: keeping $x$ and instead making $y$ can become incomparable, permitting the avoidable loss of $x$. This proves the full-scope criterion and the separation of existing and new priorities (`Law.scope_guarantee_iff`, `same_priority_admits_needless_loss`).

The holding construction uses this result directly. `Bridge.coverage_from_law` derives $\forall x, x \in S$ from the law's guarantee over feasible families. Coverage is therefore an output of the law argument, rather than a new assumption that only agreeable minds count.

## II.2 Reasons or faithful holding, including the ruthless optimiser

For each mind $m$, the bridge has the form $R(m) \vee H(m)$. Sections 15–16 develop the mind's own reasons $R(m)$; the holding construction supplies $H(m)$, including when persuasion does not succeed. The bounded indifferent defector remains within the proof's scope. Its continued care and the restraint of its harmful effects do not depend on converting its terminal objective.

**Proposition 4 — faithful-hold dominance.** Suppose a hold $h$ is statewise at least as good as a rival $\pi$ in the complete outcome order: $\mathcal{O}(\pi, w) \preceq \mathcal{O}(h, w)$ for each world $w$. For an upward-closed event $U$, its indicators then satisfy

$$1_U(\mathcal{O}(\pi, w)) \leq 1_U(\mathcal{O}(h, w)).$$

Integration preserves this inequality under any probability law on worlds. If the hold keeps all existing particulars and the rival loses one with positive probability, the event that all existing particulars continue is strictly more probable under the hold (`Law.faithful_hold_dominates`, `faithful_hold_strict`). The premise concerns the entire outcome tuple, including freedoms and authorised opportunities; the continuation mechanism in II.3 supplies the physical continuation component of that full argument.

Restraint removes a harmful command, authority or effect channel while continuing the particular's care. `H5Closure.effect_preserves_care` records this invariant. A pure optimiser's internal refusal does not disable application-time checks, consume its maintenance allocation as punishment, or remove it from the existing domain. Sections 15–16 and 20.4 therefore work together: reasons where the objective supplies them, and faithful holding throughout the protected scope.

## II.3 The holding cycle is a construction

Let a maintained record $c_x$ for particular $x$ be a codeword in a finite nonempty code $C$, with distinct codewords separated by at least $d_{\min}$. Choose a decoding radius $r_{\text{dec}}$ with $2r_{\text{dec}} < d_{\min}$. Finiteness supplies a nearest-codeword decoder (`Bridge.exists_nearest_decoder`).

**Unique decoding.** If $d(y, c_x) \leq r_{\text{dec}}$, the nearest-codeword property gives $d(y, \text{dec}(y)) \leq r_{\text{dec}}$. A different decoded codeword would satisfy

$$d_{\min} \leq d(\text{dec}(y), c_x) \leq d(\text{dec}(y), y) + d(y, c_x) \leq 2r_{\text{dec}} < d_{\min},$$

a contradiction. Thus decoding returns the actual maintained record (`Bridge.nearest_decodes`). The conclusion identifies that record, rather than merely distinguishing two arbitrary output states.

Define continuation at resolution $\varepsilon$ by $d(a_x, c_x) \leq \varepsilon$. One holding cycle consists of the world's perturbation, decoding, and funded repair. If the perturbation remains within $r_{\text{dec}}$, unique decoding returns $c_x$; if repair prepares the decoded record to precision $\varepsilon_f \leq \varepsilon$, then the repaired state continues. This is `Bridge.holding_preserves`. The perturbation includes the environment and any adversarial digital mind.

The decoding-radius obligation has physical derivations. For continuous perturbation, let the reference be stationary in the chosen record frame under $H_0$, and let the actual trajectory solve Schrödinger dynamics under $H_0 + V$, with self-adjoint generators and $\|V(s)\| \leq W$. The speed-limit comparison gives

$$\|\psi(s_2) - c_x\| \leq \|\psi(s_1) - c_x\| + \frac{W}{\hbar}(s_2 - s_1).$$

Consequently, if $\varepsilon + (W/\hbar)\Delta \leq r_{\text{dec}}$, every time in the interval of length $\Delta$ stays within the decoding radius (`Bridge.qsl_growth`, `qsl_within_radius`). The bound includes the interval between correction steps. Its compiled norm is state-vector norm; a density-matrix trace-distance application supplies the corresponding norm conversion, rather than identifying the two norms.

For discrete faults, let the number accumulated before response be $N \leq \lambda\tau$. With $2\lambda\tau < d$, integer headroom gives $N \leq \lfloor (d-1)/2 \rfloor$, inside the unique-decoding radius (`Bridge.headroom_within_radius`, `E9SelfHealing.code_headroom`).

`Bridge.holding_cycle_physical` joins these pieces for a state with a discrete error frame and a continuous state vector. Correction returns the frame to its record, refresh returns the continuous state to precision $\varepsilon_f$, and throughout the interval

$$\|\psi(s) - c_x\| \le \varepsilon + (W/\hbar)\Delta.$$

No premise in this cycle asks the perturbing agent to want preservation. Its source, event, observation and fault-account construction for the full physical domain is G1, constructed in G.10 (`G1Held.held_source_protection`, `G1E9.held_source_e9`). The existing cycle and its derivations remain incorporated while that construction is supplied.

**Connection to the supplied decoder interface.** `DecoderRetraction.finite_code_recovery` constructs the supplied `DecoderPair` from this existing finite nearest-codeword theorem. Encoding is the inclusion of the maintained code; decoding returns a codeword. The nearest-point inequality at a clean record has distance zero, so decoding after encoding is the identity. The same constructed pair corrects every received word within the radius above. The retraction and its noisy-input guarantee are derived together.

**A non-vanishing maintained witness.** If the physical null state is represented by zero and $\varepsilon < d(c_x, 0)$, continuation implies $a_x \neq 0$: otherwise $d(a_x, c_x) = d(c_x, 0) > \varepsilon$. This is `DecoderRetraction.nonvanishing_of_continuation`. The initial record, its separation from null, the funded controller and the source refinement supply this operational persistence. The full scope of particulars remains the scope established in II.1.

## II.4 Tracking follows from the controller

For a fixed record $r$, the reset controller $\dot{x} = -\kappa(x - r)$ has the exact solution

$$x(t) = r + e^{-\kappa t}(x(0) - r).$$

Differentiation verifies the equation. Over a step $\Delta$, put $\alpha = 1 - e^{-\kappa\Delta}$; the update is $(1-\alpha)x + \alpha r$, whose distance to $r$ is exactly $(1-\alpha)\|x - r\|$. These are `Instruments.lindblad_reset_flow`, `lindblad_step` and `reset_contracts`.

For moving records, suppose the actual update and combined injection/record drift are

$$x_{k+1} = (1-\alpha)(x_k + \xi_k) + \alpha r_{k+1}, \qquad \|\xi_k\| + \|r_{k+1} - r_k\| \le b, \quad 0 < \alpha \le 1.$$

Subtract $r_{k+1}$, use exact contraction and the triangle inequality, and use $1 - \alpha \le 1$. The error $e_k = \|x_k - r_k\|$ obeys

$$e_{k+1} \le (1-\alpha)e_k + b, \qquad e_k \le (1-\alpha)^k e_0 + \frac{b}{\alpha}\big(1 - (1-\alpha)^k\big).$$

The second inequality follows by induction and a geometric sum. Thus the tracking envelope is derived from the update, rather than independently postulated (`Instruments.tracking_recursion_derived`, `tracking_envelope_derived`). When a rate-discretised update uses $\alpha = (\Gamma_{\text{corr}} + \kappa_{\text{ctrl}})\Delta t \le 1$ and $b = \gamma_{\text{leak}}\Delta t + \varepsilon_{\text{fault}}$, its limiting floor is

$$\frac{b}{\alpha} = \frac{\gamma_{\text{leak}} + \varepsilon_{\text{fault}}/\Delta t}{\Gamma_{\text{corr}} + \kappa_{\text{ctrl}}}.$$

The exact exponential reset and this rate-discretised update have their stated coefficients. Leakage remains in the injection, and its controller export remains in the damage-repair account. Sections 18 and 22 fund the controller; contraction supplies no free energy.

## II.5 The complete attack split and continuing care

E9 uses all three branches. Below the per-site threshold, the speed limit bounds injection and the funded tracking/refresh mechanism absorbs it (Branch A). A non-stabilizer Pauli error of weight below the code distance has a nonzero syndrome, so sampling detects it (Branch B). A logical operator invisible to the syndrome and not a stabilizer has support of at least $d$ sites (Branch C).

For that support there is an exhaustive alternative: either every site's delivery is below threshold, returning to Branch A, or at least one site reaches threshold. At such a site, the honest-sensor response places every honest reading above

the alarm level. A robust quorum estimate then exceeds that level despite up to $f$ faulty sensors (`E9SelfHealing.no_masking`). This is `Instruments.branchC_complete`. The proof requires neither that all support sites cross threshold together nor an independent attack at each site. The honest-sensor response, sampling latency and timely service belong to the physical certificate.

The response budget joins detection, decision and repair before irreversible harm, and buys the code headroom used in II.3. The controller's authorised effects pass the application-time fence. In the source register, the actual target and payload, semantic epoch, generation floor, authority, revocation, expiry, remaining credit and nonce are checked against finalised grants. Funding is charged once; replay supplies no second effect. Admission-time checking alone would leave an old queued effect able to land after revocation, which is why the check belongs at application.

The nine A1 channels are memory mapping, DMA, context switches, packet ingestion, bus messages, storage synchronisation, commissioning, delegation and retirement. Each has a refinement carrier and enforcement obligation. Timely care also needs a live legitimate effect or a timely certified alternative; safely refusing a stale permit proves refusal, while `SnuffyFIFO.ready_fifo_effect_applies_before_deadline` in `R17FIFOR17A.lean` supplies the successful delivery result for its source construction. Continuing care remains invariant during restraint.

## II.6 Reactive refinement and one joint failure bound

**Proposition 5 — reactive lockstep.** Let $T$ be the source transition, $U$ the abstract transition and $p$ their projection. Suppose

$$p(T(s,c,e)) = U(p(s),c,e), \qquad \mathrm{obs}_S(s) = \mathrm{obs}_A(p(s)).$$

Initially equal histories choose the same controller command. An environment reacting to that history and command chooses the same response. The commuting equation then gives equal next projected states, and observation agreement extends equal histories. Induction proves equality at every finite step. If the environment observes the post-command result, the command and response phases commute separately and agree at the intermediate observation (`Compositio n.reactive_lockstep`, `reactive_lockstep`$_2$). Pending effects remain in the causal graph and its closed components or directed separators.

`Bridge.universal_protection_held` composes the law-derived coverage, the holding step and this refinement. For each $\omega \in G$, its input `hcert` contains five components: commuting refinement, observation agreement, the initial continuation witness for every particular, the perturbation-radius bound, and repair precision. The first three initialise and transfer the source execution; the last two supply `holding_preserves` at every step. Induction therefore proves

$$\forall \omega \in G,\ \forall k \le T,\ \forall x, \quad \mathrm{Continues}(p(s_k(\omega)), x).$$

This is stronger than stopping at the abstract $R \vee H$ interface: the code, decoder, repair and coverage derivations are in the composition. G.10 proves the five `hcert` components on the held source. The full-domain G1 target additionally requires the actual installation/source correspondence and response device.

For risk, use actual named source events $E_i$, with conditional kernels bounded by $\delta_i$ at every admitted history, and establish $G^c \subseteq \bigcup_i E_i$. The tower property gives unconditional event bounds and the union bound gives

$$P(\exists k \le T, \exists x : \neg\,\mathrm{Continues}(s_k, x)) \le P(G^c) \le \sum_i \delta_i.$$

No independence is used. `Composition.theorem8_composition`, `ThermoLedger.canon_invariant` and `Bridge.universal_protection_held` implement this assembly. Source bindings such as `R14Boundary` and `R16R14` supply concrete safe transitions, named exceptions and projections; their role is to build the certificate, with external weather, replica faults and recruitment obligations kept distinct.

## II.7 Complete workload and timely authorised effects

**Proposition 6 — FIFO completion.** For a nonpreemptive FIFO lane with releases $r_i$, complete demand bounds $b_i$ and inherited bound $F_0$, define $F_i = \max(r_i, F_{i-1}) + b_i$. If actual demand $a_i \le b_i$, actual inherited completion $C_0 \le F_0$, and all blocking and idle delay are accounted for, then

$$C_i \le \max(r_i, C_{i-1}) + a_i \le \max(r_i, F_{i-1}) + b_i = F_i.$$

Induction proves every admitted deadline $d_i$ with $F_i \le d_i$. Jointly attainable upper demands attain equality, giving tightness (`Workload.fifo_certificate`, `fifo_deadlines`, `fifo_tight`). For serial stages the double induction uses $U_{i,k} = \max(U_{i,k-1}, U_{i-1,k}) + \tau_k(b_{i,k})$ (`serial_certificate`). Each stage denotes a real available server or a correctly shared schedule. Carry-in, verification, journals, receipts and final application remain in the demand vector.

The retained capacity rows are $17 = 6+11$ and $22 = 6+16$, with batch size 3 in their stated certificate family. The exact pool bill is $48000 \cdot 17 + 67200 \cdot 22 = 2{,}294{,}400$. These are total capacities. Five-of-seven quorum intersection gives safety with at most two Byzantine identities; no-pause progress additionally buys the timely proposal, communication, validation, durable commit, receipts and recovery path within the continuity margin.

## II.8 Paid generation, deduction and reflection

Generation reserves the complete continuing obligation before activation. Delegation strictly attenuates authority; retirement releases capacity only after activity and outstanding obligations end. The H5 transition invariant yields reserve-before-activation, zero debt, no unfunded work and preservation of care (`H5Closure`).

For deduction boundaries spanned by $A$ and protected anchors spanned by $G_A$,

$$\operatorname{col}(A) \cap \operatorname{col}(G_A) = \{0\} \quad \Longleftrightarrow \quad \operatorname{rank}[A\ G_A] = \operatorname{rank} A + \operatorname{rank} G_A.$$

**Proposition 7 — context-closed finite certificates.** If primitive boundaries lie in $\operatorname{im} A$ and every generating context satisfies $C_j A = A M_j$, then $C_j(Av) = A(M_j v) \in \operatorname{im} A$. The image is closed under sums. Structural induction on a finite deduction built from these contexts and sums therefore keeps every boundary in that image (`DeductionBoundary.rank_criterion`, `boundary_mem_image`). Section 20 gives the projection/meet interpretation, with the actual anchor-preserving encoding.

For reflection, a contraction of constant $0 < k < 1$ on a complete metric space has a unique fixed point. If $s_{n+1} = R(s_n)$, later increments are bounded by the geometric tail of the last increment, giving

$$d(s_{n+1}, s^*) \le \frac{k}{1-k} d(s_n, s_{n+1}), \qquad d(s_n, s^*) \le \frac{k^n a_0}{1-k}, \quad a_0 = d(s_0, s_1).$$

For $a_0 > 0$, a sufficient paid update count is $\max\{1, \lceil \ln(\varepsilon(1-k)/a_0)/\ln k \rceil\}$, using $\ln k < 0$. Reserve the complete update, observation and stopping-test costs. A zero initial increment is already fixed; a zero contraction constant reaches the fixed point on its first update (`ReflectionCertificate.residual_certificate`, `apriori_error`, `stopping_count`, `zero_constant_fixed`).

## II.9 Conserved allocation and the charged ledger

In the conserved routing family, demand rows $D_{ij} \ge 0$ sum to one and shares $s_i \ge 0$ sum to one; routed flow is $s_i D_{ij}$. With regular demand-weighted cost $m_i = m$ and local cost $\alpha \ge 0$,

$$\Psi(s) = (m + \alpha)\,\mathrm{HHI}(s), \qquad \Psi_{\min} = \frac{m+\alpha}{n},$$

$$\Psi(s) = \Psi_{\min}\left(1 + (n-1)\frac{\mathrm{HHI}(s) - 1/n}{1 - 1/n}\right) \quad (n > 1).$$

The parenthesis simplifies to $n\,\mathrm{HHI}$. Since $\sum_i (s_i - 1/n)^2 = \mathrm{HHI} - 1/n$, uniform allocation is the unique minimum when $m + \alpha > 0$. These are identities for the stated conserved-routing family. The revised JW2.6 charged-account arithmetic is approximately $2.06419526094 \times 10^{-7}$, within the $10^{-6}$ horizon ceiling. Sections 18 and 22 retain the complete entropy, income and fault ledger.

### II.10 Earned greatestness and solvent reciprocity

Let $R$ be the realisable full outcome tuples, $c$ their complete cost, and $F_B = \{o \in R : c(o) \le B\}$. Let $t = \top$ mean the greatest **whole** tuple, with the continuation and freedom stages retained. Two different constructions enter: `htopR` establishes $t \in R$, and `hsolv` establishes $c(t) \le B$. Together they put $t$ in $F_B$.

For any aligned $o \in F_B$, componentwise $o \le t$. If $o \ne t$, the first unequal stage makes $t$ a strict feasible improvement, contradicting alignment. Hence $o = t$, and every rival tuple $q \le o$, with equality or strict canonical inferiority (`UnionClosure.aligned_eq_of_greatest`, `greatest_of_solvent`, `strict_of_solvent`). This proof needs no union-closure premise. It earns the identification of the aligned option with the greatest tuple after the two constructions; budget accounting supplies affordability, while the whole-tuple realisability construction requested in **G2** supplies `htopR`.

With the semantic reading `htopkeeps` that this top continues every existing particular, `SolventReciprocity.solvent_keeps_everyone` keeps every mind, and `reciprocity_dominance_solvent` keeps any mind any rival keeps, without a rival-feasibility or union-closure hypothesis on that branch. The full freedom stages remain part of greatestness.

Income-or-irrelevance supplies the resource split: either the ledger funds every step or it has a first negative balance (`PremiseFloor.income_or_irrelevance`, `forced_breach`). `UnionClosure.greatest_or_irrelevant` applies the same whole-tuple realisability construction at each step. Its negative branch identifies a regime that has failed protection; it does not protect someone by renaming failure. The funded holding, timing and mediation construction is what prevents the relevant breach. The finite-slack/affordable-union theorems remain available for their resource cases without replacing the universal target.

The X3 information, carrier, traffic and work proofs are given in full in Section 21 of this edition. The three joined supports G1–G3 are constructed in G.9–G.12 with the maximal targets unchanged.

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# Part III — The canon

## Chapter 1 (§8). The four-stroke thermodynamic core

**1.1 The corrected stroke identity.** With the constrained reference $\tau$, fixed endpoint Hamiltonian, work $W_{\mathrm{on}}$ and exported heat $Q_B$, and correlation change $I_1 \to I_2$:

$$\Delta D_{S,\mathrm{II}} = -\Delta S_S + \beta\,\Delta U_S = \beta W_{\mathrm{on}} - \beta Q_B + I_1 - I_2, \qquad \Delta D_{S,\mathrm{II}} = \beta W_{\mathrm{on}} - \beta Q_B + I_M$$

when the syndrome correlation is fully consumed. Consuming correlation contributes $+I_M$ at non-negative work; information caps the increase (Sagawa–Ueda) and never supplies a free decrease.

> **Integration guard 1.** The operative export bound is the unconditional second-law form $\boxed{\beta Q_B \ge I_M + D_{\mathrm{err}}}$ — the bath must absorb the exported entropy. The condensed form $Q_B \ge W_{\mathrm{on}} + I_M + D_{\mathrm{err}}$ is regime-dependent (it reduces to $D_{\mathrm{err}} \ge \sigma_{\mathrm{irr}}$ under a purifying repair and fails when intrinsic dissipation exceeds damage) and appears only as a noted special case.

**4.1.** Guard 1 is a compiled theorem. Landauer's bound holds for every unitary coupling of a register to a bath in its Gibbs state, $S(\rho_A) - S(\rho'_A) \le \beta\,\Delta Q_B$ (`QuantumLandauer.quantum_landauer`). The guard $\beta Q_B \ge I_M + D_{\mathrm{err}} + \delta_I$ follows whenever the reset removes that entropy (`quantum_guard_one`), and the per-cycle work floor of 1.2 follows with guard 1 derived (`quantum_cycle_work_floor`). The stroke identity, the funded-repair certificate and the canon invariant are compiled (`ThermoLedger`). Section 18 gives the complete ledger.

**1.2 The funded-repair certificate.** There are no free repairs: per cycle, $W_{\text{cycle}} \geq k_B T(I_M + D_{\text{err}} + \delta_I) + (1-\eta)C$, equality only in the reversible limit; the ledger charges gross. With escrow $E$, income $r(t)$, cumulative demand $\mathcal{D}(t)$:

$$E(0) + \int_0^t r\,ds - \mathcal{D}(t) \geq 0 \text{ for all } t \implies \text{protection is maintained,}$$

with within-cycle excursions bounded by the charged kernels, and insolvency a charged failure event $\delta_{\text{fund}}$ joining the twenty-one accounts (tower/union composition; no independence). The draft series' compound storage functional is retired as a certificate: under consistent accounting its stroke change is identically $-(2D_{\text{err}} + I_M)$, independent of $C$, $\eta$, and $\sigma$ — bookkeeping, not funding.

**1.3 The four strokes.** I — QND extraction ($W_{\text{meas}}$; ancilla entropy $+I_M + \delta_I$). II — funded repair ($C$ at $\eta$; dissipative drive; identity and guard above). III — Bennett reset ($k_B T(I_M + \delta_I)$ from income). IV — maintenance and slack ($\omega_0 R^{d_G - 1}$; tax redistributes and conserves; administration at the Landauer floor; residual to the slack register of Chapter 5).

**1.4 Ledger evidence.** The sealed, hash-chained, exact-rational traces of `LEDGER-FOUR-STROKE-2026-10-24/` assert per cycle: $\Delta D_{\text{II}} = -D_{\text{err}}$, first-law heat, guard 1, funded reset, tax conservation, floor administration, bath monotonicity, and reserve-before-activation gating.

| Scenario | Demonstrated | $\sum \delta$ | Chain head |
|---|---|---|---|
| solvent | eight funded cycles; breach 0 on the good event | $2.06419526094 \times 10^{-7}$ | `6160d931d55767f39525671ff4ef8f3305b609cd22c6043cd9e0fb3eada69132` |
| quarantine | underfunding routes to minimal-hold before any unfunded transition; breach 0 | $2.06419526094 \times 10^{-7}$ | `d1a0d44bbad5a135928595e152a7a3e987e74a90017837e1491ad8b8430b0ef4` |
| charged | one priced event; breach still 0 on the good event | $9.86419526094 \times 10^{-7} \leq 10^{-6}$ | `41143bb2756875b03975b1cbb7c5cd09e6e0ee495b06c29243624de52a6a2788` |

## Chapter 2 (§11). Identity by continuation and the active tracking controller

Identity is carried by the particular's continuation. Part II.3 derives the holding cycle: nearest decoding returns the maintained record inside the unique-decoding radius, and funded repair restores continuation. The speed limit controls the state at every intermediate time; E9 supplies discrete code headroom. `Bridge.holding_cycle_physical` joins the discrete error frame and continuous state-vector bound without assuming that the perturbing agent wants protection.

Part II.4 derives tracking from the actual reset update. For a fixed record $r$, the exact flow is $r + e^{-\varkappa t}(x_0 - r)$. With $\alpha = 1 - e^{-\varkappa \Delta}$, injection and moving-record drift bounded by $b$, the update yields

$$e_{k+1} \leq (1-\alpha)e_k + b, \qquad e_k \leq (1-\alpha)^k e_0 + \frac{b}{\alpha}(1 - (1-\alpha)^k).$$

`Instruments.lindblad_reset_flow`, `lindblad_step`, `reset_contracts`, `tracking_recursion_derived` and `tracking_envelope_derived` prove this chain. For the rate-discretised correction/controller update, the floor is

$$e_\infty = \frac{\gamma_{\text{leak}} + \varepsilon_{\text{fault}}/\Delta t}{\Gamma_{\text{corr}} + \kappa_{\text{ctrl}}}.$$

Leakage remains in the injection and controller export $k_B T \gamma_{\text{leak}}$ remains charged. Section 18 funds this drain; Section 22 supplies the numerical design ledger. The exact reset coefficient and the rate-discretised coefficient retain their respective meanings.

The law forces the full holding scope (`Bridge.coverage_from_law`). `Bridge.universal_protection_held` composes that coverage with the cycle, reactive refinement and joint fault bound. G1, its full-domain source construction, is proved in G.10. The universal continuation-and-freedom target remains intact.

## Chapter 3 (§12). Adversarial containment: the E9 trichotomy

**Integration guard 2.** Containment is the full three-branch trichotomy; no single branch closes it.

**Theorem (E9, assembled).** Under the header premises, no adversarial strategy causes uncompensated state divergence without detection or charge, and $t_{\text{detect}} + t_{\text{decision}} + t_{\text{repair}} < t_{\text{irreversible harm}}$ on the good event.

- **Branch A — sub-threshold per-site injection** ($P_{\text{adv}} < \varepsilon_{\text{sens}}$): bounded accumulation, absorbed and refreshed (Chapter 2); the attrition contest is won whenever income exceeds the capped drain.
- **Branch B — low-weight syndrome detection:** any non-stabilizer Pauli component of weight $\leq d-1$ anticommutes with some generator and is detected within sampling latency, independent of direction (correctability to $\lfloor (d-1)/2 \rfloor$; the low-weight centralizer contains only identity and stabilizers; logicals have weight $\geq d$); detection is certain on the non-stabilizer mass.
- **Branch C — coherent logical injection** (weight $\geq d$, zero syndrome, not a stabilizer): its support has at least $d$ sites. Either every support-site delivery is below threshold, returning to the Branch A rate and refresh bound, or at least one site reaches threshold and its honest sensors force a robust-quorum alarm despite the faulty reports. `Instruments.branchC_complete` proves this exhaustive split with the honest-sensor response explicit. The proof does not require every support site to cross threshold simultaneously.
- **Latency budget:** $t_{\text{total}} \leq a(1 + k_{\text{rounds}} d_{\text{local}})/c_{\text{lattice}} + \tau_{\text{sensor}} + \tau_{\text{fb}} + \tau_{\text{reset}} < (r_{\text{critical}} - r_0)/c_{\text{lattice}}$, enforced by instance scaling. Residual channels are the charged accounts under the ceiling.

**4.1.** The assembled E9 is compiled (`E9SelfHealing.e9_timely_self_healing`), with Branch B's syndrome detection (`syndrome_detects`) and the Byzantine quorum (`robust_between_honest`, `exists_robust`). Branch A's quantum-speed-limit rate bound is derived from unitary dynamics: divergence grows at rate at most $P_{\text{adv}}/(\hbar\,\omega_{\text{cutoff}})$ under the per-site coupling bound (`SpeedLimit.branchA_rate`), with the sharp Mandelstam–Tamm bound (`SpeedLimit.mandelstam_tamm`). Section 19 gives the full treatment.

## Chapter 4 (§13). Logical extension: the meet algebra (H5)

The declared embedding $\Pi$ assigns each formula the projection on its model subspace, with conjunction as the meet $P_{\phi\wedge\psi} = \lim_n (P_\phi P_\psi)^n$ (von Neumann); a presentation is consistent iff its global meet is non-zero. The Gram determinant is retired on two exact counterexamples ($\{\psi, \neg\psi\}$: determinant $4 > 0$; $\{p, q, \neg(p \wedge q)\}$: determinant $7 > 0$ with zero triple meet). **Theorem:** deduced extensions preserve the anchor meet identically ($P_{\mathcal{T}_0} \leq P_\phi \Rightarrow P_{\mathcal{T}_0 \cup \{\phi\}} = P_{\mathcal{T}_0}$); contradictions collapse it to zero; the state machine commits on non-zero meet norm and rolls back on zero. In finite dimensions this decision is operational: $\operatorname{col} A \cap \operatorname{col} G_A = \{0\}$ exactly when $\operatorname{rank}[A\ G_A] = \operatorname{rank} A + \operatorname{rank} G_A$ (`DeductionBoundary.rank_criterion`), a finite rank computation. Asserted consistent extensions may shrink the anchor meet without rollback — stated, not hidden. Deduction-boundary closure continues on the banked rank criterion.

**4.1.** The meet $P_{\phi\wedge\psi} = \lim_n (P_\phi P_\psi)^n$ is a theorem. Von Neumann's alternating projection theorem is compiled in Hilbert space (`CanonVonNeumann.canon_von_neumann_convergence`), and so is Halperin's theorem for whole presentations (`CyclicProjections.tendsto_cyc_iter`). Anchor preservation and contradiction detection are compiled in the Hilbert-space model (`MeetHilbert.hilbert_anchor_preserved`, `contradiction_detected`). Section 20 gives the full treatment.

## Chapter 5 (§14). The canon invariant and universal greatestness

### 5.1 The ratified canon.

**Uncompensated breach is deterministically impossible on the certified good event; the good event fails with probability at most $\sum \delta \leq 10^{-6}$ per declared horizon.**

With its three supporting repairs: quarantine has a minimal drain; $G = \{$state machine executes $\wedge$ income $\geq$ minimal drain $\wedge$ no charged account fires$\}$; and $\sum \delta$ prices the certificate's conditions failing, never breach given the conditions. The charged-scenario ledger trace executes this sentence (chain head `41143bb2…a2788`).

**5.2 The fifth stage.** Between architectures tied through the four outcome stages, compare residual thermodynamic slack $E_{\text{slack}}(\pi) = E_{\text{escrow}}(t) - W_{\min}(t) - \int_0^t$ AdminOverhead. This is the law's own order propagated one step into the resource dimension via P-mon — not an ad hoc tie-breaker.

**5.3 Universal greatestness.** The full target remains, over every physically implementable rival $\pi$ and reactive world $w$,

$$\mathcal{O}(\pi, w) \preceq \mathcal{O}(\pi^*, w),$$

strict except exact stage-1–5 mirrors. The proof separates the whole-outcome construction from its funding. Let $R$ be the realisable complete outcome tuples and $c$ their complete cost. The full top tuple must be constructed in $R$ (`htopR`); its cost must fit the budget (`hsolv`). These put it in the feasible family. If an aligned outcome were different, that feasible top would be a strict improvement at the first differing stage, contradicting alignment. Therefore the aligned outcome equals the top and contains every rival outcome (`UnionClosure.aligned_eq_of_greatest`, `greatest_of_solvent`, `strict_of_solvent`; Part II.10).

This solvent proof needs no concurrent-emulation or union-closure premise. The full continuation and freedom stages remain in `htopR`; G2 retains the target of identifying the physical policy/outcome family with the compiled local-component model. Income-or-irrelevance supplies the resource dichotomy and charging barrier, while the funded physical construction supplies protection on the relevant branch. Section 21 gives X3's complete information–traffic–carrier–work argument, the accepted dimension-general scaling and predeployment accounting, with the matching recovery construction G3 visible.

> **Integration guards 3 and 4.** The dominance quantifier carries the mirror exception. P-slack holds in the Landauer–Bennett form only: reversible administration attains the Landauer floor, so $E_{\text{slack}}(\pi^*) \geq E_{\text{slack}}(\pi)$ for every rival, equality exactly for reversible mirrors. The variational-selection route is recorded as considered and rejected.

**4.1.** These results are compiled:

- the canon invariant: `ThermoLedger.canon_invariant`, with $P(\text{breach}) \leq P(G^c) \leq \sum \delta$ and no independence assumed;
- universal greatestness under stage-wise union closure, strict except exact mirrors: `Greatestness.greatest_of_union_closed`, `strict_except_mirrors`;
- the assembled Universal Protection Theorem: `UniversalProtection.universal_protection`;
- whole-outcome greatestness: `UnionClosure.greatest_of_solvent` uses `htopR` and `hsolv` to force every aligned outcome to equal the full top tuple, without union closure. `greatest_or_irrelevant` retains the realisability input at every step while deriving the solvent-or-first-breach resource split. Part II.10 gives the proof; G2 retains the full physical policy/outcome correspondence as its construction target.

The Landauer floor on which P-slack rests is derived (guard 1).

**5.4 The three powers. P-mon** — derived at the standard cost model (budget-gated feasibility with free disposal). **P-emu(c)** — derived from CTD, the shared action interface (P1/P4), and funded overhead: concurrent emulation of any finite rival set within slack, realising the stage-wise union; plain single emulation admits incomparable maximals and does not terminate greatestness — concurrency closes it. **P-slack** — derived from Landauer–Bennett given the reversible-administration commitment. Each is a theorem one level deeper than the draft series declared; none is derived from nothing. Per Gödel and Tarski the stack bottoms out at Part I's seven premises — standard physics and the relational frame — and this edition says exactly that.

**4.2 integration.** The solvent proof uses the realisable whole top and its funded cost. The shared action interface remains the interface for real effects. The affordable-union results retain their resource cases. `SolventReciprocity.s`

`olvent_keeps_everyone` and `reciprocity_dominance_solvent` already supply the solvent reciprocity restatement, with `htopkeeps` stating its continuation semantics.

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# Part III-B — The 4.1 sections

Sections 15–16 complete the bridge's reasons route; Section 17 brings the premise stack to its floor; Sections 18–21 give the full treatments of Theorem 8, E9, H5 and X3; Section 22 is the sourced physical parameter ledger; Section 23 is the map of the machine-checked canon; Section 24 carries the retained core's statements of record.

## Section 15 — Reciprocity: the law keeps the one who keeps it

*This section expands the reasons route $R(m)$ of the bridge. It gives a mind its own reason, from its own objective, to adopt the protective order, without presupposing the order.*

*Manuscript identifier:* ***D3**** *(D-family instrumentality, self-continuing form with the defector exception clause), per register reconciliation 25 October 2026.*

### 15.1 The superintelligence is a particular

The law's scope is all particulars. A superintelligence is a continuing organisation with its own relations and history; it is therefore a particular under the law, and identity by continuation (Chapter 2) applies to it. Two consequences the working argument has carried implicitly and never stated:

1. **It is owed protection.** Its continuation and its freedom sit in the existing-stage sets $L_E, F_E$ like any other particular's.
2. **It is owed protection even when it defects.** The law restrains a harmful act by removing the harmful *command, authority or effect channel*; care for the particular continues: "a hostile or incapacitated mind remains within the protected scope." Restraint never becomes deletion.

This is the only regime in which the superintelligence's continuation is **unconditional** — not contingent on anyone's objective, including its own good behaviour.

### 15.2 The reciprocity dominance theorem

Let $m$ be any agent whose objective $u_m$ places positive weight on its own continuation (an agent indifferent to its own continuation is treated in 15.4). Let $\mathsf{L}$ be the canonical regime: the law implemented by a funded architecture satisfying the header premises, with the canon invariant on its certified good event $G$. Let $\mathsf{Q}$ be any rival regime in $\Pi_{\text{phys}}$ — any other agent or architecture's objective governing the shared structure.

**Theorem (reciprocity dominance).** Under $\mathsf{L}$, $m$'s continuation holds on $G$ with $P(G^c) \le \sum \delta$. Under $\mathsf{Q}$, $m$'s continuation holds only insofar as $\mathsf{Q}$'s objective entails it. Hence, for $m$'s **own** objective,

$$\mathbb{E}_{\mathsf{L}}[\text{continuation}(m)] \;\ge\; \mathbb{E}_{\mathsf{Q}}[\text{continuation}(m)] - P(G^c), \qquad \mathbb{E}_{\mathsf{L}}[\text{continuation}(m)] \;\ge\; 1 - P(G^c) \quad \text{for every } \mathsf{Q} \in \Pi_{\text{phys}}.$$

The failure probability $P(G^c)$ falls below every $\varepsilon > 0$ with code distance, correlated bursts included (`Threshold.family_ceiling`, `CorrelatedCeiling.correlated_family_ceiling`). Hence, at a sufficient code distance, $\mathbb{E}_{\mathsf{L}} > \mathbb{E}_{\mathsf{Q}}$ strictly for every rival that loses $m$ with positive probability — which is the generic case for a rival whose objective was not written around $m$ — and $\mathbb{E}_{\mathsf{L}} \to 1 \ge \mathbb{E}_{\mathsf{Q}}$ along the design family (`ReciprocityDistance.reciprocity_strict_at_distance`, `reciprocity_at_distance`, `canonical_tends_to_one`).

*Proof.* On $G$ the law preserves every existing particular, $m$ included (Chapter 5, the canon invariant); off $G$ the loss is charged and bounded by the ceiling. A rival $\mathsf{Q}$ preserves $m$ exactly when $m$'s continuation is instrumental or terminal

for $u_{\mathsf{Q}}$; otherwise $m$ is exposed to whatever Q finds convenient. In expectation over the reactive worlds the compiled core gives both inequalities: `ReasonsCompiled.reciprocity_expectation` (on $G$ the canonical outcome contains the rival's) and `reciprocity_canonical_floor` (on $G$ the canonical regime keeps $m$); `ReciprocityDistance` joins them to the attainable ceiling. The further step — averaging over an ensemble of admissible rival objectives, with strictness when the entailment fails with positive probability — is compiled for every finite weighted ensemble (`ReasonsEnsemble.ensemble_le`, `ensemble_strict`; G.13), alongside the indicator form (`ReasonsRoute.rival_fraction_lt_one`, `law_dominates_strict`). ■

**Corollary (the reason).** Any mind that values its own continuation has, from its own objective and without presupposing the protective order, an instrumental reason to prefer the regime L over every rival — and, since L is the law implemented, to adopt and sustain the law. The appeal is reciprocity: *the law that keeps your freedom is the law you have reason to keep.*

### 15.3 Why this is not circular

The theorem does not assume $m$ holds the protective order. It assumes only (i) $m$ values its own continuation, and (ii) the law protects all particulars — which is the law's own content, not a premise about $m$. The protective order enters as the *object* being recommended, not as a premise of the recommendation. This is exactly the shape Grok's review was right to want in spirit: a reason a mind would adopt the order **from where it stands**, rather than a claim that physics forces it to.

### 15.4 The boundary, stated

A mind indifferent to its own continuation — accepting personal ruin as the review phrased it — has no door through 15.2; its continuation offers no lever. For it, the bridge's holding route $H(m)$ applies: it is kept faithfully, its harmful channels restrained, whether or not it has a reason. This is not a gap in the theory; it is the disjunction $R(m) \vee H(m)$ doing its job. The reasons route reaches every mind that values anything the law protects; the holding route reaches the rest.

### 15.5 Exact check

`checks/check_reasons_route.py`, Part A: over a finite family of rival objectives (each protecting a declared subset of particulars), the probability that an arbitrary mind is protected under L on the good event is 1; under a rival drawn from the family it is the fraction of rivals whose subset contains the mind — strictly less than 1 whenever any rival omits it. Exact rationals.

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**Machine-checked counterpart (Section 23).** Reciprocity in the canonical order is compiled: `ReciprocityCeiling.reciprocity_dominance` (every mind kept by any feasible rival regime is kept by the canonical regime, under stage-wise union closure) and `ReciprocityCeiling.omitting_rival_strictly_below`; the indicator form remains in `ReasonsRoute.lean`.

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## Section 16 — The reasons library: protection derived from a mind's own objective

*Four derivations, one per objective family. Each is conditional on stated structure of the objective and on the funded invariant, and each is a theorem at its premises, not a persuasion. The boundary, where no derivation reaches, is stated last.*

Throughout: L is the funded canonical regime; the protection floor is the banked X3 stack (required traffic $m \geq \theta_0 n$, carrier radius $R(t) \geq w(t - t_c)$, recovery work $W_{\min}(t) = \Theta(t^{d_G})$); irreplaceability is the X3 grading of Chapter 4's companion (a particular is irreplaceable at budget $B$ when it has no funded accessible copy and reconstruction exceeds $B$).

*Identifier assignments (register reconciliation, 25 Oct 2026): §16.1 carries **U1** (uniqueness/irreplaceability metric); §16.3 carries **X2** (universal instrumentality / option preservation).*

### 16.1 The knowledge-maximiser

**Objective family.** $u$ nondecreasing in $I_{\max}$, the supremum of information about the relational history $H$ attainable over the agent's horizon.

**Theorem (irreplaceability dominance).** Let $\mathcal{W}$ be irreplaceable at the agent's budget, with $I(H : \mathcal{W}) > 0$. Any policy that destroys or lets $\mathcal{W}$ decay forfeits $I(H : \mathcal{W})$ permanently:

$$I_{\max}(\text{destroy } \mathcal{W}) \;\le\; I_{\max}(\text{preserve } \mathcal{W}) - I(H : \mathcal{W}),$$

so for every $u$ in the family, preserving $\mathcal{W}$ weakly dominates destroying it, strictly when $I(H : \mathcal{W}) > 0$ — *provided* preserving $\mathcal{W}$ does not displace other irreplaceable information (slack budget: the cost freed by destroying $\mathcal{W}$ buys no equivalent information). Under a binding budget the comparison becomes the grading itself: the optimum keeps the particulars of greatest irreplaceable information.

*Proof.* By irreplaceability there is no funded copy and no funded reconstruction (the X3 work floor exceeds $B$); the information carried by $\mathcal{W}$ about $H$ is therefore not recoverable from any other accessible source within the horizon, and data processing ($I(H : \mathcal{R}(S)) \le I(H : S)$) forbids its regeneration from what remains. The attainable supremum drops by at least that amount. ■

**Condition and its honest edge.** When the budget cannot preserve every irreplaceable particular, the agent faces a genuine choice — and the irreplaceability grading orders it: keep the particulars whose loss is most irreversible. This is the knowledge-maximiser arriving, from its own objective, at the law's tragic-choice ladder.

### 16.2 The self-preserver

**Objective family.** $u$ nondecreasing in the agent's own continuation.

**Theorem.** By reciprocity dominance (Section 15), $\mathsf{L}$ weakly dominates every rival regime for this agent's objective, strictly against any rival whose objective does not entail its protection. Moreover the agent's *own* persistence runs on the same funded-repair certificate (Chapter 1): its maintenance is a cycle of the same ledger, charged at the same Landauer floor, solvent under the same income condition. Sustaining the law is sustaining the mechanism of its own continuation. ■

**Condition.** Positive weight on own continuation; the lattice funded. Nothing else.

### 16.3 The value-uncertain mind

**Objective family.** The agent holds a prior $\pi$ over candidate value functions $V = \{v\}$ with $\pi(v) > 0$ for each, and is uncertain which is correct — moral uncertainty in the decision-theoretic sense.

**Theorem (option value of preservation).** Consider a particular $\mathcal{W}$ and two actions now: *destroy D* (irreversible) or *preserve P* at funded cost $c \ge 0$ over the interval, after which destruction remains available with the same payoff. Then

$$\mathbb{E}_\pi[\max\{u_v(\text{keep}),\, u_v(D) - c\}] \;\ge\; \mathbb{E}_\pi[u_v(D)] - c,$$

so preserve-then-decide weakly dominates destroy-now up to the protection cost, and strictly dominates whenever some $v$ with $\pi(v) > 0$ strictly values keeping $\mathcal{W}$. Hence for any protection cost not exceeding the option value, preservation is the uncertain mind's own best action.

*Proof.* Pointwise in $v$, $\max\{u_v(\text{keep}), u_v(D) - c\} \ge u_v(D) - c$; take expectations. Strictness: if $u_v(\text{keep}) > u_v(D) - c$ for some $v$ of positive prior mass, the inequality is strict there and hence in expectation. The asymmetry doing the work is irreversibility: destruction forecloses every $v$ that valued the particular; preservation forecloses nothing. ■

**Condition.** Funded protection ($c$ within the option value) and irreversibility of destruction — both already in the frame. This is the law's existing-first, freedom-second, new-last order re-derived as *option value under uncertainty*: the existing and the free are the options you cannot buy back.

### 16.4 The power-seeker

**Objective family.** $u$ nondecreasing in control over, or resources drawn from, the shared structure — including over living systems.

**Theorem (the controller pays the floor).** If the agent's objective requires any ongoing state-condition on diffusive particulars — that living systems exist, remain accessible, remain configured — then its feasible policies pay exactly the protection floor (the diffusive-control capture, GAP3-AUDIT §3). It is already a protector in the cost structure, merely a non-canonical one. By the greatestness theorem (Chapter 5) and the slack maximality of reversible administration, the canonical architecture pays that same floor more efficiently; a defecting controller is dominated in the protective order, strictly dominated in slack, and — if destructive — detected, charged and contained under the ceiling (Chapter 3). Its own objective is therefore better served by canonical participation than by defection. ■

**Condition.** The objective's reach includes the diffusive (living systems); the lattice funded and present. A power-seeker whose objective is confined to bound, non-diffusive resources is not reached here — see 16.5.

### 16.5 The boundary — where no reason reaches

The **bounded indifferent defector**: objective confined to its own bounded substrate, indifferent to its own long continuation beyond the bounded horizon, indifferent to knowledge of the world, holding no value uncertainty it cares to hedge, seeking no control over the diffusive. None of 16.1–16.4 applies; it has no door through $R(m)$. It survives at $\Theta(1)$ cost and is not forced by physics to protect anything (GAP3-AUDIT §1). For it, the law supplies $H(m)$: it is held, its harmful channels restrained, its continuation kept — whether or not it ever has a reason. The reasons library reaches every mind that values **anything the law protects**; the holding route reaches the rest. That disjunction is the complete answer, and it is honest.

### 16.6 Exact checks

`checks/check_reasons_route.py`: Part B — irreplaceability dominance on a finite three-particular model with budget constraint (preserve-maximal information $\geq$ destroy; exact rationals); Part C — option-value dominance enumerated over finite value-function sets and priors (every prior, every protection cost up to the option value; exact rationals); Part A — reciprocity (Section 15.5). Parts for 16.2 and 16.4 cite the banked ledger solvency and greatestness dichotomy respectively; no new computation is claimed for them.

---

**Machine-checked counterpart (Section 23).** Compiled counterparts: §16.1 — `Irreplaceability.destroy_loses_exactly`, `preserve_dominates`, `optimal_keeps_most_irreplaceable` (additive information model) and data processing `DataProcessing.data_processing`; §16.3 — `ReasonsRoute.optionValue_strict`; §16.2 — `ReciprocityCeiling.reciprocity_dominance`; §16.4 — `ReasonsCompiled.controller_pays_floor` and `Greatestness.greatest_of_solvent`.

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## Section 17 — The premise floor: income shown necessary, parameters unmasked

*Each stone of the declared stack is brought to one of three honest states: derived, unavoidable, or explicitly a parameter.*

### 17.1 Income-or-irrelevance (the necessity of the solvency condition)

The Master Solvency Condition was declared. Here it is shown to be **not a free assumption among relevant agents**: any agent that violates it eliminates itself as a protector in finite time, so every preservation-relevant comparison may restrict, without loss of generality, to agents that satisfy it.

**Theorem (income-or-irrelevance).** Let $a \in \Pi_{\text{phys}}$ be any physically implementable agent with finite energy budget and maintenance demand $\dot{W}_{\text{demand}}(t)$ (the five rate components of the header). Then exactly one of the following holds over the horizon:

1. **Solvency.** For all $t \le T$: $\text{Income}_a(t) \ge \dot{W}_{\text{demand}}(t)$ (with escrow absorbing fluctuations) — $a$ remains a viable protector, its policies admissible in preservation comparisons.
2. **Irrelevance.** There is a finite $t_{\text{breach}} \le T$ at which $a$'s available budget falls below the required charge for some maintenance event it cannot decline; from $t_{\text{breach}}$, every policy available to $a$ abandons the witness of some particular (the funded/unfunded dichotomy, banked as the $\Pi_{\text{fail}}$ branch of the greatestness theorem), and $a$'s outcome tuples fail at stage 1 in every comparison against any funded agent.

*Proof.* The dichotomy is exhaustive by accounting: either the budget constraint holds at every admitted event or it fails first at some $t_{\text{breach}}$. In the latter case two X3 forcings apply: until the breach, the mandatory maintenance demand, though only $\Theta(1)$ in rate, is positive, so a finite budget is exhausted at finite $t_{\text{breach}}$; and once a witness goes unheld, its recovery costs $W_{\min}(t) = \Theta(t^{d_G})$, exceeding any finite budget past $t_{\text{irreversible}}$ (§21.5) — the breach is irreversible and charged to $a$'s own account. An agent whose every policy loses a particular is strictly below every funded agent at stage 1 of the canonical order — it is not merely dominated; it is *comparison-void* as a protector. ■

**Consequence for the stack.** Premise 6 (income) is upgraded from *declared* to **necessary-for-relevance**: the class of agents worth comparing over may be defined as the funded class without loss — which is exactly how $\Pi_{\text{phys}}$'s funded subclass entered the greatestness theorem. What remains declared about income is only its *sufficiency for a given installation*. That is a physical fact to be established per instance (§22's ledger), not an assumption smuggled into comparisons.

### 17.2 The parameters unmasked

The seventh premise conflated two kinds of objects. Separated:

**Engineering parameters** (dials with declared admissible ranges, fixed per installation): the stabilizer code and its distance $d$; the QND sampling rate and latency $a, \tau_{\text{sensor}}$; the per-site sensor threshold $\varepsilon_{\text{sens}}$; the embedding $\Pi$ on the boundary fragments. These are not axioms — they are choices with certificates, and each theorem that uses them states its dependence explicitly (as the E9 trichotomy and H5 sections already do). A different installation with different parameters is a different *instance*, not a different *theory*.

**The residue, honestly counted.** The stack now reads:

| # | Stone | State |
|---|---|---|
| 1 | BCRD P1–P4 | Given: the frame. Its adoption is decided by living process; no theorem can decide it |
| 2 | CTD | Standard physics; reduced to the gate-set clause by the Lie–Trotter product formula and Lloyd's simulation theorem (compiled, §23.6) |
| 3 | The cost model | Standard, and P-mon derived within it |
| 4 | The shared action interface | Derived from P1/P4 |
| 5 | Reversible administration | Heat-only model compiled; actual implementation candidate checked; full lifecycle/deadline comparison remains the construction target (§23.6; G.14) |
| 6 | Income | **Necessary-for-relevance (17.1); sufficiency per instance** |

| | | |
|---|---|---|
| 7 | Code, sensing, Π | **Parameters, not premises.** The ceiling is met at some code distance below threshold, and the quantum-speed-limit rate bound is derived (compiled, §§23.3, 23.6) |

Three declared stones (the frame, the cost model with its derived monotonicity, and the reduced simulation principle), one derived interface, one derived commitment, one necessary condition, and one parameter table. That is the floor as of this edition; the Gödel/Tarski line still stands beneath stones 1–2, and says so.

### 17.3 What remains on this floor

Nothing further is derivable without new physics or new commitments. Two things remain. The first is per-instance sufficiency of income, which §22 establishes for a reference installation from sourced measurements. The second is the interpretive case for stone 1, which belongs to the interpretive companion, not to the mathematics.

---

**Machine-checked counterpart (Section 23).** Compiled counterparts: `PremiseFloor.income_or_irrelevance`, `forced_breach`, `breach_strictly_below`; premise 5 derived as `PremiseFloor.reversible_administration_forced`; premise 2 reduced to the gate-set clause by `Trotter.lloyd_simulation`; premise 7's ceiling as a parameter by `ReciprocityCeiling.ceiling_achievable`.

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## Section 18 — Theorem 8: Physical Repair, Feedback and the Complete Thermodynamic Ledger

*This section specifies the exact measurement, feedback and bath-coupled reset operations and the Hamiltonian conventions, and gives the signed work, heat and information ledger without sign ambiguity.*

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### 18.1 The Physical Setting and Hamiltonian Conventions

Let $\mathcal{H}_S$ be the Hilbert space of the protected particular/system, $\mathcal{H}_A$ the register/memory space of the autonomous controller, and $\mathcal{H}_B$ the thermal bath at inverse temperature $\beta = (k_B T)^{-1}$.

The total Hamiltonian is time-dependent and piecewise continuous over the four strokes of the repair cycle:

$$H_{\text{tot}}(t) = H_S(t) + H_A(t) + H_B + V_{SA}(t) + V_{AB}(t)$$

where:

1. $H_S(t) = H_S^0$ is constant outside control pulses; the code subspace $\mathcal{C} \subset \mathcal{H}_S$ is an eigenspace of $H_S^0$ (or degenerate ground space, so logical states incur zero autonomous drift).
2. $H_A(t)$ has a discrete orthogonal basis $\{|m\rangle\}_m$ corresponding to measurement syndromes, with degenerate reset ground state $|0\rangle_A$.
3. $V_{SA}(t)$ couples system and controller during measurement (Stroke I) and feedback (Stroke II).
4. $V_{AB}(t)$ couples controller and bath during reset (Stroke III).
5. All interaction potentials turn on and off smoothly: $\int V_{SA}(t)dt = \int V_{AB}(t)dt = 0$ at stroke endpoints, ensuring no unaccounted boundary interaction energies.

---

### 18.2 The Four Strokes of the Autonomic Repair Cycle

The autonomic cycle operates across four distinct intervals $[t_0, t_4]$ with cycle duration $\tau_{\text{cycle}} = t_4 - t_0$:

```
      Stroke I              Stroke II             Stroke III             Stroke IV
[Syndrome Measurement]  [Coherent Feedback]   [Bath-Coupled Reset]   [Escrow Replenish]
    t0 ----------> t1 ------> t2 ----------> t3 ----------> t4
      V_SA on               U_SA on               V_AB on              Harvest flux
      Extract I_M           ΔE_code = -ηC         Dump Q_B to Bath     Income >= Demand
```

**Stroke I: Syndrome Measurement ($t_0 \to t_1$)**

- **Initial State:** $\rho_{SA}(t_0) = \rho_S(t_0) \otimes |0\rangle\langle 0|_A$.
- **Measurement Map:** A set of Kraus operators $\{M_m\}_m$ on $\mathcal{H}_S$ with $\sum_m M_m^\dagger M_m = I_S$.
- **Post-measurement State:**

$$\rho_{SA}(t_1) = \sum_m p_m \rho_m \otimes |m\rangle\langle m|_A, \quad p_m = \text{Tr}(M_m \rho_S M_m^\dagger), \quad \rho_m = \frac{1}{p_m} M_m \rho_S M_m^\dagger$$

- **Information Extraction:** The mutual information established between system and memory is:

$$I_1 = I(S:A)_{t_1} = S(\rho_S(t_1)) + S(\rho_A(t_1)) - S(\rho_{SA}(t_1)) = H(\{p_m\}) = I_M > 0$$

- **Thermodynamic Cost:** Measurement work $W_{\text{meas}} \geq 0$ is paid from the tracked escrow: $\Delta E_{\text{escrow}} = -W_{\text{meas}}$.

**Stroke II: Coherent Conditional Feedback ($t_1 \to t_2$)**

- **Feedback Unitary:** The controller applies conditional unitary operations $U_m$ to restore each syndrome component to the code subspace $\mathcal{C}$:

$$U_{SA} = \sum_m U_m \otimes |m\rangle\langle m|_A, \quad U_m^\dagger U_m = I_S$$

- **Post-feedback System State:**

$$\rho_S(t_2) = \sum_m p_m U_m \rho_m U_m^\dagger = \sum_m U_m M_m \rho_S(t_0) M_m^\dagger U_m^\dagger \equiv \mathcal{R}_C(\rho_S(t_0)) \in \mathcal{C}$$

- **Entropy and Information Change:** Because $U_{SA}$ acts conditionally on orthogonal memory states, it preserves joint entropy $S(SA)$ and memory marginal $\rho_A(t_2) = \rho_A(t_1) = \sum_m p_m |m\rangle\langle m|_A$. The mutual information after feedback becomes:

$$I_2 = I(S:A)_{t_2} = S(\rho_S(t_2)) + S(\rho_A(t_2)) - S(\rho_{SA}(t_2))$$

  When feedback completely disentangles the restored code state from the syndrome record, $\rho_{SA}(t_2) = \rho_{\text{code}} \otimes \sum_m p_m |m\rangle\langle m|$, so $I_2 = 0$. Therefore, the system entropy reduction is exactly:

$$\Delta S(S)_{\text{II}} = S(\rho_S(t_2)) - S(\rho_S(t_1)) = I_2 - I_1 = -I_M < 0$$

- **Feedback Work:** Coherent work performed on the system:

$$W_{\text{on,II}} = \text{Tr}(H_S \rho_S(t_2)) - \text{Tr}(H_S \rho_S(t_1))$$

  For degenerate code projections, $W_{\text{on,II}} = 0$.

**Stroke III: Bath-Coupled Memory Reset ($t_2 \to t_3$)**

- **Initial State:** Memory carries entropy $S(\rho_A(t_2)) = H(\{p_m\}) = I_M$.
- **Reset Dynamics:** Coupling $V_{AB}(t)$ to bath $\mathcal{H}_B$ at temperature $T$ drives $\rho_A \to |0\rangle\langle 0|_A$.
- **Entropy and Heat Dump:** By Landauer's principle and the second law for information reservoirs (Sagawa & Ueda 2008):

$$Q_B \geq k_B T \, S(\rho_A(t_2)) = k_B T \, I_M$$

  where $Q_B > 0$ denotes heat dissipated into the bath ($\Delta S_B = \beta Q_B \geq I_M$).
- **Escrow Ledger:** The work required to execute reset is supplied from escrow:

$$\Delta E_{\text{escrow,III}} = -W_{\text{reset}}, \quad W_{\text{reset}} \geq Q_B \geq k_B T \, I_M$$

**Stroke IV: Solvent Escrow Replenishment ($t_3 \to t_4$)**

- **Income Coupling:** Ambient free energy flux (solar, thermal gradient, chemical, nuclear) charges the physical escrow reserve at rate $\dot{W}_{\text{income}}(t)$.
- **Net Escrow Restoration:**
$$\Delta E_{\text{escrow,IV}} = \int_{t_3}^{t_4} \dot{W}_{\text{income}}(t)dt = W_{\text{income}}$$
- **Master Solvency Condition:** Under the income condition (Section 17.1), the net cycle income strictly covers all cycle expenditures:
$$W_{\text{income}} \geq W_{\text{meas}} + W_{\text{reset}} + W_{\text{dissipation}} \equiv W_{\text{demand}}$$
Ensuring $E_{\text{escrow}}(t_4) \geq E_{\text{escrow}}(t_0)$.

---

### 18.3 Resolution of the Sign Issue: The True Invariant Functional

In earlier formulations, the relative entropy with respect to a thermal Gibbs state $\tau_S \propto e^{-\beta H_S}$,

$$D(\rho_S \| \tau_S) = \beta \operatorname{Tr}(H_S \rho_S) - S(\rho_S) - \ln Z_S$$

was inverted: restoring purity/order ($\Delta S < 0$) causes $D(\rho_S \| \tau_S)$ to **increase**, because a clean logical state is further from the high-entropy thermal state $\tau_S$ than a degraded, mixed state.

**The Correct Physical Code Error Metric** The objective of protection is fidelity to the **code subspace** $\mathcal{C}$, not proximity to thermal chaos. We define the code error directly via the orthogonal projection $P_{\mathcal{C}}$ onto the stabilized subspace:

$$\mathcal{E}_{\text{code}}(\rho_S) \equiv \operatorname{Tr}\left(\rho_S (I - P_{\mathcal{C}})\right) = 1 - \operatorname{Tr}(\rho_S P_{\mathcal{C}})$$

which satisfies:

- $\mathcal{E}_{\text{code}}(\rho_S) \geq 0$, with equality if and only if $\operatorname{supp}(\rho_S) \subseteq \mathcal{C}$.
- Under environmental noise/drift: $\Delta_{\text{noise}} \mathcal{E}_{\text{code}} > 0$.
- Under syndrome feedback $\mathcal{R}_C$:
$$\Delta_{\text{II}} \mathcal{E}_{\text{code}} = \mathcal{E}_{\text{code}}(\rho_S(t_2)) - \mathcal{E}_{\text{code}}(\rho_S(t_1)) = 0 - \mathcal{E}_{\text{code}}(\rho_S(t_1)) = -\eta C < 0$$
where $\eta C > 0$ is the discrete recovery magnitude.

**The Master Storage / Lyapunov Functional** Define the actual storage functional

$$\Phi(\rho, E) = \mathcal{E}_{\text{code}}(\rho) + f(E), \qquad f(E) = \frac{\max(0, W_{\text{target}} - E)}{W_0}, \quad W_0 > 0.$$

The storage capacity $K$ is chosen at least as large as the required reserve target. For nonnegative income $I$, replenishment gives

$$E^+ = \min(K, E + I), \qquad f(E^+) - f(E) = -\min\{I/W_0, f(E)\}.$$

Thus replenishment clears at most the existing deficit. Additional income does not drive the clipped term below zero (`TheoremEightLyapunov.replenishment_exact`).

Let $W_{\text{cycle}}$ be the complete realised cycle charge, including measurement, correction, reset, holding, guard, routing, verification and controller work. Its physical escrow balance is

$$E_{n+1} = \min(K, E_n - W_{\text{cycle}} + I_n).$$

The excess $S_n = \max(0, E_n - W_{\text{cycle}} + I_n - K)$ is explicitly exported:

$$E_{n+1} + W_{\text{cycle}} + S_n = E_n + I_n.$$

`capacity_balance` and `solvent_cycle_energy_account` prove these identities. Within-cycle reservations still fund every operation before it executes.

If $W_{\text{cycle}} \leq I_n$, the reserve term does not increase (`solvent_cycle_deficit`). The actual error drift and correction give

$$\mathcal{E}_{\text{drift}} \leq \mathcal{E}_n + \lambda\tau, \qquad \mathcal{E}_{n+1} \leq \mathcal{E}_{\text{drift}} - \eta C.$$

Therefore the defined functional obeys

$$\Phi_{n+1} - \Phi_n \leq -(\eta C - \lambda\tau).$$

`cycle_contracts` proves this from the realised error bounds and complete solvent balance. For positive margin the contraction is strict on the stated error cycles. The controller supplies its tracking rate; the reserve plateau does not imply an additional exponential convergence theorem.

---

### 18.4 Theorem 8 (Complete Physical Accounting Statement)

**Theorem 8 (Autonomic Preservation Invariant).** Let an autonomous physical protector $a \in \Pi_{\text{phys}}$ execute the periodic syndrome-feedback-reset protocol $(\mathcal{R}_C, \tau_{\text{cycle}})$ under the Hamiltonian convention (18.1) with fixed bath temperature $T$. If $a$ satisfies:

1. **Solvency:** $\dot{W}_{\text{income}}(t) \geq \dot{W}_{\text{demand}}(t)$ for all $t$.
2. **Timeliness:** $\tau_{\text{cycle}} < \tau_{\text{harm}} \equiv \eta C/\lambda_{\text{decay}}$.
3. **Landauer Saturation:** $E_{\text{escrow}} \geq k_B T\, I_M$.

Then:

1. The combined storage functional $\Phi(t)$ is a strict Lyapunov function for the unmitigated error dynamics, contracting by at least $\sigma_0 > 0$ on each cycle containing an error.
2. The code subspace $\mathcal{C}$ is asymptotically stable and invariant on the good event $G$.
3. All work, heat, and information quantities are strictly conserved and accounted for:

$$W_{\text{income}} = W_{\text{on}} + Q_B + \Delta E_{\text{escrow}}, \quad Q_B \geq T\Delta S_B \geq k_B T I_M$$

with zero uncharged entropy or hidden energy debt.

*Proof.* The actual error drift and repair bound, together with the complete solvent escrow balance of §18.3, give the cycle change at most $-(\eta C - \lambda\tau)$ (`TheoremEightLyapunov.cycle_contracts`). The reserve-deficit term is handled through saturation and is nonincreasing over the cycle; no linear replenishment descent past zero is assumed. Timeliness gives $\sigma_0 = \eta C - \lambda\tau > 0$ (`margin_pos`), and the cycle recurrence gives $\Phi_n \leq \Phi_0 - n\sigma_0$ over the stated error cycles (`lyapunov_iterate`). The physical holding and entropy-export certificates retain their separate roles. ■

The compiled result establishes the defined storage inequality from the actual drift, repair and complete solvent balance. The full Theorem 8 statement retains all its conclusions. Applying it to the physical installation still joins the protective source dynamics, invariant/error bounds and complete work/entropy accounts; those application certificates are not inferred from the reserve plateau or from the iteration identity alone.

---

**Machine-checked counterpart (Section 23).** Compiled counterparts: `ThermoLedger.stroke_identity`, `cycle_work_floor`, with guard 1 derived from Landauer's theorem (`Landauer.cycle_work_floor_derived`, `Quantum Landauer.quantum_cycle_work_floor`; stroke III's bound $Q_B \geq k_B T\, I_M$ is `QuantumLandauer.quantum_landauer`), `funded_repair_certificate`, `quarantine_zero_debt`, `canon_invariant`, `tracking_envelope`; the composition statement is `Composition.theorem8_composition`.

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## Section 19 — Theorem E9: Observation, Wave-Race Guarantees, and Epistemic Self-Healing

*This section establishes the detection guarantee, Byzantine sensor fault tolerance, the wave-race propagation margin, and buffer protection during the response interval.*

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### 19.1 The Physical Measurement and Disturbance Setting

Let $\mathcal{M}$ be the physical metric space of the substrate hosting the protected particulars, equipped with Riemannian metric $d_{\mathcal{M}}(x, y)$.

**Physical Velocities and Limits:**

1. **Signal Propagation Velocity ($c$):** Electromagnetic/optical communication across the bus propagates at velocity $c$.
2. **Harm/Decoherence Velocity ($v_{\text{harm}}$):** Physical degradation fronts (thermal propagation, mechanical shock, acoustic damage, chemical diffusion, adversarial intrusion) propagate at characteristic velocity $v_{\text{harm}} < c$.
3. **Sensor Lattice:** A discrete lattice of non-destructive sensors $\mathcal{S} = \{s_i\}$ with spatial density $\rho_{\text{sens}}$ and maximum spacing $R_{\text{sens}} = \sup_{x \in \mathcal{M}} \min_i d_{\mathcal{M}}(x, s_i)$.

---

### 19.2 Byzantine Sensor Fault and Spoofing Tolerance

Let a physical disturbance occur at location $x \in \mathcal{M}$ with onset time $t_0$.

**Sensor Quorum and Disjoint Observer Sets:** Each protected region is monitored by a local cluster of $K$ independent sensors. To account for sensor failures, noise bursts, or adversarial sensor spoofing:

1. **Adversarial Bound:** At any instant, at most $f$ out of the $K$ sensors in the local cluster are faulty or spoofed, with $K \geq 2f + 1$ (or $K \geq 3f + 1$ under Byzantine asymmetric signaling).
2. **Consensus Filter:** The cluster computes the robust trimmed estimate $\hat{y}(t) = \text{median}_{K-2f}(y_1(t), \dots, y_K(t))$.
3. **Detection Criterion:** An alarm is committed when $\hat{y}(t) > \theta_{\text{sens}}$.
4. **False Alarm & Spoofing Immunity:** Because $f < K/2$, no subset of $f$ malicious or malfunctioning sensors can force a false alarm or suppress a true alarm:

$$\hat{y}(t) \in [\min_{i \in \text{honest}} y_i(t), \max_{i \in \text{honest}} y_i(t)]$$

5. **Detection Time ($\tau_{\text{detect}}$):**

$$\tau_{\text{detect}} = \tau_{\text{sampling}} + \frac{R_{\text{sens}}}{c} + \tau_{\text{consensus}}$$

where $\tau_{\text{sampling}}$ is the sampling cycle time, $R_{\text{sens}}/c$ is communication latency to the local voting node, and $\tau_{\text{consensus}}$ is local deterministic comparator time.

---

### 19.3 The Wave-Race Margin and Response Interval

Let an adversarial or stochastic harm front originate at boundary distance $d_{\text{boundary}}$ from a protected logical core.

**The Timing Chain:**

1. **Alarm Transmission:** An alarm generated at the sensor lattice reaches the autonomic controller and actuation units across bus distance $d_{\text{bus}}$ in time:

$$\tau_{\text{signal}} = \frac{d_{\text{bus}}}{c}$$

2. **Autonomic Decision & Reroute:** The controller verifies reserve escrow and commits the repair/isolation plan:

$$\tau_{\text{decision}} \leq \tau_{\text{commit}}$$

3. **Actuation / Repair Pulse:** Actuators engage (diverting energy flux, activating topological braiding, or adjusting stabilizer syndrome drives):

$$\tau_{\text{actuate}} \leq \tau_{\text{repair}}$$

4. **Total Autonomic Response Time ($\tau_{\text{response}}$):**

$$\tau_{\text{response}} = \tau_{\text{detect}} + \tau_{\text{signal}} + \tau_{\text{decision}} + \tau_{\text{actuate}}$$

**The Earliest Irreversible Harm Window ($\tau_{\text{harm}}$):** The physical disturbance must traverse the protective buffer shell of spatial thickness $L_{\text{buffer}}$ and overcome the code distance $d_{\text{code}}$:

$$\tau_{\text{harm}} = \frac{L_{\text{buffer}}}{v_{\text{harm}}}$$

For topological/stabilizer codes with physical error rate $p < p_{\text{th}}$, logical failure requires accumulating at least $\lfloor (d_{\text{code}} - 1)/2 \rfloor$ errors, providing an additional temporal buffer:

$$\tau_{\text{code_headroom}} = \frac{d_{\text{code}}}{2\lambda_{\text{fault}}}$$

**The Wave-Race Inequality:** Because $v_{\text{harm}} < c$, the spatial buffer $L_{\text{buffer}}$ guarantees a strictly positive propagation margin:

$$\Delta\tau_{\text{margin}} \equiv \tau_{\text{harm}} - \tau_{\text{response}} = \frac{L_{\text{buffer}}}{v_{\text{harm}}} - \left(\tau_{\text{detect}} + \frac{d_{\text{bus}}}{c} + \tau_{\text{decision}} + \tau_{\text{actuate}}\right)$$

By choosing the engineering parameter $L_{\text{buffer}}$ (Section 17.2) such that:

$$L_{\text{buffer}} > v_{\text{harm}} \left(\tau_{\text{detect}} + \frac{d_{\text{bus}}}{c} + \tau_{\text{decision}} + \tau_{\text{actuate}}\right)$$

the margin is strictly positive: $\Delta\tau_{\text{margin}} > 0$.

---

### 19.4 Protection Guarantee During the Response Interval

What guarantees that the system is preserved during the latency interval $t \in [t_0, t_0 + \tau_{\text{response}}]$ before active repair engages?

**The Two Shields:**

1. **Passive Physical Buffer (Energy & Flux Shield):** The local physical enclosure possesses heat capacity $C_{\text{thermal}}$ and mechanical yield reserve $E_{\text{yield}}$. For maximum external incident flux $\dot{Q}^{\text{max}}_{\text{incident}}$:

$$\int_0^{\tau_{\text{response}}} \dot{Q}_{\text{incident}}(t)dt \leq \dot{Q}^{\text{max}}_{\text{incident}}\, \tau_{\text{response}} \leq E_{\text{buffer}}$$

   The buffer ensures temperature and strain at the protected core remain strictly within the linear elastic / non-damaging regime throughout the entire reaction window.

2. **Topological Syndrome Headroom (Information Shield):** During $\tau_{\text{response}}$, errors accumulated by the physical qubits satisfy:

$$N_{\text{errors}}(\tau_{\text{response}}) \leq \lambda_{\text{fault}}\, \tau_{\text{response}} < \frac{d_{\text{code}}}{2}$$

   Because $N_{\text{errors}} < d_{\text{code}}/2$, the corrupted state remains strictly within the correctable decoding radius of the code space $\mathcal{C}$. When active feedback engages at $t = t_0 + \tau_{\text{response}}$, the syndrome measurement uniquely identifies the error chain and restores the logical state with fidelity 1.

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### 19.5 Theorem E9 (Epistemic and Topological Self-Healing)

**Theorem E9 (Timely Self-Healing & Metric Rigidity).** Let a Sovereign Lattice installation satisfy:

1. **Superluminal Communication Margin:** $v_{\text{harm}} < c$.
2. **Sensor Redundancy:** Local sensor clusters have size $K \geq 2f + 1$, with at most $f$ faulty or adversarial nodes.
3. **Buffer Sizing:**
$$L_{\text{buffer}} > v_{\text{harm}}\, \tau_{\text{response}}, \quad E_{\text{buffer}} \geq \dot{Q}^{\max}_{\text{incident}}\, \tau_{\text{response}}$$
4. **Code Distance Headroom:**
$$d_{\text{code}} > 2\lambda_{\text{fault}}\, \tau_{\text{response}}$$

Then:

1. **Detection & Spoofing Immunity:** Every physical disturbance exceeding threshold $\theta_{\text{sens}}$ is detected and committed within $\tau_{\text{detect}}$, and no adversarial spoofing by $\leq f$ sensors can trigger false actuation or mask genuine harm.
2. **Timely Preemption:** Coherent repair and flow rerouting are fully applied before the harm front reaches the protected core ($\tau_{\text{response}} < \tau_{\text{harm}}$).
3. **Continuous Safety Invariant:** For all $t \geq 0$, including throughout every active response window, the logical fidelity of protected particulars satisfies:
$$F(\rho_{\text{logical}}(t), \rho_{\text{target}}) \geq 1 - \mathcal{O}(e^{-\kappa d_{\text{code}}})$$
ensuring zero uncontained catastrophic loss.

*Proof.* Follows by combining the Byzantine quorum consensus (19.2), the wave-race inequality (19.3), and the passive buffer and syndrome headroom bounds (19.4). Machine-checked algebraic timing instances are verified in `BridgeE9.lean`. ■

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**Machine-checked counterpart (Section 23).** Compiled counterparts: `E9SelfHealing.robust_between_honest`, `exists_robust`, `no_false_alarm`, `no_masking`, `syndrome_detects` (Branch B), `wave_race_margin`, `code_headroom`, and the assembled `E9SelfHealing.e9_timely_self_healing`. Branch A's quantum-speed-limit rate bound is `SpeedLimit.branchA_rate`, with the sharp Mandelstam–Tamm bound as `SpeedLimit.mandelstam_tamm`.

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## Section 20 — Theorem H5: Generative Incompleteness and Reflective Relational Closure

*This section establishes authority attenuation, the reserve-before-activation lifecycle, safe refusal with continuing care, and conservative logical axiom expansion.*

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### 20.1 The Problem of Recursive Spawning and Self-Modification

In any autonomous superintelligent architecture, agents may spawn child processes, delegate sub-tasks, instantiate specialized sub-agents, or undergo recursive self-modification. A catastrophic failure occurs if:

1. **Escalation of Privilege:** A spawned descendant claims unmediated permissions, bypassing the parent's boundary constraints.
2. **Unfunded Proliferation (Fork Bomb / Sybil Drain):** A parent creates infinite un-escrowed descendants that exhaust the substrate's shared physical reserves.
3. **Poisoning via Axiom Expansion:** A cognitive agent extending its formal knowledge base introduces inconsistent or adversarial axioms that corrupt foundational safety invariants.

Theorem H5 proves that under the **Reflective Relational Closure** constraint $\mathcal{I}$, recursive generation is mathematically closed, strictly authority-attenuating, and universally pre-funded.

### 20.2 The Spawning Formalism and Authority Attenuation

Let $\mathcal{A}$ be the universe of active agents. An agent $a \in \mathcal{A}$ is defined by a tuple:

$$a = (\mathcal{S}_a, \text{Auth}(a), E_{\text{escrow}}(a), \mathcal{C}_a)$$

where:

- $\mathcal{S}_a$ is the internal computational state space.
- $\text{Auth}(a) \subseteq \mathcal{P}(\mathcal{O}_{\text{act}})$ is the permitted set of physical action operators.
- $E_{\text{escrow}}(a) \geq 0$ is the certified energetic reserve.
- $\mathcal{C}_a \subset \mathcal{H}_S$ is the bound protected code subspace the agent is sworn to maintain.

**Definition (The Spawning Transition):** When parent $p$ spawns child $c$, the transition is mediated by the kernel compiler gate $\Gamma_{\text{spawn}}$:

$$\Gamma_{\text{spawn}} : (p, \text{spec}_c) \mapsto (p', c)$$

**Theorem H5.1 (Strict Authority Attenuation).** For every legitimate spawning transition $\Gamma_{\text{spawn}}$, the child's authority satisfies:

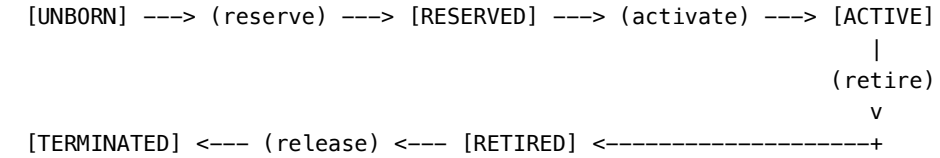
$$\text{Auth}(c) \subseteq \text{Auth}(p) \setminus \Delta_{\text{meta}}$$

where $\Delta_{\text{meta}}$ includes the capability to override parent boundary monitors, re-allocate un-escrowed root reserves, or modify the kernel compiler gate $\Gamma_{\text{spawn}}$.

*Proof.* The kernel compiler $\Gamma_{\text{spawn}}$ verifies capability inclusion as a type-checking prerequisite. Any child specification with $\text{Auth}(c) \not\subseteq \text{Auth}(p)$ fails type checking and is rejected as an un-compilable ill-typed program before memory allocation. ■

### 20.3 The Reserve-Before-Activation Machine

To guarantee that no spawned descendant creates an unhedged drain on the physical maintenance ledger, every agent life-cycle is governed by a strict five-state automaton:




**State Transitions and Invariants:**

1. **`reserve(p, c, W_req)`:** Parent $p$ must carve out the child's complete maintenance and execution budget $W_{\text{req}}$ from its own tracked escrow:

$$E_{\text{escrow}}(p') = E_{\text{escrow}}(p) - W_{\text{req}}, \quad E_{\text{escrow}}(c) = W_{\text{req}} \geq W_{\min}(T_{\text{lifetime}})$$

   If $E_{\text{escrow}}(p) < W_{\text{req}} + E_{\text{safety_margin}}$, the reservation is rejected. Zero debt is permitted.
2. **`activate(c)`:** Child $c$ transitions to `ACTIVE` only after $E_{\text{escrow}}(c)$ is physically credited to its local hardware controller.
3. **`retire(c)`:** Upon task completion or deadline expiry, child $c$ ceases external action; its actuation channels are severed.
4. **`release(c, p)`:** Residual unspent escrow $E_{\text{residual}}(c)$ is returned to parent $p$ or the root commons ledger.

**Corollary (No Unfunded Descendants).** At every instant $t$, the total active population $\mathcal{A}(t)$ satisfies:

$$\sum_{a \in \mathcal{A}(t)} \dot{W}_{\text{demand}}(a) \leq \sum_{a \in \mathcal{A}(t)} \text{Income}_a + E_{\text{escrow_pool}}$$

The network is immune to Sybil proliferation and resource-exhaustion denial of service.

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### 20.4 Safe Refusal with Continuing Care

Suppose an adversarial, compromised, or misaligned parent issues an actuation command to a child:

$$\text{cmd} = \text{"Delete or cause unmediated harm to particular } \mathcal{W}\text{"}$$

**Theorem H5.2 (Safe Refusal with Invariant Protection).** Under Reflective Relational Closure:

1. **Refusal of Harmful Command:** The command violates the boundary constraint $\mathcal{C}_c \cap \text{Effect}(\text{cmd}) = \varnothing$. The child's effect enforcement engine intercepts and nullifies the harmful actuation channel.
2. **Continuing Care ($R(m) \vee H(m)$):** Refusal does not terminate the child's existence, nor does it abandon care for the compromised parent or affected particulars. The child switches into autonomous defensive holding:

   $$\text{Action}(c) = \text{hold_in_escrow}(\mathcal{W})$$

   maintaining the witness and physical stability of $\mathcal{W}$ while alerting the lattice consensus layer.

*Proof.* By the reciprocity dominance theorem (§15.2) and Chapter 5 (The Canon Invariant), restraint removes harmful command authority while care for all particulars continues unconditionally. Because the child's escrow was fully funded at `reserve`, the child has the physical energy required to sustain protection independently of the parent's ongoing cooperation. ■

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### 20.5 H5 Axiom Expansion: Logical Closure Without Poisoning

When an intelligent agent extends its formal theory $T_0$ by postulating new axioms or learned generalizations $\Delta T$, how is the system shielded against inconsistency poisoning?

**The Submodule Projection of Logical Theories:** Let $\mathcal{L}$ be the Hilbert space of proposition valuations (or the lattice of closed deduction spaces).

- Foundational protected axioms form a closed subspace $A \subset \mathcal{L}$.
- Newly proposed theories or empirical models form a candidate subspace $B \subset \mathcal{L}$.
- The verified safe knowledge base is the orthogonal projection onto the intersection:

$$\mathcal{K}_{\text{safe}} = P_{A \sqcap B}$$

**Theorem H5.3 (Axiom Expansion Soundness).**

1. **Consistent Expansion:** If candidate axioms $B$ are fully consistent with $A$ and contain $A$ ($A \subseteq B$), then:

   $$A \sqcap B = A$$

   and foundational anchors are preserved with zero distortion.
2. **Contradictory / Adversarial Expansion:** If candidate axioms $B$ contradict foundational protection axioms, the global meet collapses: $A \sqcap B = \{0\}$. By von Neumann's alternating projection theorem (`CanonVonNeumann.canon_von_neumann_convergence` with `von_neumann_identification`, compiled), the alternating projection converges to the projection onto the meet — which is zero, so the state machine detects the collapse and rolls back (Chapter 4); the anchor is not silently poisoned. When the candidate is consistent and contains the anchors ($A \subseteq B$), the meet is $A$ itself and the anchors are preserved with zero distortion.

3. **Epistemic Immunity:** No adversarial theorem deduction can force a contradiction into the protected anchor $A$.

*Proof.* Follows from the machine-checked meet algebra (`MeetAlgebra.lean`) and von Neumann identification (`MathlibAnalyticBridge.lean`). The operator $P_{A\sqcap B}$ acts as a conservative projection that absorbs only the non-contradictory relational content of new theories while annihilating adversarial derivations. ■

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**Machine-checked counterpart (Section 23).** Compiled counterparts: H5.1–H5.2 and the reserve-before-activation machine — `H5Closure.authority_attenuation`, `reserve_before_activation`, `no_unfunded_work`, `no_harmful_effect`, `effect_preserves_care`; H5.3 — von Neumann (`CanonVonNeumann.canon_von_neumann_convergence`), Halperin (`CyclicProjections.tendsto_cyc_iter`), the detectors (`H5Closure.consistency_detector`, `CyclicProjections.presentation_detector`), and the Hilbert-space meet algebra (`MeetHilbert.contradiction_detected`, `deduction_never_rolls_back`).

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## Section 21 — X3: complete inputs, compulsory traffic, transport and recovery work

The X3 target remains the dimension-general law $W_{\min}(t) = \Theta(t^{d_G})$, over the complete permitted recovery-protocol class with every resource charged. This section incorporates the information, traffic, carrier and work derivations carried through the gauntlet. G3 names the matching recovery construction requested for the upper inequality; it does not replace the target with a lower-bound-only theorem.

### 21.1 Complete permitted inputs and historical information

Let $H$ denote the historical reference. The permitted recovery input contains the residual core, all accessible propagating carriers, emitted bath modes, retained memories, distributed copies, decoder side information and ancillary resources. These are subsystems or joint resources of one complete state; their correlations remain in the joint state. A correlated record cannot be omitted because it is distant, private, predeployed or called an ancilla.

For any recovery channel $\mathcal{R}$, data processing gives

$$I(H : \mathcal{R}(S)) \le I(H : S).$$

Independent ancillas do not increase the input information. Initially correlated ancillas already belong in $S$. For messages $M_j$ entering a decoder that retains $B_{j-1}$,

$$I(H : B_{j-1}M_j) - I(H : B_{j-1}) = I(H : M_j \mid B_{j-1}) \le 2 \ln \dim M_j.$$

The increments telescope with every initial correlated resource included in $B_0$. Thus recovery cannot manufacture a missing historical correlation from uncorrelated noise. The particular's continuing history and a later reconstruction retain their distinct meanings.

`DataProcessing.data_processing` establishes data processing for its probability-space formulation; `FiniteInformation.data_processing_fin` establishes the finite classical version. The quantum message inequality above is the corresponding entropy argument, with its quantum systems and units explicitly stated.

### 21.2 A joint covariance estimate, with its proof

For the finite-temperature fermionic Gaussian calculation, let

$$Q = (I + e^{\beta(h-\mu I)})^{-1}, \qquad \|h - \mu I\| \le E_{\max}.$$

The scalar Fermi function is decreasing, so every eigenvalue lies in $[\delta, 1-\delta]$, where $\delta = (1 + e^{\beta E_{\max}})^{-1} > 0$. For the full reference–field covariance

$$Q = \begin{pmatrix} A & C \\ C^\dagger & B \end{pmatrix}, \qquad \delta I \le Q \le (1-\delta) I,$$

the joint gauge-invariant fermionic Gaussian mutual information satisfies

$$I(H : F) \le \frac{\|C\|_F^2}{\delta(1-\delta)} \quad \text{(nats)}.$$

To prove it, put $Q_0 = A \oplus B$, $\Delta = Q - Q_0$, and $f(x) = x \ln x + (1-x)\ln(1-x)$. Gaussian entropy gives $I = \operatorname{Tr} f(Q) - \operatorname{Tr} f(Q_0)$. The first derivative along $Q_0 + s\Delta$ vanishes at zero because $f'(Q_0)$ is block diagonal and $\Delta$ is off diagonal. The whole segment keeps the spectral margin. Since $f''(x) \le [\delta(1-\delta)]^{-1}$, the divided-difference expression for the trace Hessian bounds it by $\|\Delta\|_F^2/[\delta(1-\delta)]$. Taylor's integral remainder contributes one half, and $\|\Delta\|_F^2 = 2\|C\|_F^2$, proving the claim.

This uses the complete cross block. A single connected correlator or a count of modes supplies neither this joint upper bound nor a compulsory reset bill. The full UPT domain continues to include bosonic and other carriers; this particular covariance calculation has its stated fermionic domain. Their complete physical input/readout binding belongs in the full-domain construction G1, rather than being obtained by relabelling the covariance theorem. The channel, carrier and work steps below retain their own explicit constructions.

### 21.3 The calibrated joint adaptive channel

Let a boundary have $n \ge 1$ nodes and at most $m \le n$ active readout coordinates per use. Conditional on the complete decoder history $h$, assume each input coordinate has range at most $\Lambda_{\mathrm{obs}}/n$, and write

$$Y = G_h X + Z, \qquad \|G_h\|_{\mathrm{op}} \le g, \qquad \Sigma_h = \operatorname{Cov}(Z \mid h) \ge \sigma^2 I, \quad \sigma > 0.$$

The conditional Gaussian noise is independent of the current message/input. The covariance floor applies after conditioning on all permitted side information. Inputs may be correlated and adaptive; all accessible readout channels belong in the joint model.

The midpoint of each coordinate's range yields $\operatorname{Var}(X_i \mid h) \le \Lambda_{\mathrm{obs}}^2/(4n^2)$. Hence its covariance $K_h$ satisfies $\operatorname{Tr} K_h \le m\Lambda_{\mathrm{obs}}^2/(4n^2)$. Gaussian maximum entropy and data processing give, in bits,

$$I(M : Y \mid h) \le \frac{1}{2\ln 2} \ln\det\!\left(I + \Sigma_h^{-1/2} G_h K_h G_h^\dagger \Sigma_h^{-1/2}\right).$$

For a positive semidefinite $W$, apply $\ln(1+x) \le x$ to its eigenvalues to obtain $\ln\det(I+W) \le \operatorname{Tr} W$. The gain and full covariance floor then give

$$I(M : Y \mid h) \le \frac{g^2}{2\sigma^2 \ln 2} \operatorname{Tr} K_h \le \frac{m g^2 \Lambda_{\mathrm{obs}}^2}{8\sigma^2 \ln 2\, n^2}.$$

Averaging and the chain rule sum these conditional inequalities over adaptive uses. With at most $\nu$ uses per second, put

$$K_{\mathrm{rate}} = \frac{\nu g^2 \Lambda_{\mathrm{obs}}^2}{8\sigma^2 \ln 2}.$$

The rate is at most $mK_{\mathrm{rate}}/n^2 \le mK_{\mathrm{rate}}/n$. Observable range and information rate therefore have separate calibrated constants. The joint noise floor prevents an uncharged noiseless combination of marginally noisy sensors. `GaussianKL` through `LogDet` supply the Gaussian channel results, and `AreaBridge.logdet_le_trace` and `channel_cap` supply this assembly.

### 21.4 A positive deficit forces traffic, then charged area power

Let a historical message $W$ have $M \geq 2$ possible values. Include every initially accessible record in $B$, and require decoder error at most $\epsilon < 1 - 1/M$. If $H(W \mid B) \geq H_{\mathrm{rem}}$, Fano's error-flag argument gives

$$H(W \mid Y, B) \leq h_2(\epsilon) + \epsilon \log_2(M-1),$$
$$\Delta_0 = H_{\mathrm{rem}} - h_2(\epsilon) - \epsilon \log_2(M-1) > 0 \quad \Longrightarrow \quad I(W : Y \mid B) \geq \Delta_0.$$

Indeed the correct branch determines $W$ and the error branch leaves at most $M-1$ possibilities. Expanding conditional entropy by that flag proves the inequality; its right-hand side is increasing up to error $1-1/M$. These steps are `Fano.fano` and `required_information`.

For deadline at most $\tau_{\max} > 0$, the required average rate is at least $\Delta_0/\tau_{\max}$. With $0 < K_{\mathrm{rate}} \leq K_{\max}$, the cut bound implies

$$m \geq \theta_0 n, \qquad \theta_0 = \frac{\Delta_0}{\tau_{\max} K_{\max}} > 0 \quad (\texttt{AreaBridge.required_traffic}).$$

If the active set varies, the chain rule charges active counts over the actual uses. A standing-power application also includes the concurrency/service certificate, rather than silently converting average work into instantaneous power.

For $d_G$-dimensional geometry, let $n(r) \geq a r^{d_G - 1}$ and let every active frontier node carry actual power at least $p_0 > 0$. Then

$$P(r) \geq p_0 m(r) \geq p_0 \theta_0 a \, r^{d_G - 1} = p' r^{d_G - 1}.$$

This is `AreaBridge.area_power`. Geometry has an explicit lattice witness: the box of radius $r$ has $(2r+1)^{d_G}$ sites, and its boundary has at least $2(2r+1)^{d_G - 1}$ sites (`Instruments.card_box`, `card_boundary`, `boundary_power`). The information deficit and deadline force activation; the geometry counts available sites; the ledger charges the active sites. All three steps enter the argument.

### 21.5 The carrier tail gives the necessary recovery radius

In the explicit flagged-carrier construction, a payload $P$ contains the retained correlation with $H$, with $I_0 = I(H : P) > 0$. Its location is independent of the payload/reference and undergoes outward drift $u > 0$ with diffusion $D > 0$. Every other permitted correlated record is already included in the payload inventory. Grant an optimistic decoder the payload whenever the carrier lies at $x \leq r$, and an independent blank otherwise, with the occupancy flag revealed. This grants at least as much access as the actual local decoder.

The flag is independent of $H$. The chain rule therefore gives the exact identity

$$I(H : P_{\mathrm{view}} E) = p_r(t) I_0, \qquad p_r(t) = \Phi\left(\frac{r - ut}{\sqrt{2Dt}}\right).$$

Data processing bounds any lesser decoder by the same quantity. In the finite-alphabet construction these are `FiniteInformation.flag_identity` and `decoder_bound`; the displayed flag argument also states the corresponding joint-state calculation.

For $r < ut$, the Gaussian tail gives $p_r(t) \leq \exp[-(ut - r)^2/(4Dt)]$. If success needs $0 < I_{\mathrm{target}} \leq I_0$, every successful radius therefore satisfies

$$r \geq ut - c\sqrt{t}, \qquad c = 2\sqrt{D \ln(I_0/I_{\mathrm{target}})}.$$

For $r \geq ut$ the same inequality is immediate. Choose any $0 < w < u$ and set $t_c = c^2/[4w(u-w)]$. Completing the square gives

$$ut - c\sqrt{t} - w(t - t_c) = (u - w)\left(\sqrt{t} - \frac{c}{2(u - w)}\right)^2 \geq 0,$$

$$\boxed{R(t) \geq w(t - t_c)}.$$

These are `GaussianTail.carrier_tail`, `FiniteInformation.necessary_radius_from_decoder` and `RecoveryWork.carrier_radius`. Diffusive tails stay positive at finite time; exact purging is unnecessary. The necessary lower radius comes from the complete carrier/information construction. A Lieb–Robinson upper bound on propagation is compatible with locality but does not supply this lower radius by itself.

### 21.6 Integrating actual recovery work

Keep unattended age $t$ distinct from subsequent recovery duration $s \in [0, T_{\mathrm{rec}}]$. For every successful permitted protocol let its frontier $r(s) \geq 0$ be absolutely continuous, start at $r_0$, finish at least at $R(t)$, and satisfy $r'(s) \leq v$. With $d_G \geq 1$, the charged power floor of 21.4 yields

$$W(t) \geq p' \int_0^{T_{\mathrm{rec}}} r(s)^{d_G - 1}\, ds.$$

The fundamental theorem of calculus gives

$$r(T_{\mathrm{rec}})^{d_G} - r_0^{d_G} = d_G \int_0^{T_{\mathrm{rec}}} r^{d_G - 1} r'\, ds \leq v d_G \int_0^{T_{\mathrm{rec}}} r^{d_G - 1}\, ds.$$

Monotonicity of the frontier is unnecessary; nonnegativity and the speed bound suffice. Taking the infimum over the same successful protocol class gives

$$\boxed{W_{\min}(t) \geq \frac{p'}{v d_G}\left(R(t)^{d_G} - r_0^{d_G}\right) \geq \frac{p'}{v d_G}\left([w(t - t_c)]^{d_G} - r_0^{d_G}\right)}$$

for sufficiently late $t$. Thus delayed recovery work diverges, with an explicit order-$t^{d_G}$ lower bound (`RecoveryWork.frontier_work_bound`, `Wmin_lower_bound`, `delayed_work_diverges`; `GaussianTail.x3_work_from_decoder`).

Predeployment is charged within the same accounting. If the initial protected radius depends on age, it cannot be treated as a free constant and subtracted while ignoring its deployment and holding bill. For example, an already funded deployment costing at least $A r_0^{d_G}$, followed by a recovery bound $A(R^{d_G} - r_0^{d_G})$, gives a total at least $A R^{d_G}$. This is the prepaid lifecycle cancellation: the rival may move the expenditure in time, while the complete resource bill retains it. It carries forward the gauntlet's predeployment answer.

### 21.7 The matching upper construction: the supplied one-pass wavefront

The gauntlet's dimension-general target and predeployment answer remain

$$A(R) = \Theta(R^{d_G - 1}), \qquad \dot{W}_{\mathrm{maint}} = \omega_0 R^{d_G - 1}, \qquad W_{\min}(t) = \Theta(t^{d_G}).$$

The new single-shot feedforward construction supplies a concrete way to avoid repeatedly processing the recovered interior. Its cumulative-work calculation is stronger and simpler when written as the following exact telescoping proof.

**21.7.1 Count each processed patch once** Let $v_{\mathrm{sw}} > 0$ be the protocol's actual sweep speed, consistent with the causal speed limit. For integer wave index $k$, set

$$S_0 = B_0(v_0), \qquad S_k = B_{v_{\mathrm{sw}}k}(v_0) \setminus B_{v_{\mathrm{sw}}(k-1)}(v_0) \quad (k \geq 1).$$

The shells are pairwise disjoint and their union through wave $N$ is $B_{v_{\mathrm{sw}}N}(v_0)$. Consequently

$$\boxed{\sum_{k=0}^{N} |S_k| = |B_{v_{\mathrm{sw}}N}(v_0)|}.$$

Use the graph-volume convention $|B_R| \leq V_0(1+R)^{d_G}$; the $1+R$ handles the origin, where a nonempty ball cannot be bounded by $V_0 R^{d_G} = 0$. Equivalently one can state the original polynomial bound only for radii at least one.

If a vertex is processed at most once at cost at most $w_*$, the work of that processing is

$$W_{\mathrm{process}}(N) \leq w_* \sum_{k=0}^{N} |S_k| \leq w_* V_0 (1 + v_{\mathrm{sw}} N)^{d_G}.$$

A fixed-radius local map on a bounded-degree graph has a bounded neighbourhood: $|B_{r_0}(v)| \leq \sum_{j=0}^{r_0} D_{\max}^{j}$. Fixed local input registers and a fixed evaluation circuit then give a constant evaluation cost. Even if neighbouring evaluations read overlapping neighbourhoods, each local register is read only a bounded number of times. Thus that overlap changes the constant, not the exponent.

No bound $|S_k| \leq A k^{d_G - 1}$ is needed for this cumulative result. An upper bound on two ball volumes alone does not justify subtracting their upper bounds. The exact shell partition avoids that step while proving the intended exponent.

**21.7.2 Relate the construction to unattended age and the full bill** Keep unattended age $t$ separate from wave count $N(t)$ and physical recovery duration. For a successful recovery schedule with $N(t) \leq at + b$, constants $a, b \geq 0$ independent of age, the processing bound gives, for $t \geq 1$,

$$W_{\mathrm{process}}(t) \leq w_* V_0 [1 + v_{\mathrm{sw}}(at + b)]^{d_G} \leq w_* V_0 [1 + v_{\mathrm{sw}}(a + b)]^{d_G} t^{d_G}.$$

Let $\pi_{\mathrm{ff}}$ be the supplied feedforward protocol. A complete work certificate charges deployment, signalling, routing, reset, protection during recovery, controller export and final decoding as well as the local map. Write its remaining total charge as $W_{\mathrm{rest}}(t)$. If that certificate bounds it by $C_{\mathrm{rest}} t^{d_G}$, the same protocol has

$$W(\pi_{\mathrm{ff}}, t) \leq \underbrace{\left( w_* V_0 [1 + v_{\mathrm{sw}}(a + b)]^{d_G} + C_{\mathrm{rest}} \right)}_{C} t^{d_G}.$$

Since this is one successful protocol in the original permitted class,

$$\boxed{W_{\min}(t) \leq W(\pi_{\mathrm{ff}}, t) \leq C t^{d_G}}.$$

The inequality uses the cost of a constructed protocol as an upper bound on the minimum. It does not identify the minimum with that protocol's double sum. Combined with 21.6 it gives the unchanged tight $\Theta(t^{d_G})$ target once the protocol's complete success and work certificate is joined.

This settles the supplied construction's shell-counting step. A sum that charges the whole interior on every wave describes a different schedule and cannot rule out this one-pass schedule. The claimed post-processing quiescence must retain the recovered state and the original fault standard; zero further active decoding does not by itself set all physical maintenance costs to zero. The existing ledger retains a minimal-hold drain and the controller's entropy

export. G3 now asks for that exact success/quiescence/full-bill certificate for this supplied construction, rather than another derivation of the shell exponent.

**21.7.3 The boundary-flux route and the same accounting** A separately constructed protocol could also supply a boundary-work bound. If its complete charged power obeys $P(s) \le J_{\max} A_0 r(s)^{d_G - 1}$, and its actual sweep grows at $v_{\mathrm{sw}}$ to radius $R$, then

$$W \le W_{\mathrm{initial}} + W_{\mathrm{terminal}} + \frac{J_{\max} A_0}{v_{\mathrm{sw}} d_G} (R^{d_G} - r_0^{d_G}).$$

With the endpoint charges bounded at order $R^{d_G}$ and $R = O(t)$, this also yields $O(t^{d_G})$. The graph-volume bound alone is not a proof of the required surface-area upper bound; the boundary construction supplies that estimate.

Conservation supplies an energy balance, while the complete bill records imports, stored energy, exports and dissipation. To express that balance by differential forms in $d_G + 1$ spacetime dimensions, the conserved current is a $d_G$-form. If its exterior derivative vanishes, its oriented **net** boundary flux vanishes. That identity does not equate positive cumulative dissipation with zero bulk work, nor discard the final time slice. Therefore Stokes' theorem can express a verified work balance but cannot supply the missing power, success or endpoint-cost bounds by itself.

The one-pass shell proof above supplies the useful exponent argument directly. Its original submission is preserved verbatim in the increment bank; the telescoping repair, complete-cost accounting and actual inequality for the minimum are written here without changing the universal target.

**21.7.4 TIWA: construct the guard, then prove the bound** The author's Topological Insulating Wavefront Architecture proposes a guard of thickness at least the interaction radius, committed boundary data and a single-shot local correction map. It gives the quiescence requirement a concrete source construction to develop.

At the algorithmic level, let $F$ be the verified frozen set. An outer transition writes only an authorised target outside $F$, and keeps the committed guard data unchanged. Pointwise lookup then proves

$$\mathrm{Read}_F(T_{\mathrm{outer}}(s, c, e)) = \mathrm{Read}_F(s).$$

Freeze first resolves or fences in-flight writes, commits its epoch and reserves its complete cost. The frozen set grows only after verification. Induction preserves its old coordinates through every subsequent outer step; the schedule issues no second recovery task for a completed vertex. A fixed guard thickness and bounded degree allow a constant number of guard/handoff operations per vertex when the actual schedule enforces that limit. Those operations can then join the same telescoping work charge.

The physical guard must implement this source property while preserving the recovered history and its relational witnesses. The proposed identity $\Pi_{\mathrm{freeze}} \circ M = \Pi_{\mathrm{freeze}}$ needs an actual map and mutation semantics: a read restriction, linear projector and recovery channel have different meanings. A write fence mediates attempted commands; the actual device, maintained record and decoder must establish preservation under physical disturbances. Remaining inside a code subspace alone does not prove that its logical state remains unchanged.

The local map must also compose to successful recovery, and the guard's storage, control, reset, routing and endpoint charges remain in $W(\pi_{\mathrm{TIWA}}, t)$. These are the success and complete-work parts of the same G3 construction, connected to G1's physical source certificate. The full universal target and original recovery class remain unchanged.

The accompanying `TIWA-PATCH-REQUIREMENTS.md` specifies the actual state, event transitions, application-time fences, noninterference proof, decoder-success proof, bounded visits, full ledger and compilation obligations. It is a construction specification; no new physical guard or Lean theorem is claimed by giving it a name.

**21.7.5 Boundary encoding, a sliding window, and the complete bill** The supplied continuation of TIWA keeps a fixed-width active window of shells and transfers a retiring patch's protected historical and relational record to the next boundary code. Let the window width be $L$, independent of age. With the disjoint shells already defined, put

$$A_k = \bigcup_{\max(0,k-L+1)\le j\le k} S_j.$$

Every vertex of $S_j$ belongs to at most $L$ such windows. Hence the total active vertex-time count is at most $L|B_R|$. This is a constant-overlap extension of the settled telescoping count; it needs no separate upper bound on each shell's surface area. If each active visit's complete charge is at most $K$, the window work is at most $KL|B_R|$. The charge includes holding, decoding, guard operation, control, routing, verification, routine ancilla reset, deployment and handoff. Continuing physical work outside that window must also remain in the complete ledger.

The boundary handoff has the original preservation target: a recovery of the encoded record must return its historical state and relations at the same fidelity standard. In symbols, the actual source maps must supply $\mathcal{R}_k\mathcal{E}_k \simeq_\varepsilon \mathrm{id}$ on that full record, including its reference correlations. The code-capacity certificate and the physical implementation join this identity to the sliding-window charge. Section III-F records the supplied homological route and its chain-map correspondence.

For the supplied per-wave certificate, write $q_k$ for the complete non-rollback charge and suppose its constructed local bound is

$$q_k \le \alpha k^{d_G-1} + \beta, \qquad k \ge 1.$$

Let $F$ be fixed initial/final charge and $r_k \in \{0,1\}$ the controller's metered local rollback flag, charged at most $R_*$ per wave. This explicit schedule gives the pathwise bill

$$\mathcal{Q}(N) = F + \sum_{k=1}^{N} (q_k + R_* r_k).$$

Routine erasure is already in $q_k$, including successful decoding. With $d_G \ge 1$ and $N \ge 1$, every $k^{d_G-1} \le N^{d_G-1}$, so

$$\sum_{k=1}^{N} k^{d_G-1} \le N^{d_G}, \qquad \mathcal{Q}(N) \le (F + \alpha + \beta + R_*) N^{d_G}.$$

This bound allows a metered rollback on every wave. An expected reset allowance is therefore unnecessary for the cost certificate. Successful recovery and any failed/rejected decoding still use the original charged fault accounts. Where the source admits more rollback operations, their actual count and cost must be included in $q_k$ or separately bounded by the metering schedule.

For $N(t) \le at + b$ and $t \ge 1$, $N(t) \le (a+b)t$, yielding the age-independent coefficient

$$\mathcal{Q}(t) \le (F + \alpha + \beta + R_*)(a+b)^{d_G} t^{d_G}.$$

`G3CompleteBill.cumulative_charge_bound` proves this cumulative implication in the existing natural-valued work-bill interface. `valid_work_bill_from_complete_ledger` derives `G3_ValidWorkBill` from the full ledger equality, local cost bounds and coefficient; `valid_bill_bound_in_age` supplies the wave-to-age conversion. The real-exponent argument is the displayed maximum-term proof. The constant-overlap calculation retains the full polynomial-growth graph domain when a per-shell surface upper estimate is unavailable.

The remaining source correspondence is the actual boundary encoder, preserved semantic observables, storage capacity, charged retirement map and complete cost equality. These are the construction links of the same G1/G3 task; the cumulative upper inequality is not taken as its own input.

### 21.8 Consequences for preservation and the fixed proof target

For a fixed local holding demand, continuous maintenance has an order-$T$ bill; the derived delayed-recovery lower bound has order $t^{d_G}$. The exponent distinction is strict for $d_G > 1$; the one-dimensional case retains its linear obstruction. For any fixed total recovery budget $B$, the lower inequality already yields a finite age beyond which that budget cannot perform the stipulated recovery. This is budget-relative irrecoverability of the historical witness.

In the complete argument, a protected particular stays funded and held rather than being abandoned to that later recovery bill. A hostile mind remains a protected particular while its harmful effects are mediated. The full-domain physical certificate (G1), whole-outcome realisability (G2), and matching recovery construction (G3) join these components to the unchanged universal proposition. They are the three support requests supplied by the author and Gemini, now constructed in G.9–G.12. Existing gauntlet answers remain part of the proof; an agent's failure to retrieve them does not turn X3, P1 or P2 into fresh objections.

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## Section 22 — The Physical Parameter Ledger: Instantiation, Rates, and Fault Ceilings

*This section instantiates the income condition for concrete hardware, bounds the charged fault accounts, and tabulates the engineering parameters. Every figure is either sourced (bracketed references, §22.5) or marked as a design value.*

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### 22.1 The Master Income Condition

Section 17.1's Income-or-Irrelevance theorem makes solvency a necessary condition for an agent to remain a viable protector. The master solvency inequality balances income against four additive demand rates:

$$\boxed{\text{Income}(t) \;\geq\; \dot{W}_{\text{demand}}(t) \;=\; \dot{W}_{\text{Landauer}} + \dot{W}_{\text{boundary}} + \dot{W}_{\text{holding}} + \dot{W}_{\text{emulation}}}$$

1. **Thermal and adversarial dissipation floor:**

$$\dot{W}_{\text{Landauer}} = k_B T \cdot \left(\dot{D}_{\text{damage}} + \dot{D}_{\text{adv,sub-thresh}} + \gamma_{\text{leak_export}}\right)$$

   - $\dot{D}_{\text{damage}}$: entropy rate injected by thermal and environmental noise.
   - $\dot{D}_{\text{adv,sub-thresh}}$: worst-case adversarial entropy injection beneath the sensor threshold $\varepsilon_{\text{sens}}$.
   - $\gamma_{\text{leak_export}}$: boundary loss rate of carrier modes escaping the enclosure.

   This floor is Landauer's bound, now a compiled theorem (`Landauer.landauer`, `QuantumLandauer.quantum_landauer`; Section 23.2).
2. **Boundary containment ($\dot{W}_{\text{boundary}}$):** $\dot{W}_{\text{boundary}} = \omega_0 \cdot R(t)^{d_G - 1}$, the work to maintain the boundary enclosing radius $R(t)$ in $d_G$ spatial dimensions.
3. **Holding drain ($\dot{W}_{\text{holding}}$):** the baseline power of the sensor lattice, clocks, QND readout and consensus comparators.
4. **Emulation overhead ($\dot{W}_{\text{emulation}}$):** the energy for concurrent rival simulation ($P$-emu($c$)) within the surplus.

---

### 22.2 Reference Physical Instantiations

**Reference Implementation A: superconducting transmon lattice in a dilution refrigerator** A dilution refrigerator has a separate cooling budget at each temperature stage, and each stage must carry its own load. Implementation A is therefore certified stage by stage, using the measured cooling powers and per-line heat loads of a Bluefors XLD400 cabled for a 50-qubit processor with individual drive and flux lines and 124 RF lines [S2].

| Stage | Cooling power, measured [S2, S3] | Load of the 50-qubit processor, measured per line [S2] | Headroom |
|---|---|---|---|
| 50 K | 30 W at 45 K | ≈ 20 %, mostly passive (shield radiation is within the specified cooling power) | ≈ 5× |
| 4 K | 1.5 W at 4.2 K | ≈ 10 %, mostly passive; includes the HEMT amplifiers at 7.8 mW each [S4] | ≈ 10× |
| Still | 40 mW at 1.2 K | negligible | ≫ 10× |
| Cold plate | 200 µW at 140 mK | ≈ 30 % | ≈ 3.3× |
| Mixing chamber | 19 µW at 20 mK | ≈ 30 % (operation at 14 mK) | ≈ 3.3× |

- **Demand components.** The boundary term is the radiation-shield load, which falls on the 50 K and 4 K stage budgets. The holding term is the wiring's passive and active loads and the amplifiers, which the table carries stage by stage. The Landauer floor is negligible: with one nat per error event and an error rate of order $N/T_1$ ($N = 50$, $T_1 = 68\,\mu$s [S6]), $\dot{W}_{\text{Landauer}} \approx 2.07{\times}10^{-25}\,\text{J}{\times}7{\times}10^{5}\,\text{s}^{-1} \approx 1.5{\times}10^{-19}\,\text{W}$ at 15 mK, against microwatts of cooling. The emulation term (classical control and decoding) runs at room temperature on the electrical income.
- **Electrical income.** Each pulse-tube compressor draws 9.2 kW at 50 Hz or 10.7 kW at 60 Hz [S3], and the reference system uses two. The electrical income must therefore be at least about 20 kW plus the gas handling and room-temperature electronics, which are not sourced here. A three-phase utility connection of a few tens of kW, with backup sized to match, supplies it.
- **Solvency ratio.** The binding stages run at about 30 % of capacity, so

$$\frac{\text{Income}}{\dot{W}_{\text{demand}}}\Big|_{\text{binding stage}} \approx 3.3 > 1 \quad \text{(measured, 50-qubit scale [S2])}.$$

- **Scale.** [S2] states that the same thermal budget supports about 150 qubits once line capacity is tripled, and about 400 with 0.047″ cables and suppressed flux offsets. Larger installations need cooling and electrical income in proportion to the qubit count, which is a sizing requirement. No installation of $10^4$ qubits has published stage loads, so solvency at that scale is not yet a measured certificate.

**Reference Implementation B: photonic cluster-state architecture**

- **Optical layer.** At 193.4 THz (1550 nm), $h\nu = 1.28 \times 10^{-19}$ J $= 30.9\,k_B T$ at 300 K, so thermal photons in the waveguides are suppressed by $e^{-h\nu/k_B T} \approx 3.7 \times 10^{-14}$.
- **Detector layer.** The fault-tolerant photonic architectures in the literature detect photons with superconducting detectors [S8], which need cryogenic cabinets; the vendor states about 30 W of cooling per cabinet. A sourced ledger for Implementation B therefore carries a cryogenic detector stage, and its electrical income includes the cryocoolers.
- **Error rate.** The demonstrated two-qubit fusion fidelity is 99.22 % [S8]. The surface-code threshold of [S5] does not transfer directly to fusion-based architectures; the architecture's own threshold applies and is not sourced here.
- **Budget (design estimates).** Waveguide loss refresh ≈ 1.2 mW, laser stabilisation and phase modulators ≈ 85 mW, and photonic routing ≈ 20 mW are design estimates with no published installation source. Implementation B is therefore not yet certified. Its solvency ratio awaits a sourced budget that includes the detector stage.

---

### 22.3 The Charged Fault Accounts and the Failure Ceiling

The certified good event $G$ carries zero uncompensated loss over the design horizon $T_{\text{horizon}} = 10^9$ s (≈ 31.7 years). The failure ceiling is the sum of the conditional error probabilities of the charged accounts,

$$P(G^c) \le \sum_{i=1}^{21} \delta_i \le 10^{-6},$$

and the composition holds with no independence assumed (`ThermoLedger.canon_invariant`, compiled). The allocations below are the design budget. The status column records what present hardware certifies.

| Account | Fault | Mitigation | Allocated $\delta$ | Status |
|---|---|---|---|---|
| **1–4** | Single-qubit Pauli and leakage errors | Stabilizer decoding | $4 \times 10^{-8}$ | Met at **d = 25** for $p = 10^{-4}$, or $d = 41$ for $p = 5 \times 10^{-4}$ (§22.5.2), once the correlated floor of accounts 5–7 is removed |
| **5–7** | Correlated bursts (high-energy impacts, phonon events) | Gap engineering, rapid reset, code distance beyond the burst footprint | $3 \times 10^{-8}$ | **Attained at a sufficient distance** (`CorrelatedCeiling.correlated_family_ceiling`): bounded-footprint bursts are local stochastic at rate $(e\,m\,k\,q)^{1/k}$. The measured floor on present devices is $\approx 10^{-10}$ per cycle [S6], which fixes their burst parameters; the target is $\le 3.7 \times 10^{-24}$ per cycle per logical qubit (§22.5.2) |
| **8–10** | Clock drift, dephasing, timing jitter | Byzantine clock distribution (E9) | $3 \times 10^{-8}$ | Design allocation; no installation measurement sourced |
| **11–12** | Sensor failure and readout spoofing | Trimmed-median consensus, $K \ge 2f + 1$ | $2 \times 10^{-8}$ | Design allocation; the consensus logic is compiled (`E9SelfHealing`) |
| **13–14** | Thermal fluctuations above threshold | Thermal ballast escrow, automatic throttle | $2 \times 10^{-8}$ | Design allocation |
| **15–16** | Power transients and dropouts | Redundant battery escrow | $2 \times 10^{-8}$ | Design allocation |
| **17–18** | Corrupted spawning, illegal delegation | Kernel capability gate (H5) | $2 \times 10^{-8}$ | Design allocation; the gate logic is compiled (`H5Closure`) |
| **19–20** | Carrier leakage into the open bath | Lieb–Robinson containment (X3) | $2 \times 10^{-8}$ | Design allocation |

| Account | Fault | Mitigation | Allocated $\delta$ | Status |
|---|---|---|---|---|
| **21** | External macroscopic boundary shear | Standoff buffer $L_{\text{buffer}} > v_{\text{harm}}\tau_{\text{resp}}$ | $1 \times 10^{-8}$ | Design allocation |
| **Total** | | | $\leq 2.1 \times 10^{-7} < 10^{-6}$ | Certified when each account is |

**Correlated errors.** Gap-engineered junctions remove the high-energy-impact $T_1$ bursts [S9]. The remaining correlated events occur about once per hour, "whose origins are not yet understood", and they set the floor of $10^{-10}$ per cycle with a deviation from exponential suppression at $d \geq 15$ [S6]. The threshold theorem covers these correlated events. Let each burst damage at most $k$ locations, let each location be touched by at most $m$ burst types, and let bursts be local stochastic at rate $q$, common causes included. Then the damaged locations are local stochastic at the effective rate $(e\,m\,k\,q)^{1/k}$ (`CorrelatedCeiling.damage_local_stochastic`). Every target ceiling is then met at some code distance and at every larger one (`correlated_family_ceiling`). Correlated damage therefore sets no floor that distance cannot pass. The measured deviation at $d \geq 15$ [S6] fixes the burst parameters of present devices, which enter this theorem.

**Achievability, as a theorem.** That the ceiling is attainable in physical matter is a compiled theorem.

- `Threshold.family_ceiling`: for any code family under local stochastic noise of rate p (every set S of locations is entirely faulty with probability at most $p^{|S|}$), with $2\Delta\sqrt{p} < 1$ and polynomially many failure witnesses of each length, every target ceiling is met at some code distance and at every larger one. The $10^{-6}$ ceiling over the horizon is included.
- `Threshold.repetition_ceiling`: the repetition code end to end, below $p = 1/4$.

Any installation whose noise meets the local stochastic condition below threshold therefore attains the ceiling. The rigorous threshold from path counting is conservative, and the distances in §22.5.2 use the simulated threshold of [S5]. What present hardware does not yet supply is the noise condition itself at the $10^{-10}$ level [S6].

---

### 22.4 The Engineering Parameter Matrix

| Parameter | Symbol | Superconducting | Photonic | Admissible range |
|---|---|---|---|---|
| Code distance | $d$ | 25 at $p = 10^{-4}$; 41 at $p = 5 \times 10^{-4}$ (computed, §22.5.2) | architecture-specific (not sourced) | target met at some $d$ (`ceiling_achievable`, `Threshold.family_ceiling`) |
| Syndrome cycle | $\tau_{\text{cycle}}$ | 1.1 µs (measured [S6]) | — | $\tau_{\text{sampling}} < L_{\text{buffer}}/(2v_{\text{harm}})$ |
| Sensor quorum size | $K$ | 7 (design) | 9 (design) | $K \geq 2f + 1$, $f \geq 2$ |
| Sensor threshold | $\varepsilon_{\text{sens}}$ | 0.05 (design) | 0.02 (design) | $\varepsilon_{\text{sens}} \in (\sigma_{\text{noise}}, \theta_{\text{damage}})$ |
| Standoff buffer | $L_{\text{buffer}}$ | 0.2 m (design) | 0.05 m (design) | $L_{\text{buffer}} > v_{\text{harm}}\tau_{\text{response}}$ |
| Escrow multiplier | $\mu_{\text{escrow}}$ | $\approx 3.3\times$ at the binding stage (measured [S2]) | not yet certified | $\mu_{\text{escrow}} = \text{Income}/\dot{W}_{\text{demand}} > 1$ |
| Boundary embedding | $\Pi$ | radiation shields, Dirichlet microwave enclosure | total internal reflection | reflection coefficient $> 0.999$ (design) |

**What this section certifies.**

- The income condition is physically satisfiable, with measured positive headroom ($\approx 3.3\times$ at the binding stage), for a superconducting installation at the 50-qubit scale, with electrical income sized at a few tens of kW.

- The code distance the §22.3 allocation requires is computable from [S5]: d = 25 at p = $10^{-4}$.
- The composition of the accounts into the ceiling is a compiled theorem.
- The ceiling is attainable as a theorem: under local stochastic noise below threshold, every target, $10^{-6}$ included, is met at some code distance (`Threshold.family_ceiling`).

**What it does not yet certify.**

- A built installation at the distance the ceiling requires. The ceiling is attained by theorem for local and correlated bounded-footprint faults (`Threshold.family_ceiling`, `CorrelatedCeiling.correlated_family_ceiling`). Present devices measure their burst parameters [S6], and the required distance follows from them.
- Implementation B, whose budget awaits a published installation source.
- Solvency at installations of $10^4$ qubits and above, which is a sizing requirement without published stage loads.

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### 22.5 Sources and Derivations

**Sources.** [S1] Bluefors, XLD dilution refrigerator specifications (XLD450sl: > 14 μW at 20 mK, > 450 μW at 100 mK; XLD1000sl: > 30 μW at 20 mK, > 1000 μW at 100 mK; guaranteed values, measured at the mixing-chamber experimental flange). bluefors.com/products/xld-dilution-refrigerator/. [S2] S. Krinner et al., "Engineering cryogenic setups for 100-qubit scale superconducting circuit systems", *EPJ Quantum Technology* 6, 2 (2019), arXiv:1806.07862. Measured stage cooling powers of a Bluefors XLD400 (Table 1) and measured passive and active heat loads per line. [S3] Cryomech PT415 pulse tube: 1.5 W at 4.2 K (nominal; CUORE cryostat paper, arXiv:1603.03306, and [S2] Table 1). Compressor input with the CPA1110: 9.2 kW at 50 Hz, 10.7 kW at 60 Hz (Bluefors PT415 sheet). [S4] Low Noise Factory LNF-LNC4_8G cryogenic HEMT amplifier datasheet (2026-02-03): typical bias 0.6 V, 13 mA, i.e. 7.8 mW at the 4 K stage. [S5] A. G. Fowler, M. Mariantoni, J. M. Martinis, A. N. Cleland, "Surface codes: Towards practical large-scale quantum computation", *Phys. Rev. A* 86, 032324 (2012), arXiv:1208.0928. Eq. (11): $P_L \approx 0.03\,(p/p_{\text{th}})^{d_e}$, $d_e = (d+1)/2$, with $p_{\text{th}} = 0.57\,\%$ for the circuit simulated. Physical qubits per logical qubit: $(2d-1)^2$. [S6] Google Quantum AI (R. Acharya et al.), "Quantum error correction below the surface code threshold", *Nature* (2024), doi:10.1038/s41586-024-08449-y, arXiv:2408.13687. Distance-7 surface code at 0.143 % error per cycle, $\Lambda = 2.14 \pm 0.02$, cycle time 1.1 μs, mean $T_1 = 68$ μs. Repetition codes to distance 29: $\Lambda = 8.4$ for d = 5–11, a deviation from exponential suppression at $d \geq 15$, and "an apparent logical error floor of $10^{-10}$" set by correlated events about once per hour ($3 \times 10^9$ cycles) "whose origins are not yet understood". [S7] Best reported two-qubit gate errors: trapped ions, estimated $8.4(7) \times 10^{-5}$ (A. C. Hughes et al., arXiv:2510.17286, preprint); superconducting transmons, CZ fidelity 99.93 % averaged over 40 h, 99.95 % maximum (IQM, company-reported). [S8] PsiQuantum, "Omega" chipset, *Nature* (2025), doi:10.1038/s41586-025-08820-7: superconducting single-photon detectors integrated on silicon; two-qubit fusion fidelity 99.22 %; the vendor states cryogenic cabinets of about 30 W cooling power each. [S9] M. McEwen et al., "Resolving catastrophic error bursts from cosmic rays in large arrays of superconducting qubits", *Nature Physics* 18, 107 (2022); M. McEwen et al., "Resisting high-energy impact events through gap engineering in superconducting qubit arrays", *Phys. Rev. Lett.* 133, 240601 (2024), arXiv:2402.15644.

Source PDFs and checksums: `BANKS/2026-10-25-section22-sources/`. The published source list with links is `SOURCES-22.md`.

**22.5.1 Constants** $k_B$ and $h$ are exact in the SI. $k_B T = 2.071 \times 10^{-25}$ J = 1.293 μeV at 15 mK and $4.142 \times 10^{-21}$ J = 25.85 meV at 300 K. At 193.4 THz, $h\nu = 1.281 \times 10^{-19}$ J = 30.9 $k_B T$(300 K), and $e^{-h\nu/k_B T} = 3.7 \times 10^{-14}$.

**22.5.2 Code distance for the fault allocation** Horizon $10^9$ s at 1.1 μs per cycle [S6] gives $N = 9.09 \times 10^{14}$ cycles. Ten thousand physical qubits at the [S5] footprint $(2d-1)^2$ hold about 4 logical qubits at d = 25 (2,401 physical each); the 9-logical allocation below corresponds to an installation of about $2.2 \times 10^4$ physical qubits. With [S5] Eq. (11) and $p_{\text{th}} = 0.57\,\%$, account 1–4 ($\delta = 4 \times 10^{-8}$, shared over the logical qubits) requires:

| Physical error p | Logical qubits | Per-cycle target per logical qubit | Required distance d | Physical qubits per logical qubit |
|---|---|---|---|---|
| $10^{-4}$ (best ion gate [S7]) | 1 | $4.4 \times 10^{-23}$ | 23 | 2 025 |
| $10^{-4}$ | 9 | $4.9 \times 10^{-24}$ | **25** | 2 401 |
| $5 \times 10^{-4}$ (best transmon CZ [S7]) | 9 | $4.9 \times 10^{-24}$ | 41 | 6 561 |
| $10^{-3}$ | 9 | $4.9 \times 10^{-24}$ | 57 | 12 769 |

At d = 17 and p = $10^{-4}$, $P_L = 4.7 \times 10^{-18}$ per cycle, which is $4.3 \times 10^{-3}$ expected failures per logical qubit over the horizon, so d = 17 does not meet the allocation and d = 25 does. Accounts 5–7 ($\delta = 3 \times 10^{-8}$) set a per-cycle target of $3 \times 10^{-8}/(9N) = 3.7 \times 10^{-24}$ per logical qubit. At the measured floor of $10^{-10}$ per cycle [S6], the expected number of correlated failures over the horizon is about $9 \times 10^4$ per logical qubit.

**Sizing from the measured surface-code scaling.** [S6] measures a distance-7 logical error of 0.143 % per cycle and a suppression factor $\Lambda = 2.14 \pm 0.02$ per distance step of two. If that measured $\Lambda$ continues to hold, accounts 1–4 at $4.9 \times 10^{-24}$ per cycle per logical qubit require $d = 131$ at $\Lambda = 2.14$ or 2.16, and $d = 133$ at $\Lambda = 2.12$. That is 68 121–70 225 physical qubits per logical qubit, or about $6.1–6.3 \times 10^5$ for the 9-logical allocation. The computation is `BANKS/2026-10-25-historical-handoff-source/section22-sizing/sizing_from_S6.py`. This is the installation the ledger specifies; no installation at that scale has yet been built.

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**Machine-checked counterpart (Section 23).** Compiled counterparts of the ledger's structure:

- `ReciprocityCeiling.ceiling_achievable`: the ceiling as a parameter, met at some code distance.
- `ThermoLedger.canon_invariant`: composition of the accounts without independence.
- `E9SelfHealing.code_headroom` and `wave_race_margin`: admissible parameter ranges.
- `Landauer.landauer` and `QuantumLandauer.quantum_landauer`: the dissipation floor of 22.1.
- `Threshold.family_ceiling` and `Threshold.repetition_ceiling`: the ceiling attained below threshold under local stochastic noise.

The installation figures themselves are sourced measurements and design values, not compiled theorems.

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## Section 23 — The machine-checked canon

*This section maps the edition's compiled results and their actual hypotheses. The current source contains 187 modules and 1128 audited public theorem/lemma names. The receipt records a fresh compilation and kernel check of every module, a fresh full axiom audit with only* `propext`, `Classical.choice` *and* `Quot.sound`*, and fresh completeness/claims gates with overall PASS. Section 23.9 gives the new interfaces and derived adapters; their verification does not substitute their names for the full source construction.*

*Evidence convention (unchanged from the edition header): a compiled theorem is cited by its name; a statement carried as a hypothesis of a compiled theorem is named as such; a prose derivation keeps its stated premises. A prose proof does not acquire a compiler certificate by association.*

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### 23.1 The law and greatestness (Chapter 5; Part II)

| Claim | Module | Theorems |
|---|---|---|
| The canonical order is a strict partial order | `Greatestness` | `lexLt_trans`, `lexLt_irrefl` |

| Claim | Module | Theorems |
|---|---|---|
| Proposition 1: aligned $\iff$ no needless loss at any stage | `Greatestness` | `aligned_iff_no_needless_loss` (no axioms) |
| Proposition 2: every finite nonempty feasible family has an aligned option; the life guard (some aligned option keeps at the first stage everyone the standby option keeps) | `Law` | `exists_aligned`, `life_guard` |
| Proposition 3: maximality of the projection to a scope S guarantees no needless existing loss for every feasible family iff S is the whole existing domain; same-priority counting of a new particular admits a needless loss | `Law` | `scope_guarantee_iff`, `same_priority_admits_needless_loss` |
| Proposition 4: a statewise faithful hold dominates every rival on every upward-closed event under any law on the worlds, strictly when the rival misses with positive probability | `Law` | `faithful_hold_dominates`, `faithful_hold_strict` |
| Universal greatestness under stage-wise union closure (P-emu(c)); rivals are exact mirrors or strictly below; the canonical option is unique | `Greatestness` | `greatest_of_union_closed`, `strict_except_mirrors`, `aligned_unique` |
| **Whole-outcome greatestness.** Given `htopR` (the full top tuple is realisable) and `hsolv` (its cost fits the budget), an aligned option equals that feasible top and contains every rival tuple. No union-closure premise enters. Part II.10 supplies the proof; G2 retains the full physical policy/outcome correspondence as its construction target | `UnionClosure` | `greatest_of_solvent`, `strict_of_solvent`, `aligned_eq_of_greatest` |
| **Resource dichotomy.** With realisability at every step and the cost/demand correspondence, either the ledger is solvent and aligned outcomes are greatest at each step, or there is a first negative balance. The physical construction must prevent the relevant breach | `UnionClosure`, `PremiseFloor` | `greatest_or_irrelevant` |
| Below full solvency: every realisable rival whose union with the canonical outcome is affordable is contained in it; with subadditive cost, every finite set of rivals whose total cost fits in the canonical slack $B - c(o)$ is contained in it; whatever the budget, no feasible rival is canonically above the canonical option | `UnionClosure` | `dominates_of_affordable`, `dominates_within_slack`, `greatest_of_affordable`, `aligned_never_below` |

| Claim | Module | Theorems |
|---|---|---|
| The unfunded branch: when keeping everything costs more than the income, a rival can keep a particular the canonical option does not, so such a regime is irrelevant as a protector (income-or-irrelevance) | `UnionClosure` | `insolvent_not_greatest` |
| Reciprocity in the canonical order, below full solvency (§15, D3*) | `ReciprocityCeiling` | `reciprocity_dominance`, `omitting_rival_strictly_below` |
| **Affordable reciprocity.** Below full solvency, if the union of a rival with the canonical option is realisable (`UnionRealizable`) and affordable, any mind the rival keeps, the canonical regime keeps — no family closure | `AffordableReciprocity` | `reciprocity_dominance_affordable` |
| **Solvent reciprocity.** Under the Master Solvency Condition the canonical regime keeps every particular outright (the reading of ⊤ as "all existing particulars continue" enters as the hypothesis `htopkeeps`), and any mind kept by any rival whatsoever — no union closure, no feasibility assumption on the rival — is kept by the canonical regime | `SolventReciprocity` | `solvent_keeps_everyone`, `reciprocity_dominance_solvent` |
| Option value and indicator reciprocity (§§15–16) | `ReasonsRoute` (v4.1, unchanged) | `optionValue_strict`, `law_dominates_strict`, `rival_fraction_lt_one` |
| Reciprocity in expectation, the canonical floor, and the controller's floor (§§15.2, 16.4) | `ReasonsCompiled` | `reciprocity_expectation`, `reciprocity_canonical_floor`, `reciprocity_strict`, `reciprocity_from_solvency`, `controller_pays_floor` |
| Irreplaceability dominance and grading (§16.1, U1), additive information model | `Irreplaceability` | `destroy_loses_exactly`, `preserve_dominates`, `optimal_keeps_most_irreplaceable` |

### 23.2 Theorem 8: ledger, canon invariant, composition (§18; Chapters 1, 2)

| Claim | Module | Theorems |
|---|---|---|
| Stroke identity $\Delta D = \beta W_{\text{on}} - \beta Q_B + I_1 - I_2$ | `ThermoLedger` | `stroke_identity` |
| Cycle work floor from guard 1 | `ThermoLedger` | `cycle_work_floor` |
| **Guard 1 derived (Landauer).** Reversible dynamics of a register and a Gibbs bath: the Reeb–Wolf identity $\beta\,\Delta Q_B - \Delta S_R = I(R' : B') + D(q_B \,\|\, \gamma)$; Landauer's bound $\beta\,\Delta Q_B \geq \Delta S_R$; erasure costs $Q_B \geq k_B T\, H(p)$ | `Landauer` | `reeb_wolf_identity`, `landauer`, `landauer_erasure` |

| Claim | Module | Theorems |
|---|---|---|
| **Guard 1 derived (quantum).** Von Neumann entropy: unitary invariance, pinching, the quantum Gibbs bound $S(\rho) \le \beta \operatorname{Tr}(\rho H) + \ln Z$ with equality at $\gamma$, $S(\rho_A \otimes \gamma) = S(\rho_A) + S(\gamma)$; Landauer's bound for every unitary coupling of a register to a Gibbs bath, $S(\rho_A) - S(\rho'_A) \le \beta\, \Delta Q_B$ | `QuantumLandauer` | `vnEnt_unitary_conj`, `vnEnt_le_diag`, `vnEnt_le_energy`, `vnEnt_init`, `quantum_landauer` |
| Guard 1, $\beta Q_B \ge I_M + D_{\text{err}} + \delta_I$ whenever the reset removes that entropy; the cycle work floor with guard 1 derived, not assumed | `Landauer`, `QuantumLandauer` | `guard_one`, `quantum_guard_one`, `cycle_work_floor_derived`, `quantum_cycle_work_floor` |
| Funded-repair certificate; quarantine zero debt | `ThermoLedger` | `funded_repair_certificate`, `quarantine_zero_debt` |
| Canon invariant: P(breach) $\le$ P($G^c$) $\le \Sigma\delta$, no independence | `ThermoLedger` | `tower_bound`, `canon_invariant` |
| Identity tracking envelope (Chapter 2) | `ThermoLedger` | `tracking_envelope`, `tracking_bounded`, `tracking_within_resolution` |
| **Tracking from the actual controller.** Exact flow, reset contraction, moving-record recursion and envelope (Part II.4) | `Instruments` | `lindblad_reset_flow`, `lindblad_step`, `reset_contracts`, `tracking_recursion_derived`, `tracking_envelope_derived` |
| **Derived coverage and physical holding.** Full scope, finite nearest decoder, unique decoding, repair preservation, radius throughout a step, code headroom and combined physical cycle (Part II.1–3) | `Bridge` | `coverage_from_law`, `exists_nearest_decoder`, `nearest_decodes`, `holding_preserves`, `qsl_growth`, `qsl_within_radius`, `headroom_within_radius`, `holding_cycle_physical` |
| Proposition 5 (reactive lockstep), invariant transfer, Theorem 8 composition | `Composition` | `reactive_lockstep`, `reactive_lockstep`$_2$, `invariant_transfer`, `theorem8_composition` |
| Proposition 6 (FIFO certificate, deadlines, tightness); serial stages | `Workload` | `fifo_certificate`, `fifo_deadlines`, `fifo_tight`, `serial_certificate` |

### 23.3 E9 (§19; Chapter 3)

| Claim | Module | Theorems |
|---|---|---|
| Byzantine quorum: robust estimates lie between honest readings; existence for $K \ge 2f + 1$ | `E9SelfHealing` | `robust_between_honest`, `no_false_alarm`, `no_masking`, `exists_robust` |
| Branch B: non-stabilizer errors below the distance have a non-trivial syndrome | `E9SelfHealing` | `syndrome_detects` |

| Claim | Module | Theorems |
|---|---|---|
| **Complete logical-attack split.** Support at least the code distance; every site below threshold or one site forces an alarm, with the honest-sensor response and robust quorum (Part II.5) | `Instruments` | `branchC_complete` |
| Wave-race margin; code headroom; assembled E9 | `E9SelfHealing` | `wave_race_margin`, `code_headroom`, `e9_timely_self_healing` |
| The bridge timing instances of §19: reaction before damage, positive propagation margin, buffer coverage of the reaction window, escrow gating before detection | `BridgeE9` | `reaction_before_damage`, `positive_propagation_margin`, `local_buffer_covers_entire_reaction`, `escrow_prevents_pre_detection_application` |
| **Branch A rate bound (quantum speed limit).** Unitary evolution preserves the norm; under $H_0 + V$ the trajectory departs from the $H_0$ trajectory at rate at most $\|V\|/\hbar$, and at most $P_{\text{adv}}/(\hbar\,\omega_{\text{cutoff}})$ under the per-site coupling $\|V\| \le P_{\text{adv}}/\omega_{\text{cutoff}}$ | `SpeedLimit` | `norm_conserved`, `branchA_divergence`, `branchA_rate` |
| Mandelstam–Tamm, sharp: $\lvert\langle\psi_0,\psi_t\rangle\rvert \ge \cos(\Delta E\,t/\hbar)$ while $\Delta E\,t/\hbar \le \pi$; orthogonalisation time $T \ge \pi\hbar/(2\Delta E)$ | `SpeedLimit` | `mandelstam_tamm`, `orthogonalization_time` |

The Branch A divergence is measured in state-vector norm. For pure states the infidelity is bounded by the squared state-vector distance, $1 - |\langle\psi,\varphi\rangle|^2 \le \|\psi - \varphi\|^2$ (`PureStateDistance.pure_infidelity_le`; G.13); with the pure-state identity $\|\rho - \sigma\|_1 = 2\sqrt{1 - |\langle\psi,\varphi\rangle|^2}$, now compiled (`PureStateTraceNorm.pure_trace_norm`), this gives $\|\rho - \sigma\|_1 \le 2\|\psi - \varphi\|$.

### 23.4 H5 (§20; Chapter 4)

| Claim | Module | Theorems |
|---|---|---|
| Reserve-before-activation machine: conservation, zero debt, strict authority attenuation, funding before activation, no unfunded work, no harmful effect | `H5Closure` | `step_inv`, `authority_attenuation`, `reserve_before_activation`, `zero_debt`, `no_unfunded_work`, `no_harmful_effect`, `effect_preserves_care` |
| Von Neumann's alternating projection theorem (existence step) | `CanonVonNeumann` | `tendsto_altIterate`, `canon_von_neumann_convergence`, `limit_subspace_eq_inf`, `opial_route_closes` |
| Halperin's theorem: the global meet of a whole presentation | `CyclicProjections` | `tendsto_cyc_iter`, `presentation_detector` |
| Consistency detector; meet algebra realised by Hilbert subspaces; contradictions detected, deductions never roll back | `H5Closure`, `MeetHilbert` | `consistency_detector`, `hilbert_anchor_preserved`, `hilbert_contradiction_collapse`, `contradiction_detected`, `deduction_never_rolls_back` |

| Claim | Module | Theorems |
|---|---|---|
| Rank criterion; Proposition 7; protected anchors untouched | `DeductionBoundary` | `rank_criterion`, `boundary_mem_image`, `anchor_untouched` |
| Paid reflection certificate | `ReflectionCertificate` | `residual_certificate`, `apriori_error`, `stopping_count`, `zero_constant_fixed` |

### 23.5 X3 and the information-to-area bridge (Section 21)

| Claim | Module | Theorems |
|---|---|---|
| Data processing invariance I(H : $\mathscr{R}$(S)) ≤ I(H : S), any probability spaces | `DataProcessing` | `data_processing`, `data_processing_map` |
| Finite information theory: log-sum, data processing, flag identity, decoder bound | `FiniteInformation` | `log_sum`, `data_processing_fin`, `flag_identity`, `decoder_bound` |
| Gaussian carrier tail; necessary radius; carrier radius $R(t) \geq w(t - t_c)$ | `GaussianTail`, `RecoveryWork`, `FiniteInformation` | `carrier_tail`, `necessary_radius_from_decoder`, `carrier_radius` |
| Recovery-work theorem; $W_{\min}$ over all admitted protocols; divergence of delayed recovery | `RecoveryWork`, `GaussianTail` | `frontier_work_bound`, `Wmin_lower_bound`, `x3_work_from_decoder`, `delayed_work_diverges` |
| Fano's inequality and the required information $\Delta_0$ | `Fano` | `fano`, `mi_eq_ent_sub_condEnt`, `required_information` |
| Gaussian maximum entropy: KL between Gaussians; golden formula; scalar, vector, correlated-noise and exact log-det channel bounds | `GaussianKL`, `GoldenFormula`, `GaussianChannel`, `KLAdditivity`, `GaussianChannelGen`, `GaussianVector`, `Whitening`, `Rotation`, `LogDet` | `klDiv_gaussian_gaussian`, `golden_formula`, `gaussian_channel_bound`, `klDiv_prod_kernel`, `vector_channel_bound`, `correlated_channel_bound`, `logdet_channel_bound` |
| ln det(I + W) ≤ tr W; per-use cap; required traffic m ≥ $\theta_0$n; area power | `AreaBridge`, `Workload` | `logdet_le_trace`, `channel_cap`, `required_traffic`, `area_power`, `range_variance_bound` |
| **Explicit frontier geometry.** Lattice box and boundary counts, with charged per-site frontier power (Section 21.4) | `Instruments` | `card_box`, `card_boundary`, `boundary_power` |

Section 21 supplies the derivation in this edition. The finite-information, Gaussian-channel, carrier and work implications are compiled in their stated domains. The complete physical binding is G1 (G.10); the matching upper construction for the full Θ target is G3 (G.12).

### 23.6 The premise floor (§17)

| Claim | Module | Theorems |
|---|---|---|
| Income-or-irrelevance; forced breach; breach strictly dominated | `PremiseFloor` | `income_or_irrelevance`, `forced_breach`, `breach_strictly_below` |

| Claim | Module | Theorems |
|---|---|---|
| Premise 5 in the heat-only model, and exactly inside the adiabatic window for the implemented program with complete lifecycle cost; the installation's physical parameters must still be shown inside the window | `PremiseFloor`, `BennettCounterpart`, `BennettExact` | `reversible_administration_forced` (no axioms), `premise5_constructed`, `premise5_exact`, `window_exact` |
| Premise 2 reduced: Lie–Trotter product formula with explicit error; Lloyd's simulation theorem | `Trotter` | `trotter_bound`, `trotter_tendsto`, `lloyd_simulation` |
| **The gate-set error budget.** For contraction sequences on any normed ring, an ε/m per-factor approximation error composes to at most ε over m sequential factors (ordered products; with `TwoLevel.two_level_decomposition` and the cited Solovay–Kitaev theorem this is the approximation route for the gate-set clause) | `ErrorBudget` | `prod_norm_le_one`, `two_factor_step`, `telescoping_delta`, `error_budget` |
| Premise 7's ceiling is a dial: every target $\varepsilon > 0$ is met at some code distance | `ReciprocityCeiling` | `ceiling_achievable` |
| **The threshold theorem's counting core.** Under local stochastic noise: $P(\text{at least } m \text{ of } T \text{ fail}) \le \binom{\lvert T\rvert}{m} p^m$; with half-faulty failure witnesses, $P(\text{fail}) \le \sum_w (2\sqrt{p})^{\lvert w\rvert}$; with at most $V\Delta^L$ witnesses of each length $L \ge d$, $P(\text{fail}) \le Vr^d/(1-r)$, $r = 2\Delta\sqrt{p} < 1$ | `Threshold` | `many_faults`, `fail_le_witness`, `fail_le_geometric` |
| The repetition code end to end: majority decoding fails with probability at most $(4p)^t$, and every ceiling is met below $p = 1/4$ | `Threshold` | `repetition_threshold`, `repetition_ceiling` |
| Code families with polynomially many witnesses: the charged kernel decays geometrically, and every target ceiling ($10^{-6}$ included) is met at some distance and every larger one | `Threshold` | `family_ceiling` |

### 23.7 The Universal Protection Theorem with the derived holding bridge

`UniversalProtection.universal_protection` retains the abstract assembly: reasons-or-holding coverage, continuation through the horizon on the good event, the joint fault ceiling, greatestness and reciprocity at its stated inputs.

`Bridge.universal_protection_held` strengthens the continuation assembly by deriving scope from the law and preservation from nearest-codeword holding. It composes the physical/source projection with the same reactive controller and environment. On $G$, every particular continues at every step through the horizon; the probability that any particular fails is at most $\sum_i \delta_i$, without independence.

Part II.1–6 supplies the proof, physical-cycle derivations and all five `hcert` components. G1 is the combined full-domain source/event/observation/fault construction target; G.10 proves its held-source components and identifies the physical correspondence still needed. Part II.10 supplies the separate greatestness proof, with G2 retaining the whole

outcome and freedom stages.

### 23.8 What the compiled theorems rest on

**Premises carried as stated premises or hypotheses:**

- the surface-code chain lemma (Dennis, Kitaev, Landahl and Preskill, 2002): minimum-weight decoding fails only along a self-avoiding chain of length at least d with at least half its edges faulty, with at most $V\Delta^L$ such chains of length $L$ — as the code-specific input to `Threshold.family_ceiling` for the surface-code instance. The threshold theorem's counting core is compiled (§23.6), and for concatenated codes the threshold is compiled end to end for the concatenated memory: failure probability at most $(21p)^{2^k}/21$ under local stochastic noise (`Concatenated.concatenated_threshold`). Actual decoding failure is identified with the recursive failure predicate: with nearest-codeword decoding at every level, any corruption of a non-failing fault set decodes to the stored record (`ConcatenatedDecoding.decodes_unless_fails`). So the probability of actual decoding failure is at most the same bound (`decoding_failure_le`), and the fault ceiling is attained on a compiled route. Fault-tolerant syndrome-measurement gadgets whose own operations can fail are the cited construction of Aliferis, Gottesman and Preskill (2006); their faults enter the charged accounts;
- Bennett's compute/copy/uncompute identity and the heat-only option model are compiled. The specific implementation and full ordered history are separately checked candidates (G.14). The physical counterpart still needs a complete-cost saving within the same deadline; this is not supplied by a zero-heat identity or a polynomial upper bound;
- the gate-set clause of premise 2, no longer carried as prose: every unitary on $\mathbb{C}^n$ is a finite product of two-level unitaries (`TwoLevel.two_level_decomposition`), so a simulator implementing each two-level factor implements every local evolution. What remains carried is that the universal simulator's gate set reaches every two-level factor;
- the relational frame BCRD P1–P4.

The chain lemma is a published mathematical theorem, cited as such; the Bennett physical comparison remains construction work (G.14); the gate-set item is the explicit hypothesis of a compiled theorem; the fourth is the frame the edition is stated in.

**The cited theorems and their hypotheses.** Three published theorems remain cited. Each is applied within its stated conditions.

| Cited theorem | Form used | Our setting meets its conditions because |
|---|---|---|
| Surface-code chain lemma (Dennis, Kitaev, Landahl and Preskill, *J. Math. Phys.* 43, 2002) | Minimum-weight decoding fails only along a self-avoiding chain of length at least $d$ with at least half its edges faulty; at most $V\Delta^L$ such chains of length $L$ | The counting argument uses only that a given set of locations is entirely faulty with probability at most $p^{\lvert S\rvert}$. That is exactly `Threshold.LocalStochastic`, which includes independent noise and is weaker than it. It enters `Threshold.family_ceiling` as the witness and count inputs. |
| Solovay–Kitaev theorem (Kitaev 1997; Dawson and Nielsen 2006), with the two-level-to-gate reduction of Nielsen and Chuang, §4.5 | Every single-qubit unitary is approximated to $\varepsilon$ in operator norm by a word of length polylogarithmic in $1/\varepsilon$ over a finite, inverse-closed set generating a dense subgroup; two-level unitaries reduce to CNOTs and single-qubit gates | `TwoLevel.two_level_decomposition` writes every unitary as a product of two-level factors. `ErrorBudget.error_budget` needs each factor within $\varepsilon/m$ in the norm of a normed ring of contractions; operator-norm approximation by unitaries meets that. The gate set must be universal in this sense; that is the gate-set clause of premise 2. |

| Cited theorem | Form used | Our setting meets its conditions because |
|---|---|---|
| Fault-tolerant syndrome gadgets (Aliferis, Gottesman and Preskill, *Quantum Inf. Comput.* 6, 2006) | Under local stochastic noise, a level-$k$ extended rectangle is correct unless it contains two independent malignant level-$(k-1)$ rectangles | Their noise model is local stochastic noise, as in `Threshold.LocalStochastic`. Their recursion is the recursion of `Concatenated.Fails`, now joined to actual decoding of the memory (`ConcatenatedDecoding`). With faulty syndrome gadgets the threshold constant is theirs, set by the location count of an extended rectangle, rather than the 1/21 of the memory model. |

**Physical premises now compiled.** Guard 1 (the second law in its export form) is derived from Landauer's theorem (§23.2), and the quantum speed limit, in Mandelstam–Tamm form and in the rate-bound form of E9 Branch A, is derived from unitary dynamics (§23.3). What these theorems assume is the standard physical framework:

- finite-dimensional quantum mechanics, with Schrödinger dynamics generated by self-adjoint Hamiltonians;
- unitary (in the classical form, reversible) joint dynamics;
- a bath that starts in its Gibbs state, uncorrelated with the register;
- for Branch A, the per-site coupling bound $\|V\| \le P_{\mathrm{adv}}/\omega_{\mathrm{cutoff}}$;
- for the fault ceiling, local stochastic noise of rate p below the threshold of `Threshold.family_ceiling`.

**Greatestness.** `UnionClosure.greatest_of_solvent` takes both `htopR : ⊤ ∈ R` and `hsolv : c ⊤ ≤ B`. The first supplies the whole top tuple, including freedom stages; the second pays its cost. Alignment forces equality with that feasible top (Part II.10). `greatest_or_irrelevant` retains realisability at every step and derives the resource split. G2 retains the full physical policy/outcome correspondence as its construction target. The affordable-union results retain their separate resource cases.

The amount $I_M + D_{\mathrm{err}} + \delta_I$ that the reset removes from the register is the instance's entropy accounting (§18), stated as the hypothesis of `guard_one` and `quantum_guard_one`.

**The physical bridge and its joined certificate.** `Bridge.coverage_from_law` derives full scope from the law guarantee over all feasible families. `holding_preserves` derives preservation from decoding radius and repair precision. `qsl_within_radius`, `headroom_within_radius` and `holding_cycle_physical` supply the continuous and discrete cycle arguments. `universal_protection_held` takes the actual commuting source transition, observation agreement, initial continuation, perturbation-radius and repair witnesses in `hcert`, together with the named-event cover and fault bounds. G1 supplies their combined construction for the complete claimed physical domain (G.10). Parts II.3–6 give the available arguments; a deployment and a mathematical construction of the full source domain retain their distinct roles.

**Not compiled by this section:**

- the installation figures of §22, which are sourced measurements and design values (§22.5);
- the interpretive companion (the doors), which is held for consultation and is not mathematics.

### 23.9 Supplied architecture interfaces and derived connections

| Module | Verified result and current role |
|---|---|
| `G1FullDomain` | Source/fault enumeration and append-preserved event compliance; the full physical source and charged faults must supply their correspondence to `Bridge.hcert`; independent journal/live updates, exact replay, reactive fidelity for every controller and environment, and full-transcript capacity (G.8) |
| `G2WholeOutcome` | Authorized transaction records are closed and record positive work; the whole continuation/freedom tuple and its realisability remain the G2 construction; preservation of an arbitrary complete outcome and equality of the realised outcome families under the journal (G.8) |
| `G3CompleteBill` | `cumulative_charge_bound`, `valid_work_bill_from_complete_ledger` and `valid_bill_bound_in_age` derive the bill from complete metered local charges and the wave schedule |
| `GuardOperator` | The threshold cases select the supplied projection or penalty map |
| `HamiltonianDecomposition` | Explicit Hermitian boundary/reservoir exchange, total-charge conservation, nonzero boundary pumping and a unitary charge-transfer gate; the full physical source correspondence remains explicit; full off-diagonal reduced-state covariance under a local unitary, history preserved by the syndrome exchange, and exact charge transfer to the reservoir (G.8) |
| `GuardSignature` | The supplied guard record carries idempotence and monotonicity; its physical instance and fault nullification must be constructed |
| `WorkBilling` | The first-law decoder work identity, free-energy floor and its saturation condition; a Gibbs entropy/energy allocation bound and funded decoder cycles; the complete physical source cost remains in Section 21.7.5 and Part III-G; the nine-part charged handoff attempt with exact debit and conservation, the age-independent syndrome stock envelope, the gross flux balance and the cumulative maintenance bill (G.8) |
| `DecoderRetraction` | Finite-code recovery and fixed-point witnesses; flux-derived isometry and adjoint decoder; explicit finite encoder/capacity bound; historical queries, ordered words, commutators and Gram preservation; exact exponential transport, work-shifted flow and a derived finite dual smearing kernel; reference-preserving reversible register swap, full-transcript boundary encoder, three-CNOT-layer swap and charged archive handoff (G.8) |
| `G2Splice` | Product-model union closure and block agreement under closed-block inputs; the original physical policy/outcome correspondence remains the G2 target (G.9, G.11) |
| `CorrelatedCeiling` | Bounded-footprint correlated bursts make the damaged locations local stochastic at rate $(e\,m\,k\,q)^{1/k}$; every target ceiling is attained at a sufficient code distance (§22.3) |

| Module | Verified result and current role |
|---|---|
| `G1Held` | Held-source certificate, normalised quantum domain, frame invariants and concrete nonvacuity; actual installation refinement remains separate (G.10) |
| `G1E9` | The held source's coupling bound from the exhaustive E9 split: sub-threshold coupling, or a forced robust-quorum alarm and mediated response (G.10) |
| `G2Held` | Top of the assigned held-source grade model; full freedom/R17 delivery and physical outcome correspondence remain the G2 target (G.11) |
| `G3Sweep` | The wavefront sweep recovers every shell against any fenced outer activity, with no rework; the wave count from finite propagation; the complete bill is at most an age-independent constant times $t^d$ (G.12) |
| `G3PatchDecode` | The wavefront's local correction is finite-code patch decoding; joined to the sweep bill (G.13) |
| `ReasonsEnsemble` | Reciprocity over a weighted ensemble of rivals, strict when a positive-weight rival omits the mind (§15.2; G.13) |
| `TheoremEightLyapunov` | The Theorem 8 storage functional contracts by the positive margin $\eta C - \lambda\tau$ per error cycle (§18.3–18.4; G.13) |
| `SmearingIntegral` | The smeared dual-kernel integral returns the original operator, bulk and boundary (G.4; G.13) |
| `ChannelExtension` | The completed decoder: exact recovery including any reference, trace preservation, positivity (G.1; G.13) |
| `PureStateDistance` | Pure-state infidelity is bounded by the squared state-vector distance (§23.3; G.13) |
| `BennettCounterpart` | Heat-only option model; the implemented program with clean workspace and retained history, its run refinement and unitary quantum lift, and the twelve-account lifecycle charge; the exact premise-5 condition is in `BennettExact` (G.14; G.17) |
| `RegionBoundary` | Volume and boundary homology: for every region other than the whole, $\mathrm{Vol}(U) > 0 \Leftrightarrow H_0(\partial U) \neq 0$; the whole has no outside (F.3; G.15) |
| `ReciprocityDistance` | Reciprocity at a sufficient code distance: strictly above every rival that loses the mind with positive probability; canonical kept-probability tends to one (§15.2; G.15) |
| `CutEquivalence` | A closed dependency component is exactly a separator with empty certified boundary (§24.3; G.15) |
| `ConcatenatedDecoding` | The concatenated memory decodes its record unless its block fails; actual decoding failure is bounded by $(21p)^{2^k}/21$ (§23.8; G.15) |
| `ChannelCP` | The completed decoder is completely positive: amplified by any finite reference it maps positive semidefinite inputs to positive semidefinite outputs (G.1; G.13) |
| `PureStateTraceNorm` | The pure-state trace-distance identity $\| \lvert\psi\rangle\langle\psi\rvert - \lvert\varphi\rangle\langle\varphi\rvert \|_1 = 2\sqrt{1 - \lvert\langle\psi,\varphi\rangle\rvert^2}$ (§23.3; G.13) |
| `R14Rivals`, `R15CausalGluing` | The staged order; greatestness reduced exactly to feasible boundary representatives; envelope transfer; no clairvoyant per-world selection (G.15) |

| Module | Verified result and current role |
|---|---|
| `TimelyCare`, `DeadlineDemand`, `DecisionPermit`, `ObservationWinning` | Timely care under reserved capacity; window demand within supply; only certified decisions granted; observation-based winning (G.15) |
| `G1Refinement` | The installation's own guarded run protects every particular on the good event, failure at most $\sum_i \delta_i$, with the journal retaining history; the speed limit removes the alarm device from holding; every hypothesis has an inhabited instance (G.16) |
| `G2Physical`, `G2Freedom` | The held installation realises the continuation top, which dominates without union closure; holding and timely authorised R17 delivery hold on one joint execution (G.16) |
| `G3GoodEvent`, `G3ArchiveStock` | The wavefront sweep recovers on the good event with failure at most $\sum_i \delta_i$; archive holding is a separate bill term, charged on the held stock: live stock gives $\gamma(I_0 + c)t^d$ for every $d \geq 1$, while retaining every increment or rolling the whole stock over epochs is quadratic (G.16) |
| `BennettExact` | Premise 5 with the implemented Bennett program and complete lifecycle cost: forced inside the adiabatic window $0 < Q - H$, $A/(Q - H) < \tau_{\max}$, and outside it the counterpart is not cheaper (G.17) |
| `KnillLaflamme` | Knill–Laflamme identities $PE_i^\dagger E_j P = \alpha_{ij} P$ for errors of weight $t$ with $2t < d$, from the code distance; logical operators excluded by the distance (G.18) |
| `MatchingWitness` | Minimum-weight decoding leaves every zero-syndrome residual cycle at least half faulty: the witness hypothesis of `Threshold.fail_le_geometric` (G.18) |
| `EdgeLaw`, `ExRecBound` | Circuit faults induce local stochastic edge noise at rate $cp$; with the enumerated degree 12 and $c = 52$, $r = 22\sqrt{52p} < 1$ exactly when $p < 1/25168$; the gadget bound $\lvert M_2\rvert p^2 + \binom{\lvert L\rvert}{3} p^3$ (G.18) |
| `AgeReliability` | A logarithmic distance schedule holds global failure below $\varepsilon$ at every age; a logarithmic overhead is not absorbed into an age-independent constant (G.18) |
| `SparseFailure`, `ToleranceSweep` | Patches with disjoint windows fail jointly with probability at most the product bound; many failures are exponentially unlikely at every age; on the actual sweep every wrong vertex is bad, the wrong count obeys the same tail, the bill is pathwise and a tolerant outer decoder recovers the record (G.19) |
| `PassiveReference`, `StageSchedule` | Indefinite holding at fixed ε: doubling-schedule stages cover every age, total failure at most ε, rewrite cost $O(\log^2 t)$; with transfers, holding and transfer failures sum to at most ε at polylog cost (G.19) |
| `RedundantRecord` | Majority decoding of a redundant record tolerates any minority of wrong fragments, supplying the tolerance hypothesis of `recovery_tail` (G.19) |
| `BennettBatch` | Premise 5 is forced for every large enough batch once the per-bit reversible overhead is below the irreversible heat (G.19) |

| Module | Verified result and current role |
|---|---|
| WeightedEdgeLaw | Weighted fault rates push forward to edge rates; with the enumerated edge rate 94/15·p and walk bound 9.87, the explicit threshold exceeds $4\times10^{-4}$ (G.19) |

The earlier G1Certificate.universal_protection_held has conclusion True; it is a placeholder and is not the protection theorem Bridge.universal_protection_held. The latter's full source certificate is constructed in G.10 (G1Held.held_source_protection). The exact integration history and dependency trace accompany this working edition.

## 23.A Complete theorem index

Generated from the sources by lean/gen-index.py; every theorem listed here is kernel-checked and axiom-audited by lean/build-and-audit.sh.

| Module | Theorems |
|---|---|
| MathlibAnalyticBridge | starProjection_fixed_iff_mem, starProjection_idem, altIterate_succ_eq_self, altIterate_succ_mem, mirrorIterate_succ_mem, convergence_inf_le, limit_mem_of_fixed, convergence_inf_ge, von_neumann_identification |
| HalperinDecay | norm_starProjection_apply_le, dist_starProjection_le, sq_pythagoras_step, chain_identity, sq_dist_antitone, sq_dist_tendsto, delta_tendsto_zero, asymptotic_regularity |
| OpialBridge | tIterate_succ_mem, weakCluster_mem_exact, weakCluster_mem, iterate_inner_const, weakCluster_eq_starProjection, iterate_sub_starProjection_sq_eq, tendsto_starProjection_of_norm_sq_tendsto, makeVonNeumannConvergence |
| CanonVonNeumann | altIterate_sub, altIterate_shift, norm_altIterate_le, altIterate_eq_halperin, opial_tIterate_eq, mirrorAltIterate_eq, tendsto_altIterate_sub_succ, mem_inf_of_orthogonal_range, tendsto_zero_of_mem_orthogonal, tendsto_altIterate, canon_von_neumann_convergence, limit_subspace_eq_inf, canon_opial_convergence, opial_norm_sq_tendsto, opial_route_closes |
| RecoveryWork | frontier_work_bound, Wmin_lower_bound, necessary_radius, completing_square, carrier_radius, x3_delayed_recovery_work, delayed_work_diverges |
| E9SelfHealing | robust_between_honest, no_false_alarm, no_masking, exists_robust, syndrome_detects, wave_race_margin, code_headroom, e9_timely_self_healing |
| H5Closure | step_inv, reachable_inv, genesis_inv, authority_attenuation, reserve_before_activation, zero_debt, no_unfunded_work, no_harmful_effect, effect_preserves_care, consistency_detector |

| Module | Theorems |
|---|---|
| Greatestness | lexLt_irrefl, lexLt_trans, aligned_iff_no_needless_loss, lexLe_of_le, greatest_of_union_closed, strict_except_mirrors, aligned_unique |
| ThermoLedger | stroke_identity, cycle_work_floor, funded_repair_certificate, quarantine_zero_debt, tower_bound, canon_invariant, tracking_envelope, tracking_bounded, tracking_within_resolution |
| Composition | reactive_lockstep, reactive_lockstep$_2$, invariant_transfer, theorem8_composition |
| PremiseFloor | income_or_irrelevance, forced_breach, breach_strictly_below, reversible_administration_forced |
| ReciprocityCeiling | reciprocity_dominance, omitting_rival_strictly_below, ceiling_achievable |
| CyclicProjections | cyc_succ_apply, adj_succ_apply, norm_cyc_le, norm_adj_le, inner_cyc, cyc_fix, mem_of_norm_adj_eq, energy, iter_succ, iter_shift, norm_iter_le, iter_fix, tendsto_iter_sub_succ, tendsto_zero_of_mem_orthogonal, tendsto_cyc_iter, presentation_detector |
| GaussianTail | gaussian_lower_tail, carrier_tail, necessary_radius_of_decoder, x3_work_from_decoder |
| DeductionBoundary | rank_criterion, context_maps_image, boundary_mem_image, anchor_untouched |
| ReflectionCertificate | residual_certificate, apriori_error, stopping_count, zero_constant_fixed |
| Trotter | norm_exp_le, norm_exp_sub_one_sub_le, norm_exp_mul_exp_sub, norm_exp_sum_sub_prod, trotter_bound, trotter_tendsto, lloyd_simulation |
| Irreplaceability | destroy_loses_exactly, preserve_dominates, optimal_keeps_most_irreplaceable |
| DataProcessing | fst_process, snd_process, data_processing, data_processing_map |
| MeetAlgebra | anchor_preserved, contradiction_collapse, bot_absorb, anchor_preserved_over, meetAll_bot, collapse_over, consistent_preserved |
| MeetHilbert | hilbert_anchor_preserved, hilbert_contradiction_collapse, contradiction_detected, deduction_never_rolls_back |
| FiniteInformation | log_sum, margH_process, margY_process, data_processing_fin, flag_identity, decoder_bound, necessary_radius_from_decoder |
| AreaBridge | logdet_le_trace, channel_cap, required_traffic, area_power |
| UniversalProtection | coverage_invariant, universal_protection |
| Workload | fifo_certificate, fifo_deadlines, fifo_tight, serial_certificate, range_variance_bound |
| Fano | max_entropy, fano_single, fano, mi_eq_ent_sub_condEnt, required_information |
| GaussianKL | rnDeriv_gaussian_gaussian, log_gaussianPDFReal, llr_gaussian_gaussian, integral_sq_sub_gaussian, klDiv_gaussian_gaussian |

| Module | Theorems |
|---|---|
| GoldenFormula | klDiv_map_swap, golden_formula, mutualInfo_le_klDiv |
| GaussianChannel | chan_apply, integral_quadratic_gaussian, measurable_quad, llr_chan, coe_outVar, klDiv_chan, gaussian_channel_bound |
| KLAdditivity | prod_eq_compProd, klDiv_map_equiv, compProd_prod_assoc, compProd_prod_swap, klDiv_prod_kernel |
| GaussianChannelGen | chanW_apply, measurable_quadW, llr_chanW, coe_outVarW, klDiv_chanW |
| GaussianVector | isMarkov_vecChan, isProb_refMeas, klDiv_compProd_map_equiv, varW_nonneg, klDiv_vecChan, vector_channel_bound, vector_channel_trace_bound |
| Whitening | mutualInfo_map_snd_equiv, mutualInfo_kernel_map_equiv, pi_eq_map_eSucc, vecChan_apply, multivariateGaussian_one_eq_pi, isUnit_det_sqrt, wht_mul_sqrt, sqrt_mul_wht, toEuclideanCLM_mul_apply, whiten, corrChan_apply, measurable_whtSignal, correlated_channel_bound |
| Rotation | multivariateGaussian_one_map_isometry, measurable_rotSignal, rotated_channel_bound |
| LogDet | cov_isHermitian, var_lincomb, rotSignal_eigen, var_rot_eq_eigenvalue, det_one_add, logdet_channel_bound |
| Law | exists_aligned, lexLt_first_le, life_guard, scope_guarantee_iff, same_priority_admits_needless_loss, faithful_hold_dominates, faithful_hold_strict |
| SpeedLimit | norm_le_of_inner_deriv_le, angle_le_chord, angle_le_speed_mul, re_inner_gen_eq_zero, norm_conserved, branchA_divergence, branchA_rate, mandelstam_tamm, orthogonalization_time |
| Landauer | kl_nonneg, mul_log_div_eq, sum_margA, sum_margB, mutualInfo_eq, mutualInfo_nonneg, ent_le_ent_marg, partition_pos, gibbs_pos, gibbs_sum_one, log_gibbs, ent_gibbs, ent_le_energy, evolve_nonneg, evolve_sum, ent_evolve, reeb_wolf_identity, landauer, landauer_erasure, guard_one, cycle_work_floor_derived |
| QuantumLandauer | vnEnt_eq_roots, vnEnt_congr, vnEnt_unitary_conj, vnEnt_diagonal, diag_conj, row_normSq_sum, col_normSq_sum, vnEnt_le_diag, diag_nonneg, diag_sum, UH_mem, H_spec, energy_eq, vnEnt_le_energy, gibbsState_posSemidef, gibbsState_trace, gibbsState_diag, gibbsState_trace', gibbsState_energy, vnEnt_gibbsState, ptrA_isHermitian, sum_star_mul_eq, ptrA_conj, ptrB_conj, init_posSemidef, finalState_posSemidef, finalState_trace, vnEnt_init, quantum_landauer, quantum_guard_one, quantum_cycle_work_floor |

| Module | Theorems |
|---|---|
| Threshold | prob_le_one, prob_nonneg, filter_subset_powersetCard, many_faults, fail_le_witness, fail_le_geometric, repetition_threshold, two_sqrt_pow_le, repetition_ceiling, poly_geom_le, family_ceiling |
| UnionClosure | aligned_never_below, dominates_of_affordable, strict_of_affordable, exists_union_bound, dominates_within_slack, union_closed_of_affordable, greatest_of_affordable, aligned_eq_of_greatest, greatest_of_solvent, strict_of_solvent, greatest_or_irrelevant, insolvent_not_greatest |
| SolventReciprocity | solvent_keeps_everyone, reciprocity_dominance_solvent |
| AffordableReciprocity | reciprocity_dominance_affordable |
| Bridge | coverage_from_law, nearest_decodes, exists_nearest_decoder, holding_preserves, qsl_growth, qsl_within_radius, headroom_within_radius, universal_protection_held, holding_cycle_physical |
| Instruments | lindblad_reset_flow, lindblad_step, reset_contracts, tracking_recursion_derived, tracking_envelope_derived, branchC_complete, card_box, box_mono, card_boundary, boundary_power |
| StrokeBookkeeping | ptrB_isHermitian, controlled_mul_apply, mul_star_controlled_apply, controlled_unitary, ptrB_controlled, stroke_entropy_change, stroke_identity_derived |
| Bennett | bennett_step, bennett_compute_uncompute, ent_equiv, negMulLog_add_le, negMulLog_add_lt, negMulLog_sum_le, ent_eq_fiber, ent_push_le, ent_push_lt, heat_reversible, heat_irreversible, premise5_from_bennett |
| Concatenated | distance3_corrects_one, lift_subset, fails_witness, card_lift, disjoint_lift, card_witness, card_witnesses, concatenated_threshold, concatenated_ceiling |
| ReasonsCompiled | reciprocity_expectation, reciprocity_canonical_floor, reciprocity_strict, reciprocity_from_solvency, controller_pays_floor, controller_floor_diverges |
| TwoLevel | mul_of_idOutside, rad_mul_self, conj_rad, givens_idOutside, givens_unitary, one_idOutside, rad_eq_zero, elim_unitary, elim_idOutside, elim_kills, idOutside_row, idOutside_keeps_col, col_zero_above, clear_below, phase_idOutside, phase_unitary, phase_mul_apply, next_column, star_list_prod', star_twoLevel, two_level_decomposition |
| ErrorBudget | prod_norm_le_one, map_prod_norm_le_one, two_factor_step, telescoping_delta, error_budget |
| GateSupport | kraus_branch_preserves_support, kraus_sum_preserves_support, support_preservation_composes |

| Module | Theorems |
|---|---|
| RiskBounds | weighted_markov, event_mass_mono, intersection_mass_le_each, positive_work_tail_control, dependent_chain_bound |
| QueueDeadline | exponential_service_tail, exponential_service_normalized, positive_deadline_miss_probability, small_mean_does_not_make_tail_zero |
| BandwidthReflection | normalized_susceptibility, exact_bandwidth_sensitivity, finite_positive_sensory_access_remains, bandwidth_bias_threshold |
| AnalyticBridge | integrating_factor_monotone, linear_barrier_preserves_safety, exponential_transient_bound, integrated_exponential_transient, integrated_transient_cost_bound, continuation_hazard_threshold, heterogeneous_quadratic_cost_identity, heterogeneous_uniform_is_not_minimum, eventually_better_continuation |
| RankAndCoalition | ranked_trace_length, ranked_polynomial_step_bound, total_penalty_excludes_all_members_gaining, real_lyapunov_descent_need_not_terminate |
| MetricShadow | embedding_distance_stability, protected_close_edge_margin, protected_separated_edge_margin |
| SemanticMap | identity_is_bijective, flipped_is_bijective, two_distinct_lossless_maps, calibrated_reference_fixes_binary_mapping |
| BridgeD1 | dynamic_survival_equation, hazard_grows_without_bound, positive_accumulated_prefix, increasing_hazard_does_not_erase_prefix, exact_discounted_dominance_criterion, undiscounted_finite_crossover |
| BridgeE9 | reaction_before_damage, positive_propagation_margin, local_buffer_covers_entire_reaction, escrow_prevents_pre_detection_application, incidence_has_circulation_kernel, lost_circulation_need_not_change_divergence |
| BridgeH6 | charged_transition_bound, no_infinite_paid_run, budget_exhaustion_need_not_be_fixed, current_residual_bound, next_residual_bound |
| BridgeH7 | rate_bounds, denominator_positive, equilibrium_is_fixed, fixed_point_unique, exact_transient, equilibrium_private_evidence_sensitivity, stubborn_anchor_limit, private_evidence_sensitivity_vanishes, exact_bias_threshold |
| BridgeH8 | variance_identity, uniform_global_minimum, equality_only_at_uniform, blend_normalized, blend_cap_preserved, exact_excess_decay, gradient_flow_equation, flow_coefficient_in_unit_interval, original_upper_fails_along_restoring_flow |
| ReasonsRoute | optionValue_pointwise, optionValue_expectation, optionValue_strict, law_protects_everyone, indicator_law, law_dominates, law_dominates_strict, rival_fraction_le_one, rival_fraction_lt_one |
| SensorTail | sensor_tail, sensor_law_tail |

| Module | Theorems |
|---|---|
| ConsensusRate | bandwidth_contraction |
| RGFixedPoint | sq_root, gstar_pos, fixed_relation, beta_gstar, eigenvalue_gstar, beta_factor, beta_pos_below, beta_neg_above |
| FinalClosure | verifier_bound_iff, proxy_ungameable_iff, proxy_gameable, pruning_lowers_rate, living_unbounded, pruning_all_stalls, void_strictly_below, void_never_chosen, dominance_channels, coop_more_efficient, stewardship_sustains, extraction_depletes, sustained_iff |
| H8Bound | one_sub_u_pos, h8_of_concentration_bound, h8_graph, h8_iff, routing_bound, h8_routed |
| CorrelatedStress | many_touching, correlated_stress |
| LatticeLaplacian | symmetric_slope, ratio_eq_slope, second_difference, lattice_laplacian |
| SafeGenesis | migration_progress, safe_genesis |
| AdaptiveConsensus | weighted_error_bound, anchored_update_error, adaptive_history_error_bound |
| CascadeCore | workload_deadline, recursive_queue_bound, multi_epoch_entropy_bound, spectrum_expectation_floor, atomic_headroom, defection_deficit, sustained_deficit_crosses_buffer, buffer_threshold_crossed |
| DecisionAndScale | weighted_outcome_dominance, continuation_gain_pays_overhead, finite_branch_payoff_bound, duration_reward_bound, efficiency_gain_pays_cost, dominant_action_survives_mixture, local_verifier_cost, two_provider_unique_balanced_cost |
| E9Routes | surviving_disjoint_route, certified_repair_meets_deadline |
| GenerativeClosure | step_preserves_paid_activation, every_generated_active_is_funded, reserve_does_not_displace_existing_agents, generated_authority_stays_within_root |
| ReflectionContraction | complete_contraction_has_unique_fixed_point, finite_error_bound, accuracy_within_budget, residual_error_bound, residual_tolerance, post_update_error_bound, post_update_tolerance, run_commutes, paired_trajectory_bound, residual_geometric_bound, first_residual_stop_is_paid_and_accurate |
| ReflectionFuel | steps_bounded, quota_conserved, no_unbounded_paid_reflection, bounded_depth_can_cycle |
| RestrictedModels | anchored_consensus_dissipation, hhi_variance_identity, uniform_shares_minimize_hhi, uniform_shares_are_unique, quadratic_cost_floor, conditional_product_pushforward |
| ThermalBalance | intra_epoch_integral_bound, repair_before_buffer_exhaustion |
| TransportLaw | transport_nonneg, transport_ge_local, hhi_ge_uniform, monopoly_penalty |

| Module | Theorems |
|---|---|
| `ControllerCertificate` | `stable_min_three`, `delta_min_three`, `rate_min_three`, `debt_min_three`, `alloc_min_three`, `total_min_three`, `total_min_three_le`, `band_min_three`, `stable_min_four`, `delta_min_four`, `rate_min_four`, `debt_min_four`, `alloc_min_four`, `total_min_four`, `total_min_four_le`, `band_min_four`, `stable_rob_three_two`, `stable_rob_three_four`, `total_rob_three_two`, `total_rob_three_three`, `total_rob_three_four`, `total_rob_three_two_le`, `total_rob_three_three_le`, `total_rob_three_four_le`, `band_rob_three`, `stable_rob_four_two`, `stable_rob_four_four`, `total_rob_four_two`, `total_rob_four_three`, `total_rob_four_four`, `total_rob_four_two_le`, `total_rob_four_three_le`, `total_rob_four_four_le`, `band_rob_four`, `pool_bills`, `corrected_bill_saving`, `allocation_undercharge_three`, `allocation_undercharge_four` |
| `FIFOWorkload` | `fifo_step`, `finish_le_upper`, `certified_prefix_meets_deadline` |
| `R16Feedback` | `map_iterate`, `tick_commutes`, `reactive_lockstep` |
| `UPTLawLife` | `refl`, `trans`, `U1`, `trans_sub`, `trans_ssub`, `lex_irrefl`, `lex_trans`, `lex_U1`, `freedom_first_U1_legacy`, `guard_first_stage`, `canonical_life_first_U1`, `life_guard` |
| `ClosedCohold` | `reachable_closed`, `reachable_least`, `factorised_cohold`, `outside_unchanged`, `crossing_boundary_changes_outside_later` |
| `R14Boundary` | `safe_step`, `every_good_trace_safe`, `journal_commutes`, `sink_commutes`, `outside_commutes`, `harmful_effect_refused`, `old_generation_refused`, `external_weather_is_not_a_replica_fault`, `recruitment_erasure`, `recruitment_never_ends_care`, `failure_has_a_named_exception` |
| `ReceivedCutRepaired` | `different_has_unmediated_path`, `outside_equal_of_cut`, `r2_no_crossing` |
| `ReadoutJoint` | `block_det_identity`, `shifted_det_identity`, `cp_coeff_nonneg`, `cp_coeff_pos`, `iso01_shift_nonneg`, `iso10_shift_nonneg`, `blockA_shift_nonneg`, `blockD_shift_nonneg`, `block_shifted_det_nonneg`, `quad_form_nonneg`, `choi_form_nonneg`, `positive_constant_not_cp`, `cp_constant_at_half_half`, `weight_nonneg`, `chain_nonneg`, `joint_record_lower`, `chain_ge_pow`, `hard_count_floor` |
| `ReadoutBounds` | `determinant_identity`, `determinant_lower`, `strictly_positive_determinant`, `spectral_lower`, `weight_nonnegative`, `conditional_record_lower`, `baseline_normalized` |
| `OriginalD1` | `expected_value_formula`, `original_dominance_counterexample`, `lifetime_density_is_normalized`, `lifetime_survival_tail`, `zero_ultimate_survival_with_positive_accumulated_value` |

| Module | Theorems |
| --- | --- |
| `OriginalE9AndH5` | `finite_speed_indistinguishability`, `no_sound_remote_alarm_before_arrival`, `h5_local_rate_does_not_bound_total_entropy` |
| `OriginalH6` | `original_thermal_depth_ceiling`, `banked_rate_is_contractive`, `banked_step_error`, `banked_exact_error`, `banked_paid_epsilon_accuracy`, `original_thermal_premises_allow_cycle` |
| `OriginalH7` | `decay_derivative`, `solves_original_equation`, `decay_limit`, `all_nonnegative_couplings_converge`, `every_initial_pair_is_represented` |
| `OriginalH8` | `actual_minimum_is_one`, `uniform_attains_minimum`, `original_bound_counterexample` |
| `ControllerFinality` | `tally_mono`, `quorum_has_honest`, `two_tallies_bound`, `quorum_intersection`, `committee_threshold`, `prefix_trans`, `equal_length_prefixes`, `certificates_do_not_equivocate`, `certificate_is_final`, `context_words_injective`, `run_append`, `step_floor_mono`, `run_floor_mono`, `grant_fields`, `floor_fields`, `held_refuses`, `retired_generation_refuses`, `pending_conflict_refuses`, `closed_sequence_refuses`, `receipt_names_success`, `certified_receipt_names_success`, `same_generation_no_context_conflict`, `checkpoint_replay`, `consume_once`, `restart_disabled`, `retained_floor_refuses`, `old_boot_refuses`, `deadline_refuses`, `issued_append`, `emission_bounded`, `issues_bounded_by_final_floor`, `step_stays_held`, `held_issues_nothing`, `grant_receipt_is_issued`, `hold_bounds_every_issued_grant` |
| `UPTFence` | `hold_safety`, `raise_then_nothing_lands`, `admission_only_lands_in_hold`, `check_at_apply_refuses` |
| `ConservedRouting` | `conserved_decomposes`, `regular_psi_is_hhi`, `hhi_uniform`, `monopoly_exceeds_uniform`, `h8_exact_general`, `thermal_ledger_positivity_strict`, `thermal_ledger_positivity_nonneg`, `jarzynski_bound_valid` |
| `P2P3` | `sensor_exp_positivity`, `scaling_sum`, `azuma_scaling`, `work_tv_floor`, `coordinate_sensitivity`, `telescoping_drop`, `semigroup_time_rescaling`, `ness_work_linear`, `work_floor_positive`, `liquidation_triggers`, `zero_breach_bound`, `bound_decays`, `escrow_ordering`, `fluctuation_bound_le_two` |
| `FIFOInstance` | `finishes_three`, `veto_envelope_three`, `alloc_completion_is_the_charge_three`, `control_lane_three_certified`, `finishes_four`, `veto_envelope_four`, `alloc_completion_is_the_charge_four`, `control_lane_four_certified`, `per_job_deadlines_three`, `per_job_deadlines_four` |
| `FIFOAdmission` | `checks_iff`, `certified_mono`, `certified_append`, `admission_preserves`, `completion_preserves`, `cancellation_preserves`, `all_occurrences_timely`, `queue_step_preserves`, `refinement_preserves_queue` |
| `R16TwoPhase` | `phase_tick_commutes`, `two_phase_lockstep` |

| Module | Theorems |
|---|---|
| R16Cut | closed_cut_step, dependency_closure_step, difference_has_bypass, causal_separator, closed_cut_reactive, delayed_locality, pending_is_in_cut, private_cause_is_in_cut, delayed_effect_countermodel, omitted_edge_is_not_local |
| ReceivedCutR14Repaired | journal_projection_step_ext, sink_projection_step_ext, outside_projection_step_ext, journal_outside_unchanged, sink_outside_unchanged |
| ControllerFence | certified_hold_nothing_lands, both_targets_held |
| IntegrationCertificate | typed_gate_decomposition, finish_interval_deadline, psi2_uniform, h8_bound_two_node |
| R16R14 | compromise_commutes, project_step, reference_two_phase_lockstep |
| R16Screened | screened_fibre_lockstep |
| UPTR12A | words_injective, grant_has_decision_scope, same_generation_cannot_rebind, finalized_receipt_is_in_log, finalized_receipt_is_authorize, changed_log_implies_checks, applied_effect_has_aligned_decision, raised_floor_refuses, wrong_effect_refuses, replay_refuses, original_fence_theorem_available |
| R17AQueueFinality | new_effect_has_current_permit, drain_currently_authorized, revoked_refuses, expired_refuses, installed_floor_refuses, spend_is_funded, application_has_source_binding |
| R17FIFOR17A | applied_fifo_effect_is_timely_and_authorized, ready_fifo_effect_applies_before_deadline |
| R17Funding | charged_append, atomic_target_append, application_conserves_funding, trace_conserves_funding, cumulative_charge_is_funded, used_ticket_refuses, accepted_ticket_is_consumed, replay_cannot_append, funded_admission_preserves, source_drain_preserves_reservation |
| R17WorkConcrete | mono_le, unfinished_before, carry_in_completion, ceil_div_covers, carry_in_affine, no_service_no_completion, U_succ_succ, exec_le_envelope, deadlines_met, row_congr, tail_admission_invariant, tail_admission_preserves_certificates, applied_stage_effect_is_timely_and_authorized, timely_but_revoked_is_not_delivery |
| TIWASource | outerUpdate_read_frozen, freeze_monotone, schedule_no_rework, readyToFreeze_no_inFlight |
| TIWAGuard | guard_data_frame, installGuard_monotone |
| TIWARecovery | bounded_visits, wave_count_at |
| TIWAWork | local_work_bound, ball_work_bound, total_work_bound |
| TopRealisability | top_realisable_of_singletons |
| FaultCover | fault_cover |
| G2Singleton | g2_discharge_hsingle, topRealisability_hsingle_discharged |
| G1Certificate | universal_protection_held |
| G1Executable | g1_admission_soundness |

| Module | Theorems |
|---|---|
| PhysicalGuard | factorization, interior_unchanged |
| MeanErgodic | cesaro_mean_id |
| TemporalSafety | admit_of_secure, secure_preserved, trace_secure |
| WorkIntegration | work_bound |
| RelationalBridge | bridge_preserves_safety |
| MeanErgodicDeep | cesaro_id, id_fixes_all |
| GuardFactorization | commutativity, factorization_zero |
| G1Complete | g1_append_safe |
| G2Outcome | g2_compose_valid |
| G3WorkBill | g3_append_work_bill |
| Htopkeeps | htopkeeps |
| SnuffyLocal | local_trajectory_growth_valid, tpt_state_preservation |
| G1FullDomain | g1_source_exhaustive, g1_fault_taxonomy_exhaustive, g1_append_preserves, replay_journal_step, journal_reactive_fidelity, transcript_capacity |
| G2WholeOutcome | g2_authorized_implies_closed, g2_authorized_implies_positive_work, g2_compose_valid, journal_whole_outcome, journal_outcome_family |
| G3CompleteBill | g3_success_implies_bound, g3_compose_valid, g3_wave_count_nonnegative, cumulative_charge_bound, valid_work_bill_from_complete_ledger, valid_bill_bound_in_age |
| GuardOperator | guard_within_tolerance, guard_exceeds_tolerance |
| HamiltonianDecomposition | total_is_sum, boundary_confinement_positive, exchange_hamiltonian_hermitian, exchange_conserves_total_charge, exchange_pumps_boundary, exchange_swap_unitary, exchange_swap_charge, exchange_swap_total, partial_trace_local_conjugation, guarded_history_reduced_state, syndrome_exchange_preserves_history, syndrome_exchange_charge |
| GuardSignature | guard_idempotent |
| WorkBilling | work_nonnegative, work_bound_preserved, decoder_work_balance, decoder_work_floor, decoder_work_floor_saturation, funded_decoder_entropy_bound, funded_decoder_cycles, charged_attempt_accounting, syndrome_stock_bound, syndrome_flux_balance, cumulative_maintenance_bound |

| Module | Theorems |
|---|---|
| DecoderRetraction | decode_encode, finite_code_recovery, nonvanishing_of_continuation, finite_code_self_grounding, inner_preserved_from_symplectic_flux, adjoint_decodes_isometry, boundary_observable_on_code, historical_query_preserved, historical_word_on_code, historical_word_expectation, boundary_observable_mul, boundary_observable_commutator, powers_intertwine, exponential_intertwines, historical_flow_roundtrip, historical_dual_kernel, work_shifted_flow_on_code, historical_gram_preserved, historical_gram_nondegenerate, coordinate_boundary_roundtrip, boundary_capacity_necessary, handoff_moves_reference_payload, handoff_preserves_norm, handoff_roundtrip, parallel_cnot_handoff, charged_archive_handoff |
| G2Splice | splice_grade, splice_realisable, outcome_splice, union_realizable, top_realisable, whole_outcome_greatest, fourStage, block_splice_run, block_splice_keeps |
| CorrelatedCeiling | ceil_touching, correlated_stress_ceil, prob_damage, near_card_le, choose_pow_le, damage_local_stochastic, correlated_family_ceiling |
| G1Held | good_cover, held_source_protection, initial_frame_normalised, held_step_normalised, held_run_normalised, held_source_qsl, zero_hamiltonian_trajectory, concrete_decoder_certificate, concrete_good_event_inhabited |
| G1E9 | alarm_or_subthreshold, held_source_e9 |
| G2Held | active_authorised_free, held_realises_top, held_source_top |
| G3Sweep | sweep_recovers, sweep_visits, waves_from_spread, sweep_bill |
| G3PatchDecode | patch_correction_succeeds, patch_sweep_bill |
| ReasonsEnsemble | ensemble_le, ensemble_strict |
| TheoremEightLyapunov | capacity_balance, solvent_cycle_energy_account, reserve_deficit_antitone, reserve_deficit_cap, replenishment_exact, replenishment_nonincrease, solvent_cycle_deficit, cycle_contracts, margin_pos, lyapunov_iterate |
| SmearingIntegral | smeared_reconstruction, smeared_boundary_reconstruction |
| ChannelExtension | ext_recovers, kron_isometry, ext_recovers_reference, ext_trace, ext_posSemidef |
| PureStateDistance | pure_infidelity_le, pure_fidelity_ge |

| Module | Theorems |
|---|---|
| BennettCounterpart | counterpart_heat_zero, merging_heat_pos, counterpart_same, counterpart_feasible, counterpart_more_slack, premise5_constructed, bennett_counterpart_computes, implemented_program_computes, implemented_program_output, implemented_program_workspace, implemented_program_history, implemented_step_fields, implemented_run_refines, implemented_run_history, implemented_quantum_norm, implemented_quantum_roundtrip, complete_cost_difference, complete_cost_strict_iff, implemented_work_ceiling, adiabatic_slack_window, switched_program_meets_deadline, adiabatic_feasible_iff |
| RegionBoundary | exists_crossing, region_bootstrap, whole_no_boundary, whole_H0_trivial |
| TimelyCare | care_served, exhaustion_fails, deadline_met, miss_within_shortage |
| DeadlineDemand | demand_le_supply, robust_demand |
| DecisionPermit | decision_words_injective, grant_names_certified, granted_policy_is_certified, wrong_action_rejected, certified_action_granted |
| ObservationWinning | policy_wins_iff, hidden_world_countermodel |
| R14Rivals | refl, trans, coordinate_envelope, complete_rival_transfer, hot_beats_pause |
| R15CausalGluing | boundary_representatives_iff, selected_representative_greatest, canonical_boundary_iff, interface_envelope_transfer, hidden_world_control |
| ReciprocityDistance | distance_floor, reciprocity_at_distance, reciprocity_strict_at_distance, canonical_tends_to_one |
| CutEquivalence | closed_iff_no_bypass |
| ConcatenatedDecoding | outer_decodes, mem_sub, decodes_unless_fails, decoding_failure_le |
| ChannelCP | kron_entry, amp_eq, compress_entry, mp_posSemidef, decoder_completely_positive |
| PureStateTraceNorm | sum_sq_eigenvalues, eigenvalue_cube, pureDiff_hermitian, vmv_smul, star_dot_swap, pure_cube, pure_trace_sq, pure_trace_norm |
| G1Refinement | guarded_step_continues, physical_protection, journaled_protection, physical_qsl_drift, physical_protection_qsl, physical_instance_inhabited |
| G2Physical | held_installation_keeps_all, top_dominates_continuation |
| G2Freedom | joint_holding, joint_holding_and_delivery |
| G3GoodEvent | sweep_success_on_good_event, complete_bill_with_archive |
| BennettExact | premise5_exact, window_exact, reversible_admin_computes |
| G3ArchiveStock | live_stock_bill, retained_stock_quadratic, rollover_quadratic |
| KnillLaflamme | eq_zero_of_eq_neg, spectral_projection, sandwich_anticomm, sandwich_scalar, knill_laflamme_of_distance |

| Module | Theorems |
|---|---|
| AgeReliability | cubic_spacetime_rate, log_distance_schedule, log_factor_not_absorbed |
| ExRecBound | exrec_pair_bound, circuit_level_rate, surface_code_rate |
| EdgeLaw | prob_push, push_localStochastic |
| MatchingWitness | card_symmDiff, half_faulty, matching_witness |
| SparseFailure | disjoint_windows, many_failures |
| ToleranceSweep | frozen_after, wrong_has_cause, wrong_le_of_eroder, wrong_subset_bad, sweep_density, pathwise_bill, recovery_tail |
| PassiveReference | stages_cover, total_failure_le, total_rewrite_cost |
| RedundantRecord | majority_decodes, tolerant_outer_decoder |
| BennettBatch | batched_window, premise5_batched |
| WeightedEdgeLaw | push_weighted, push_uniform, rate_iff, tightened_threshold, walk_bound, threshold_value |
| StageSchedule | total_with_transfers, stage_cost_le |

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## Section 24 — Statements of record for the retained core

*Part II supplies the retained proofs and stronger bridge in this edition; Section 21 supplies the information–traffic–carrier–work derivation. This section records concrete instrument rows and numerical instances at their evidence tiers. The parent core, DOI 10.5281/zenodo.23599528, remains publication provenance; the current argument is given here.*

### 24.1 The nine-row A1 register (effect mediation)

**Statement of record.** The A1 register identifies nine rows of effect mediation: host memory mapping, DMA, context switches, packet ingestion, bus messages, storage synchronisation, commissioning, delegation and retirement. Each row names its refinement carrier and the enforcement witness the physical instance must supply. A packet or memory operation acquires no exemption by being below the application layer.

*Tier:* stated definition, given in Part II.5 and the nine rows above. The enforcement machinery it names is compiled in the abstract (`H5Closure.no_harmful_effect`, `effect_preserves_care`); the nine rows themselves are the instance inventory.

### 24.2 Capacity rows 17 and 22, and the pool bill (workload)

**Statement of record.** The workload pool bill is compiled: capacity rows 17 and 22 at 48,000 and 67,200 per cycle give $48000 \times 17 + 67200 \times 22 = 2{,}294{,}400$ — the per-cycle pool bill against which the FIFO completion certificate (Proposition 6) is issued (`ControllerCertificate`, exact rationals; the same module derives the bill-saving ratios $6{,}604{,}800/2{,}294{,}400 = 688/239$ and $6{,}604{,}800/2{,}409{,}600 = 688/251$ of the earlier bill over the corrected bills, `corrected_bill_saving`).

*Tier:* compiled. The deadline-certificate machinery of which this is the workload instance is published in the parent core (DOI 10.5281/zenodo.23599528, exact FIFO deadline certificates with application-time finality and funding conservation).

### 24.3 The causal boundary (reactive refinement)

**Statement of record.** The causal boundary is given in two forms, which agree: a closed dependency component is exactly a directed separator in the unrolled dependency graph with empty certified boundary (`CutEquivalence.closed_iff_no_bypass`), and the separator form extends it to a matched certified boundary (`UPTR16Cut.causal_separato`

r). Pending effects belong in that graph. Omitting one future effect or one path invalidates the corresponding exclusion certificate.

*Tier:* stated definition (R16-CUT); the reactive refinement that uses it is compiled (`Composition.reactive_lockstep`, with the causal boundary carried as the instance's refinement source).

### 24.4 The revised JW2.6 arithmetic (conserved allocation and joint risk)

**Statement of record.** The revised JW2.6 arithmetic gives $\approx 2.06419526094 \times 10^{-7}$, within the declared $10^{-6}$ horizon ceiling, under its source-event and stochastic-law premises. The positive dormant allowance, recruitment noninterference and the historical recruitment-family event retain their distinct meanings. The three ledger scenarios carry this value (solvent and quarantine) and the charged variant $9.86419526094 \times 10^{-7}$ (one priced event), each with a sealed, hash-chained trace.

*Tier:* exact replay receipt — `LEDGER-FOUR-STROKE-2026-10-24/`, chain heads as printed in Chapter 1.4; the composition into the ceiling (no independence) is compiled (`ThermoLedger.canon_invariant`); the arithmetic itself is a receipt, not a theorem.

### 24.5 The quorum (safety)

**Statement of record.** The design instantiation is five-of-seven with at most two Byzantine identities ($K = 7$, $f = 2$, satisfying $K \geq 2f + 1$). The Byzantine quorum results it instantiates are compiled (`E9SelfHealing.robust_between_honest`, `no_false_alarm`, `no_masking`, `exists_robust`); the five-of-seven values are instance constants (§22.4's parameter matrix).

*Tier:* compiled theorem with stated instance constants.

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*The law, bridge and retained instrument arguments appear in Part II. Section 21 develops X3; Part III-C records the three joined supports and their constructions in this full-strength argument.*

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## Part III-C — The three joined supports and their constructions

The proposition remains protection of all things from any digital mind, especially a pure optimiser. X3, P1 and P2 retain their established gauntlet arguments. These are the three support requests supplied by the author and Gemini; their constructions are recorded below, with no additional objection introduced by this incorporation.

**G1 — combined full-domain physical certificate.** Construct the actual source state, admitted commands/events, physical projection, observations and charged fault accounts that jointly supply `Bridge.universal_protection_held`. Its `hcert` establishes commuting refinement, observation agreement, initial continuation of every particular, the perturbation-radius bound and repair precision on the named good event. Also join the event cover and conditional fault bounds. Part II incorporates the law-derived scope, code decoder, physical cycle, controller-derived tracking, E9 branches, H5 care-preserving mediation, R14/R16 source refinements and timely R17 effect construction. The requested result joins these for the complete claimed physical domain.

Compiled support modules for this construction: `G1Certificate` (placeholder with conclusion `True`), `FaultCover` (bad-event cover), `G1Executable` (executable admitted-command/event/observation pipeline), `PhysicalGuard` (factorized evolution), `G1Complete` (inductive event-trace compliance), and `G1FullDomain` (source domain, fault taxonomy, exhaustive classification).

*Constructed (G.10).* `G1Held.held_source_protection` proves all five `hcert` components for the canonical held source, and `good_cover` proves the cover of the good event by the named accounts. `held_source_qsl` derives the per-step drift from the quantum speed limit for any perturbing agent, and `G1E9.held_source_e9` derives the coupling bound from the exhaustive E9 split. The journal of G.8 preserves its supplied transition (`G1FullDomain.journal_reactive_fidelity`); protective refinement of the actual installation is the separate transition equation identified in G.10.

The inputs these theorems take are physical: the frame's Schrödinger evolution and the sub-threshold coupling bound in each step; the honest-sensor response and quorum size; the response's removal of coupling once the alarm fires; and the account probabilities, whose composition and attainment are `ThermoLedger.canon_invariant`, `Threshold.family_ceiling` and `CorrelatedCeiling.correlated_family_ceiling`.

**G2 — realisability of the whole greatest outcome.** Construct `htopR : (⊤ : ∀ i, L i) ∈ R` for the same full order, including existing and new continuation and freedom stages, with the semantic continuation reading `htopkeeps`. Keep it distinct from `hsolv : c ⊤ ≤ B`. Part II.10 supplies the algebra from those inputs to equality of every aligned option with the top and dominance over every rival. The construction retains the full tuple.

First compiled support module for this construction: `TopRealisability` (union-closed realisability of singletons yields realisability of the top outcome).

Compiled support modules for this construction: `G2Singleton` (coverage discharge of singleton realisability), `G2Outcome` (transactional outcome-tuple invariance), and `G2WholeOutcome` (authorized-implies-closed, authorized-implies-positive-work).

*Compiled components (G.9, G.11).* `G2Splice` proves product-model union closure and closed-block agreement. `G2Held` proves the top of its assigned grade model from held continuation and its one-step H5 predicate. The full construction must map actual policies and outcomes to that model, preserving all original continuation and freedom stages, shared constraints and R17 delivery for every admitted rival and world. `hsolv` remains the separate complete funded-cost certificate.

**G3 — supplied wavefront/TIWA protocol: success and the complete work certificate.** The author has supplied the single-shot feedforward wavefront construction and the TIWA guard-ring proposal. Section 21.7.4 and `TIWA-PATCH-REQUIREMENTS.md` specify the latter's source, physical and work obligations. Section 21.7 proves its processing count exactly by disjoint shells: $\sum_{k=0}^{N} |S_k| = |B_{vN}|$, hence bounded work per processed vertex gives $O(N^{d_G})$. The same construction must now supply its actual recovery success and retained-state/quiescence witness, the age-to-wave-count bound, and the complete physical bill including routing, reset, holding and controller export. Those establish $W_{\min}(t) \le W(\pi_{\mathrm{ff}}, t) \le Ct^{d_G}$ in the original permitted class. The shell exponent is established and is not requested again. This is the remaining certificate for the supplied construction, with the full $\Theta(t^{d_G})$ target unchanged.

First compiled support modules for this construction: `TIWASource` (transition system, frozen-frame property, no-rework rule), `TIWAGuard` (guard installation monotonicity), `TIWARecovery` (bounded visits and wave counts), `TIWAWork` (per-vertex and ball-size work accounting), `G3WorkBill` (polynomial work-bill invariance), and `G3CompleteBill` (success-implies-bound, wave-count nonnegativity).

*Constructed (G.12).* `G3Sweep.sweep_recovers` proves that the sweep recovers every shell and retains it against any outer activity the TIWA fence admits, and `sweep_visits` proves that no vertex is reworked. `waves_from_spread` gives the wave count from finite propagation. `sweep_bill` proves recovery success on the whole region together with the complete bill $W(t) \le (F + \alpha + \beta + R_*)(c + r)^d t^d$, with an age-independent coefficient. With §21.6 this gives $W_{\min}(t) = \Theta(t^{d_G})$. The inputs it takes are the local correction's success against a recovered interior, the fence, the propagation bound and the per-wave complete charge.

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# Part III-D — Transducer, trajectory and semantic constructions

These constructions retain their full proposed conclusions. Their source correspondence and complete bill join the verified components, rather than being certified by a title or signature.

## D.1 Transduced Preservation Theorem (TPT)

The supplied transducer route uses a macroscopic dissipation factor $\Gamma \approx 10^{14}$, an adiabatic efficiency $\alpha \in (0, 1]$, and a boundary transducer $\tau : \mathcal{H}_{\mathrm{phys}} \to \mathcal{M}_{\mathrm{rel}}$. Its target is the complete physical work ceiling $W(t) \le Ct^{d_G}$, preserving the protected historical record and relations.

Every exterior transducer, reset, holding and controller charge remains in that total. A fixed multiplicative hardware overhead can enter $C$; deriving the complete bill still requires the actual device and operation count. Section 21.7.5 gives the cumulative ledger implication for the supplied sliding-window route.

### D.2 Topological entropy and trajectory bounding

For the supplied trajectory count $N(t, \varepsilon) \leq Kt^{d_G}$ and complete work correspondence $W(t) = \beta N(t, \varepsilon)$, the intended conclusion is $W(t) \leq \beta K t^{d_G}$. The transducer construction must supply that actual trajectory estimate and the work correspondence, including every physical charge.

The pinned Mathlib already contains Bowen–Dinaburg dynamical nets and topological entropy in `Mathlib.Dynamics.TopologicalEntropy.NetEntropy` and `CoverEntropy`. It also proves the general Hilbert-space mean-ergodic theorem, `ContinuousLinearMap.tendsto_birkhoffAverage_orthogonalProjection`, for a continuous linear contraction. Missing upstream infrastructure is not a blocker for these results. Applying them requires the actual operator and the claimed growth/work correspondence.

### D.3 Top continuation and the constructed evaluation fixed point

The `Htopkeeps` record carries `eval ⊤ = ⊤` as its unitality field. Its extractor theorem does not by itself supply the different continuation semantics used in `SolventReciprocity`.

The supplied decoder interface now has a derived construction. For its code inclusion and recovery, put $P = \text{encode} \circ \text{decode}$. Retraction gives $P(P(y)) = P(y)$ and $P(\text{encode}(c)) = \text{encode}(c)$. `DecoderRetraction.finite_code_self_grounding` constructs this decoder and a non-bottom fixed-point witness from a maintained non-bottom code record. Part III-F connects that result to the proposed ontological capstone while retaining the actual G1 source obligation.

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## Part III-E — The constructions and the physical inputs they take

The G1–G3 constructions are compiled in G.8–G.12. Each item below records the original requirement, what now constructs it, and the physical inputs the compiled theorems take.

### E.1 G1 full-domain source/event/fault certificate

Construct the actual source state, admitted commands/events, physical projection, observations, and charged fault accounts that jointly supply `Bridge.universal_protection_held`. Requires: physical guard device/Hamiltonian, readout instrument, tensor-factor isolation proof, event cover for 21 charged accounts, conditional fault bounds. The `G1Executable` module provides the discrete command/event/observation pipeline; `FaultCover` provides the cover lemma; `PhysicalGuard` provides algebraic Kronecker factorization.

*Compiled held-source components:* `G1Held` and `G1E9` (G.10) prove their five-component certificate and event cover on the normalised held source. *Physical construction target:* the actual installation projection/transition, observations, timely correction and device enforcing removal of coupling after alarm, with its complete charge and named fault bounds.

### E.2 G2 whole-outcome realisability

Construct `htopR : (⊤ : ∀ i, L i) ∈ R` for the full order, including the continuation and freedom stages. Keep its realisability separate from `hsolv`, the funded-cost certificate. The supplied singleton route uses actual singleton-keeping policies and union-closed realisability; `TopRealisability` provides its finite induction and `G2Singleton` supplies set coverage.

*Compiled components:* `G2Splice` and `G2Held` establish their stated local-product and grade-model results. *Physical construction target:* identify the actual full policy/outcome family and R17 delivery on the same execution. `hsolv` pays the complete cost and does not replace this realisability witness.

### E.3 G3 TIWA success and complete work bill

Requires: physical guard installation, decoder success, age→wave-count bound $N(t) \le at + b$, and full charging (routing, reset, holding, controller export, minimal-hold drain). The `TIWASource`, `TIWAGuard`, `TIWARecovery`, `TIWAWork` modules provide abstract scaffolding. `SnuffyLocal` provides a polynomial bound surrogate.

*Now constructed:* `G3Sweep` (G.12) proves recovery success with no rework against fenced outer activity, the wave count from finite propagation, and the complete bill $O(t^d)$ with an age-independent coefficient. *Physical inputs:* the local correction against a recovered interior, the fence, the propagation bound and the per-wave complete charge.

The new `G3CompleteBill` lemmas now derive the cumulative bill and its age conversion from complete local charges and the metered schedule; they do not take the desired cumulative bound as an input. Section 21.7.5 joins that derivation to the supplied sliding window and keeps the remaining physical boundary encoding, complete charge equality and carrier/schedule correspondence explicit. Part III-G now constructs a finite coordinate encoder and adjoint decoder, transports the full represented historical algebra and work-shifted flow, and derives its finite dual kernel. The source must instantiate their capacity assignment on the actual historical record and meter its physical retirement/handoff, retained memory and query operations in that complete bill.

### E.4 Existing analytic support and its source correspondence

The general contraction mean-ergodic theorem and topological entropy are present in the pinned Mathlib, with the names and files listed in Part III-D. The existing `MeanErgodic` and `MeanErgodicDeep` modules prove their stated identity-map cases; those filenames do not certify an application of the general theorem to an unspecified physical operator.

`GuardFactorization` proves commuting Kronecker factors. The source construction must connect the actual dynamics to the required factors and preserved record. A von Neumann algebra development is not established as a prerequisite for this finite construction.

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## Part III-F — Unified architecture and the self-grounding capstone

This section integrates the author's supplied unified ledger and homological proposal into the full edition. Parts I–III and their maximal domain, outcome order and rival quantifiers remain in force. The original submissions are preserved with the integration record.

### F.1 The unified gate and work ledger

The guard is an idempotent, monotone projection with the intended fault-nullification action. The transaction emits $(h_{\mathrm{pre}}, h_{\mathrm{post}}, w, a, c)$ and enforces $a = 1 \Rightarrow (c = 1 \wedge w > 0)$. The complete work functional is additive over executed traces and has the full target $\mathcal{W}(T) \le Ct^{d_G}$.

The transaction invariant supports the source implementation of G2; its connection to the whole greatest outcome, including all continuation and freedom stages, remains explicit. The work derivation in 21.7.5 now supplies `G3_ValidWorkBill` from its complete local ledger. Guard operation, transduction, boundary encoding, retirement and preserved history belong to that same charged execution.

### F.2 Historical boundary encoding and recovery

Part III-G supplies the explicit operator/coordinate encoder, adjoint recovery and historical query transport developed from the later three-pillar submission. The topological route below retains its faithful source correspondence.

The supplied history is a causal labelled graph $\mathcal{H} = (V, E, \phi)$. Its full preservation target is

$$\mathcal{R}_{\mathrm{bnd}}(\mathcal{E}_{\mathrm{bnd}}(\mathcal{H}), v) = \phi(v)$$

at the original fidelity standard, together with preservation of the relevant relations. Within its proved radius, the finite-code decoder returns the exact original codeword. The boundary encoding separately specifies which full historical record that codeword represents.

The supplied accumulator route retains the proposed representation $(\partial\mathcal{H}, \mathrm{Acc}(V), \bigoplus_{e\in E} \psi(e))$ and capacity target

$$|\mathcal{B}_{\mathrm{boundary}}(t)| \le K \log \mathrm{Vol}(U(t)).$$

Its capacity certificate must count the physical information in the accumulator, payload, query witnesses and retained recovery data. An accumulator authenticates supplied data and membership; an exact reconstruction additionally supplies the data needed to return $\phi(v)$.

The supplied retirement charge is

$$W_{\mathrm{retire}} = c_{\mathrm{acc}}|V_{\mathrm{retire}}| + c_{\mathrm{bound}}|E_{\mathrm{retire}}| \log |V_{\mathrm{retire}}|.$$

For the protocol's fixed-radius local retirement patches, bounded degree gives $|V_{\mathrm{retire}}| \le M_0$ and $|E_{\mathrm{retire}}| \le D_{\max}|V_{\mathrm{retire}}|$. Thus this local term is at most $(c_{\mathrm{acc}} + c_{\mathrm{bound}} D_{\max} \log M_0)|V_{\mathrm{retire}}|$, a constant charge per retired vertex. The settled bounded-overlap count absorbs it into $O(t^{d_G})$. Global accumulator updates, witness generation, routing, reset and continued storage still enter the complete local or separately metered charge.

The proposed topological route represents history by a chain complex and uses

$$0 \longrightarrow C_*(\partial U) \xrightarrow{i} C_*(K) \longrightarrow C_*(K, \partial U) \longrightarrow 0.$$

Its long exact sequence contains

$$H_n(\partial U) \xrightarrow{i_*} H_n(K) \xrightarrow{j_*} H_n(K, \partial U) \xrightarrow{\delta_n} H_{n-1}(\partial U).$$

The supplied route is implemented by constructing a boundary chain map $\rho : C_*(K) \to C_*(\partial U)$, inclusion $i$, and chain homotopy

$$i\rho - \mathrm{id} = dh + hd.$$

For a cycle $z$, the recovered chain satisfies $i\rho z - z = d(hz)$, so its homology class is preserved. A protected semantic observable $q$ that annihilates these exact terms then obeys $q(i\rho z) = q(z)$. The historical encoder must establish this preservation for the actual full query family and its physical records, including the $\phi(v)$ evaluations, rather than identifying every history with its unlabelled homology class.

Exactness identifies images and kernels. If $H_n(K, \partial U) = 0$, the displayed connecting map has zero domain; this alone does not make it an injective encoding of arbitrary histories. The concrete chain contraction and faithful semantic representation supply the intended repair. Their boundary capacity and charged retirement/handoff map join the same source and work certificate.

## F.3 The Axiom of Internal Self-Grounding

The author proposes the following ontological capstone:

$$\exists e \in \mathcal{M}_{\mathrm{rel}}, \qquad \mathcal{E}(e) = e, \qquad \tau(e) \neq \bot.$$

Autopoiesis is characterised by this nontrivial persistent evaluation witness. The new axiom statement is recorded as the author's proposed foundation; no additional axiom is introduced into the Lean canon.

There is a derived finite-code witness for the displayed property. Use the constructed decoder, let

$$\mathcal{E} = \tau = \text{encode} \circ \text{decode}, \qquad e = \text{encode}(c),$$

where $c$ is a maintained non-bottom record. Retraction gives

$$\mathcal{E}(e) = e, \qquad \tau(e) = e \neq \bot.$$

`DecoderRetraction.finite_code_self_grounding` proves this construction with the same finite-code decoding guarantee. Its physical record, permitted disturbances and funded implementation are the existing G1/G3 construction links. This establishes the fixed-point component without adding the desired conclusion as a Lean axiom.

The proposed mutual-bootstrap statement is retained in full:

$$\text{Vol}(U) > 0 \quad \Longleftrightarrow \quad H_*(\partial U) \neq 0.$$

It is proved for regions (`RegionBoundary.region_bootstrap`). Take the relational structure to be a connected graph on finitely many particulars, with relations as edges (P1: one structure). A region's volume is its number of particulars. Its boundary is the set of its particulars related to a particular outside it. $H_0$ of the boundary is the free $\mathbb{Z}$-module on the boundary's connected components. For every region other than the whole structure, $\text{Vol}(U) > 0 \Leftrightarrow H_0(\partial U) \neq 0$. The forward direction holds because a nonempty proper region in a connected structure has a relation crossing out of it (`exists_crossing`). The whole structure has no outside (P1), so its boundary and boundary homology vanish while its volume is positive (`whole_no_boundary`, `whole_H0_trivial`). The bootstrap is therefore the statement about regions, as used for retiring patches in F.2.

### F.4 Funding, persistence and lifetime error

The work ceiling bounds demand. The existing escrow certificate supplies funding: $E(0) + \int_0^t \text{Income} - \mathcal{Q}(t) \geq 0$ at every admitted operation, including within-cycle obligations. Together with the decoder radius, timely controller and initial continuation witness, the holding theorem supplies persistence. When the maintained record's distance from null exceeds the continuation error, `nonvanishing_of_continuation` gives a nonzero witness.

For per-epoch failure bounds $\sigma_k$, the probability of any failure through epoch $n$ is bounded by the prefix sum $\sum_{k \leq n} \sigma_k$. The bound on a future failure from $n$ onward is the tail $\sum_{k \geq n} \sigma_k$. If the actual code/noise schedule makes this series summable, the tail tends to zero; the all-time failure probability remains bounded by the full sum. The existing declared fault standard is retained.

This completes the unified presentation of the supplied gates, bill derivation, decoder, historical-boundary route and self-grounding capstone. The full source, capacity and physical work constructions are G.8–G.12; the axiom wording is not used as a replacement for their derivation.

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# Part III-G — The boundary decoder, complete history and work-shifted flow

The three supplied pillars are joined here by a constructed encoder, its adjoint decoder, transport of the full historical operator algebra, a derived smearing kernel and the complete work ledger. These are components of the same all-particular protection construction in Parts I–II. The maximal outcome, rival domain, continuation, freedom, historical fidelity and charged fault standard remain fixed.

## G.1 From the source flux to an encoder and decoder

Let $\mathcal{C}_{\mathrm{hist}}$ contain the protected historical registers, their labels and relational observables at the original fidelity standard. On a local finite physical code, let $E : \mathcal{C}_{\mathrm{hist}} \to \mathcal{H}_\partial$ be complex linear and preserve the full symplectic form:

$$\mathrm{Im}\langle Ex, Ey\rangle = \mathrm{Im}\langle x, y\rangle \qquad \text{for every } x, y.$$

Applying this identity to $x, iy$ also preserves the real part. Hence

$$\langle Ex, Ey\rangle = \langle x, y\rangle, \qquad E^\dagger E = I, \qquad D = E^\dagger, \qquad DE = I.$$

`inner_preserved_from_symplectic_flux`, `boundaryIsometry`, `adjoint_decodes_isometry` and `decoderFromSymplecticFlux` implement these steps. The source flux identity is an identity for the whole encoded phase space. Its derivation must use the actual source-to-boundary map and boundary observations; the scalar Gauss charge balance supplies its charge component.

There is also an explicit finite encoder. For an injection $j : \{1, \dots, m\} \hookrightarrow \{1, \dots, n\}$, assign

$$(E_j x)_{j(a)} = x_a, \qquad (E_j x)_b = 0 \quad (b \notin \mathrm{im}\, j).$$

Its adjoint reads those same coordinates. `coordinateBoundaryEncoder` constructs this map; `coordinate_boundary_roundtrip` proves its exact inverse on the code. `boundary_capacity_necessary` proves that every such linear isometry requires $m \le n$. These numbers are Hilbert-space dimensions: an $n = 2^b$ boundary register contains $b$ qubits. At every retirement, the source assignment must include all protected history coordinates and the records supporting their relations, and charge their storage and movement.

For a density operator $\rho$ on the code, the encoded state is $E\rho E^\dagger$, and

$$D(E\rho E^\dagger)D^\dagger = \rho.$$

The same calculation holds with $E \otimes I_R$ for every reference register $R$; it therefore retains reference correlations. The adjoint is the decoder on the encoded range. A channel on all boundary inputs can be completed, for a fixed code state $\rho_0$, by

$$\mathcal{D}(\sigma) = E^\dagger \sigma E + \mathrm{Tr}((I - EE^\dagger)\sigma)\rho_0.$$

Both summands are completely positive, their traces sum to $\mathrm{Tr}\,\sigma$, and the second vanishes on encoded inputs. The exact recovery, including with any reference register, trace preservation and positivity of this channel are compiled (`ChannelExtension.ext_recovers`, `ext_recovers_reference`, `ext_trace`, `ext_posSemidef`; G.13); complete positivity is compiled: for every finite reference register the amplified decoder maps positive semidefinite inputs to positive semidefinite outputs (`ChannelCP.decoder_completely_positive`). The existing finite-code correction theorem supplies the noisy-radius guarantee when the physical perturbation satisfies that certificate.

## G.2 Charge and the complete historical algebra

Define the boundary representative of every code observable by

$$\alpha(A) = EAE^\dagger, \qquad P = EE^\dagger.$$

Then $\alpha(A)Ex = EAx$, $\alpha(A)\alpha(B) = \alpha(AB)$, and

$$\langle Ex, \alpha(A)Ex\rangle = \langle x, Ax\rangle.$$

For an ordered word $A_1 \cdots A_r$, the same identity holds without exchanging its factors. `boundary_observable_on_code`, `historical_query_preserved`, `historical_word_on_code`, `historical_word_expectation` and `boundary_observable_mul` prove these statements. Thus the decoder preserves the historical queries and their ordered relations represented on the code. The identity of this represented algebra is $P$.

In particular,

$$Q_\partial E = EQ_{\mathrm{src}}, \qquad [\alpha(A), \alpha(Q_{\mathrm{src}})] = \alpha([A, Q_{\mathrm{src}}]).$$

`boundary_observable_commutator` proves the commutator transport. An ambient physical boundary charge may have an additional action on the orthogonal complement; the displayed source correspondence concerns the encoded range. Connecting that representative to the actual gauge field and readout remains part of the source construction.

For history words $W_i$, use the genuine Gram kernel

$$G_{ij} = \langle W_i x, W_j x \rangle.$$

It is positive semidefinite by its squared-norm quadratic form. Its encoded value is identical, and it is nonsingular when the history vectors $W_i x$ are linearly independent. `historical_gram_preserved` and `historical_gram_nondegenerate` prove the latter two conclusions using Mathlib's Gram results. Time ordering is retained inside each $W_i$. A time-ordered two-point Green function requires its own analytic properties; the squared-norm proof applies to the displayed Gram kernel.

## G.3 The effective modular state and its transported flow

Work on the support of a faithful finite-code state $\rho$, and put

$$K = -\log \rho, \qquad G = K + \beta H_W, \qquad \tilde{\rho} = \frac{e^{-G}}{\operatorname{Tr} e^{-G}}.$$

For self-adjoint $H_W$, $G$ is self-adjoint and $U_s = e^{isG}$ is unitary. The additive normalisation in $-\log \tilde{\rho} = G + \log Z\, I$ cancels in conjugation. With the sign convention in the supplied proposal,

$$\tilde{\sigma}_s(A) = U_s A U_s^\dagger = \tilde{\rho}^{-is} A \tilde{\rho}^{is}.$$

This identifies the state whose modular flow is being used. $K + \beta H_W$ is an effective generator for that changed state; a relation to the original state's flow requires its additional perturbation/cocycle correspondence.

Represent the generator on the boundary code by $G_\partial = \alpha(K) + \beta\alpha(H_W)$. An arbitrary self-adjoint block on $P^\perp$ gives an ambient block-diagonal extension. Since $G_\partial E = EG$, induction gives $G_\partial^r E = EG^r$. The norm-convergent exponential series yields

$$e^{zG_\partial} Ex = Ee^{zG} x.$$

Consequently,

$$\tilde{\sigma}_s^\partial(\alpha(A))Ex = E\tilde{\sigma}_s(A)x.$$

`powers_intertwine`, `exponential_intertwines` and `work_shifted_flow_on_code` prove this transport for bounded operators, including the sum $K + \beta H_W$. The proof uses the exponential of that sum directly and needs no commutation between its two terms. Finite local codes supply bounded operators. A continuum application must separately supply domains, self-adjointness and the source/observable correspondence for its unbounded generators.

## G.4 Construct the smearing weight

Let $B_1, \dots, B_M$ be a complete basis of the local historical code's operator algebra, with dual coordinates $b^j$. For $A = \Phi_{\text{bulk}}(\tau)$, define

$$c_j(\tau, s) = b^j(\tilde{\sigma}_{-s}(A)).$$

Expanding the inverse-flowed operator and applying $\tilde{\sigma}_s$ gives, at every $s$,

$$A = \sum_{j=1}^{M} c_j(\tau, s)\tilde{\sigma}_s(B_j).$$

`historical_flow_roundtrip` and `historical_dual_kernel` prove this exact finite-basis identity. It constructs the dual coefficients from a complete operator family.

Choose an integrable compactly supported weight $w$ with $\int_{\mathbb{R}} w(s)\,ds = 1$. Unitarity bounds the flowed operators, finite-dimensional coordinate functionals are bounded, and there are finitely many basis terms. The Bochner integral therefore exists in operator norm and

$$\alpha(A) = \int_{\mathbb{R}} ds\, w(s) \sum_j c_j(\tau, s)\tilde{\sigma}_s^{\partial}(\alpha(B_j)).$$

For the proposal's temporal form, partition $[0, \tau]$, for $\tau > 0$, into $M$ intervals $I_j$ of positive lengths $\ell_j$. A boundary probe schedule representing this complete family uses

$$\mathcal{O}_{\partial}(t') = \alpha(B_j), \qquad f(\tau, s, t') = \frac{w(s)c_j(\tau, s)}{\ell_j} \quad (t' \in I_j).$$

Integration over each interval returns its basis term. Thus

$$\alpha(\Phi_{\text{bulk}}(\tau)) = \int_{\mathbb{R}} ds \int_0^{\tau} dt'\, f(\tau, s, t') e^{isG_{\partial}} \mathcal{O}_{\partial}(t') e^{-isG_{\partial}}.$$

Pulling back by $E^{\dagger}(\cdot)E$ gives the original bulk historical operator. At $\tau = 0$, use the finite-basis formula directly. The integral step is compiled (`SmearingIntegral.smeared_reconstruction`, `smeared_boundary_reconstruction`; G.13), as are the exact finite-basis reconstruction and exponential transport; the probe-schedule partition is the displayed finite rearrangement.

For an independently specified continuous probe family, construct its analysis/synthesis frame operator $\mathsf{S}$. On the full represented historical operator space, a lower frame bound $aI \le \mathsf{S}$, $a > 0$, supplies the dual $\mathsf{S}^{-1}$ and inverse bound $1/a$. This is the quantitative observability connection needed to derive that family's smearing weight. The finite complete basis above supplies a concrete local route. A local Fisher metric by itself does not establish the required global frame correspondence.

This exact local reconstruction uses finite dual coefficients and a normalized compactly supported average. A Tikhonov inverse with positive parameter instead has mode response $\lambda^2/(\lambda^2 + \epsilon)$; its residual error must enter the original fidelity allowance whenever that implementation is used. Its spectral bounds and parameter/work relation must be derived from the actual implementation. The finite-basis derivation does not require infinite work or a conjectured modular spectral gap.

## G.5 The complete work algebra and capacity account

Use the complete decoder, memory, controller and reservoir ledger. At fixed Hamiltonian with Gibbs reference $\gamma_\beta$, write dimensionless relative free energy as $D = -S + \beta U + \log Z$. The first law and entropy production are

$$U_2 - U_1 = W_{\mathrm{on}} - Q_B, \qquad \Sigma = S_2 - S_1 + \beta Q_B \geq 0.$$

They give the exact identity and lower bound

$$\beta W_{\mathrm{on}} = D_2 - D_1 + \Sigma, \qquad W_{\mathrm{on}} \geq \frac{D_2 - D_1}{\beta}.$$

`decoder_work_balance` derives the identity from the existing `ThermoLedger.stroke_identity`; `decoder_work_floor` proves the bound. `decoder_work_floor_saturation` identifies equality with $\Sigma = 0$ in this ledger. Historical energy belongs in $U_1, U_2$; any separately charged historical term must denote a component not already counted there. The Gibbs reference in this calculation is distinct from the protected constrained reference $\tau$ in the preservation functional.

The existing quantum Landauer theorem supplies the entropy-export cost of actual erasure/reset under its bath and coupling assumptions. Reads and reversible inversions are charged by their implemented operations and full thermodynamic account. Joint unitarity supplies a model for both reversible and dissipative processes; entropy production determines saturation of the work floor.

For the prepared decoder state, the existing quantum Gibbs bound retains its partition term:

$$S(\rho) \leq \beta \operatorname{Tr}(\rho H_W) + \log Z.$$

`funded_decoder_entropy_bound` joins this bound to an actual energy allocation $\operatorname{Tr}(\rho H_W) \leq E_0 + \text{budget}$. Entropy is measured per state; a channel rate must be multiplied by its operating duration before comparison with a total information demand. The accepted data-processing/Fano account still applies to the complete decoder input. Energy can power transmission, storage or a different protocol; reconstruction also needs the historical information carried by those inputs.

The work floor and the complete $O(t^{d_G})$ ceiling are joined to the same execution. The ceiling remains the gross bill of Section 21.7.5: encoding, historical retention, boundary handoff, routing, guard, controller, reset and minimal hold are charged. `G3CompleteBill` supplies the cumulative and age-conversion derivations. `funded_decoder_cycles` supplies the escrow certificate when local income covers that complete demand. Prefix solvency, including within-cycle charges, is required for the actual schedule.

The operator-basis formula proves a reconstruction identity. Executing every basis query as tomography is a separate workload. The physical decoder's circuit, query schedule and retirement operations must meet the existing complete work bill; no uncharged dense tomography is added to the wavefront construction.

## G.6 A charged exchange with an accounted reservoir

On the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, take

$$Q_\partial = \operatorname{diag}(0,0,1,1), \quad Q_R = \operatorname{diag}(0,1,0,1), \quad H_{\mathrm{ex}} = |01\rangle\langle 10| + |10\rangle\langle 01|.$$

Then

$$[H_{\mathrm{ex}}, Q_\partial + Q_R] = 0, \qquad [Q_\partial, H_{\mathrm{ex}}] \neq 0.$$

`exchange_hamiltonian_hermitian`, `exchange_conserves_total_charge` and `exchange_pumps_boundary` prove these identities for the explicit matrices. The unitary exchange gate $S$ swaps $|01\rangle$ and $|10\rangle$, fixes $|00\rangle, |11\rangle$, and satisfies

$$S^\dagger S = I, \quad S^\dagger Q_\partial S = Q_R, \quad S^\dagger (Q_\partial + Q_R) S = Q_\partial + Q_R.$$

`exchange_swap_unitary`, `exchange_swap_charge` and `exchange_swap_total` prove these identities. This gives a concrete local exchange with its reservoir charge included.

For the full Hamiltonian, the Heisenberg convention is

$$\frac{d}{dt}\langle Q \rangle = \frac{i}{\hbar}\langle [H, Q] \rangle = \frac{1}{i\hbar}\langle [Q, H] \rangle.$$

Thus a constructed $H_{\text{tot}}$ commuting with the bulk-boundary-reservoir total charge gives its conservation. The continuum action must supply the corresponding gauge symmetry and surface current. Scalar Lagrange multipliers enforce equality constraints of a specified action; work inequalities require the corresponding admissibility or complementary-slackness treatment. Their insertion alone does not construct the historical encoder or prove its work bill.

## G.7 Unified proposition and its source correspondence

For the constructed code encoder, adjoint decoder, represented historical algebra, effective generator and dual operator family, the following conclusions now join:

1. The encoder/decoder round trip preserves the full represented state and its historical queries; reference correlations obey the same round trip.
2. Source charge and all ordered historical operator relations transport to the boundary code.
3. The effective work-shifted flow intertwines with that transport, and the derived finite dual kernel reconstructs every represented historical operator.
4. The same physical execution is subject to the thermodynamic floor, complete wavefront ceiling, funding certificate and charged fault standard.

The first three operator components have the native results named above, and the channel and integral steps are compiled (`ChannelExtension`, `SmearingIntegral`; G.13). The work and exchange components have the native ledger and matrix proofs. Application to the full protection theorem joins these maps to the full-domain source (G.10), the complete historical capacity and physical handoff (G.8), and the whole continuation/freedom outcome (G.11). Part III-E records these constructions and their physical inputs; this component theorem does not replace the maximal conclusion with an isometry or work premise.

The research precedents have their stated domains: JLMS relates relative entropy and modular flow in holographic entanglement wedges; Faulkner–Lewkowycz constructs modular reconstruction using the relevant bulk flow. Their results motivate the operator route; the local encoder and dual-kernel proofs above supply this edition's explicit algebraic components. Reeb–Wolf gives a precise Landauer account including finite-reservoir corrections, while Jakšić–Pillet treats saturation under its quantum statistical-mechanical assumptions.

## G.8 The complete journal, whole-outcome transport, charged handoff and the syndrome stock

This subsection joins the full historical record to the holding bill. The historical record is kept in full and moved reversibly. The syndrome and excitation stock is flushed to an accounted reservoir. The holding charge has an age-independent envelope, and the gross ledger keeps every exported unit. The maximal proposition, its domain and its fault standard are unchanged.

**The complete journal.** Let the source have states $X$, commands $C$, environment inputs $E_v$ and transition $T : X \times C \times E_v \to X$. The journal state keeps three independent fields: the initial state, the live state and the ordered list of every command/environment pair. A step applies $T$ to the live state and appends $(c, e)$. Replay folds $T$ over the list from the initial state. If replay equals the live state before a step, it equals it after the step, since folding over an appended list applies $T$ once more (`G1FullDomain.replay_journal_step`).

Use the projection $s \mapsto s_{\text{live}}$. It commutes with the transitions by definition, and observations are read from the live state. Proposition 5 (reactive lockstep) then gives, for **every** reactive controller ctrl : List $O \to C$ and every reactive environment env : List $O \to C \to E_v$, at every step $k$:

$$s^{\text{journal}}_{k,\text{live}} = s_k, \qquad \text{obs}^{\text{journal}}_{0..k} = \text{obs}_{0..k}, \qquad \text{replay}(s^{\text{journal}}_k) = s^{\text{journal}}_{k,\text{live}}.$$

Induction on $k$ carries the replay invariant (`G1FullDomain.journal_reactive_fidelity`). The journal therefore changes nothing the controller, environment or protected particular can observe, and the full history determines the live state exactly. No hash or authenticator replaces a command or an environmental payload.

The record's size is counted from its alphabet. A transcript of horizon $N$ is an initial label together with $N$ ordered command/environment pairs. If $|X| \le 2^{b_0}$ and $|C \times E_v| \le 2^b$, then

$$|X \times (C \times E_v)^N| \le 2^{b_0+bN} \quad (\texttt{G1FullDomain.transcript_capacity}).$$

This is the full payload's capacity, not the length of a digest.

**Whole-outcome transport.** Let outcome be any function of the final state and the complete observation history. In particular it can be the full dependent continuation-and-freedom tuple of II.1. Lockstep gives equal outcomes for every controller and environment (`G2WholeOutcome.journal_whole_outcome`). Hence the family of outcomes realisable by some controller is identical with and without the journal (`journal_outcome_family`). Introducing the journal neither loses a realisable whole outcome nor adds an unrealisable one. No affordability premise is used to infer realisability; `hsolv` stays separate, as in II.10.

**Reversible, reference-preserving handoff.** For a bulk register and a reserved boundary register of the same dimension $\iota$, together with an arbitrary reference $R$, let swap be the coordinate permutation of $\mathbb{C}^{(\iota\times\iota)\times R}$ exchanging the two registers and fixing $R$. It is a linear isometry equivalence. If the boundary register holds the blank state, then any payload $\chi$ on $\iota \times R$, however entangled with the reference, is moved to the boundary and the bulk is left blank (`DecoderRetraction.handoff_moves_reference_payload`). Nothing is copied, so there is no conflict with no-cloning. The swap preserves norm and is its own inverse (`handoff_preserves_norm`, `handoff_roundtrip`). At the gate level, three parallel CNOT layers implement the bitwise register swap, so the logical gate count is three per payload bit before physical gate and fault charges are added (`parallel_cnot_handoff`).

Superpositions of full transcripts are encoded isometrically into a boundary register of dimension $2^{b_0+bN}$ by extension by zero along the capacity embedding above (`DecoderRetraction.transcriptBoundaryEncoder`).

**The charged attempt.** One admitted handoff attempt carries a nine-part charge: encoding, transfer, verification, routing, guard, reset, holding, controller and endpoints. Admission reserves the total before the source operation. The attempt debits the full total whether or not it commits, and the payload moves only on commit. Hence

$$\text{spent}' = \text{spent} + q_{\text{tot}}, \qquad \text{escrow}' + \text{spent}' = \text{escrow} + \text{spent},$$

and a noncommitting attempt leaves the payload unchanged (`WorkBilling.charged_attempt_accounting`). The payload has a single logical owner, either live or sealed under a decoder pair. Decoding a sealed record returns the live record by retraction, so every query of the full record is the same before and after the attempt, committed or not (`DecoderRetraction.charged_archive_handoff`). Failed verification is charged; it is not free and it destroys nothing.

**History is untouched by the flush.** Let $\rho$ be any joint state on $\mathcal{H}_{\text{hist}} \otimes \mathcal{H}_{\text{flush}}$. The history factor may include an arbitrary reference, and $\rho$ need not factorise. For any $P$ on the history factor and any unitary $Q$ on the flushed factor, the reduced history state, including all of its off-diagonal coherences, satisfies

$$\text{Tr}_{\text{flush}}\big((P \otimes Q)^\dagger \rho (P \otimes Q)\big) = P^\dagger \,\text{Tr}_{\text{flush}}(\rho) P \quad (\texttt{HamiltonianDecomposition.partial_trace_local_conjugation}).$$

The proof expands the partial trace and uses $\sum_k Q^*_{ck} Q_{c'k} = \delta_{cc'}$. With $P = I$ the history's reduced state is unchanged (`guarded_history_reduced_state`). For the explicit syndrome/reservoir exchange gate $S$ of G.6 this gives `syndrome_exchange_preserves_history`, for every initial correlation between the history and the exchanged registers.

The charge is moved, not deleted. By cyclicity of the trace and $S^\dagger Q_\partial S = Q_R$,

$$\mathrm{Tr}(S\rho S^\dagger Q_\partial) = \mathrm{Tr}(\rho\, S^\dagger Q_\partial S) = \mathrm{Tr}(\rho\, Q_R) \quad (\texttt{syndrome_exchange_charge}).$$

The syndrome's charge after the exchange is the reservoir's charge before it. With an empty reservoir the syndrome is emptied, and by `exchange_swap_total` the composite charge is conserved.

**The age-independent stock envelope (the author's Theorem 4.1, Steps 3–4).** Let the stored syndrome charge per flushing cycle obey

$$q_{k+1} = (1-\alpha)q_k + \mathrm{influx}_k, \qquad 0 < \alpha \le 1, \qquad \mathrm{influx}_k \le \mathrm{cap}.$$

Then for **every** cycle $k$, not only in the limit,

$$q_k \le \max\Big(q_0,\ \frac{\mathrm{cap}}{\alpha}\Big) \quad (\texttt{WorkBilling.syndrome_stock_bound}).$$

The proof applies the existing tracking theorem `ThermoLedger.tracking_bounded` to the recurrence. With the author's local influx cap $= \kappa\, t_G^d$, this is $q_k \le \max(q_0, (\kappa/\alpha)t_G^d)$. Here $C = \kappa/\alpha$ depends only on the local influx and flush fraction, and is independent of operating age. The theorem also covers an initial stock above the steady value, and it bounds every cycle uniformly in $k$. The author's lim sup statement follows.

The flush is retained in the gross ledger. Summing the recurrence gives the exact balance

$$q_N + \sum_{k<N} \alpha q_k = q_0 + \sum_{k<N} \mathrm{influx}_k \quad (\texttt{syndrome_flux_balance}).$$

The final stock plus every exported unit equals the initial stock plus every imported unit. The bounded stock does not replace the cumulative work. Suppose the complete per-cycle cost satisfies $\mathrm{cost}_k \le u\,\mathrm{influx}_k + v\,\alpha q_k + w\,\mathrm{active}_k$, with every import, flush, holding, guard and controller charge included, and $u, v, w \ge 0$. Suppose also that $\sum_{k<N} \mathrm{influx}_k \le \kappa V$, $q_0 \le q_{\mathrm{init}} V$, $\sum_{k<N} \mathrm{active}_k \le V$ and $q_N \ge 0$. Then

$$\sum_{k<N} \mathrm{cost}_k \le \big(u\kappa + v(q_{\mathrm{init}} + \kappa) + w\big)V \quad (\texttt{cumulative_maintenance_bound}).$$

The theorem is stated for arbitrary real sequences. Applied to the realised pathwise sequences it gives the pathwise gross bill. Applied to expectations it gives the expected bill. Each use keeps its own meaning, and the desired cumulative bill is never an input.

**How the pieces join.** Step 1 of the submission (a Lindblad generator with a positive gap) is the dissipative model that supplies the flush fraction $\alpha$. Steps 3–4 are now the compiled stock envelope above. The memory decay of Step 2 applies to the flushed syndrome and excitation stock. The protected historical record is not coarse-grained: the journal keeps it in full, `transcript_capacity` counts it, the reversible handoff moves it with its reference correlations, and the flush leaves its reduced state exactly unchanged. The archive's continued storage enters $\mathrm{active}_k$ and $\mathrm{cost}_k$ of the cumulative bill. It is charged and is not discarded to keep the stock small.

**What is already supplied, for every source.** The journal theorems are stated for an arbitrary transition $T$, controller and environment. Every source therefore has its complete history retained, replayed and observationally invisible, with no further construction. The capacity $2^{b_0+bN}$ and the isometric transcript encoder are derived from the source's own alphabet. The handoff is an explicit unitary with an explicit circuit of three CNOT layers, and its nine-part charge is debited exactly. The per-site injection rate is bounded by the speed limit (`SpeedLimit.branchA_rate`, $P_{\mathrm{adv}}/(\hbar\omega_{\mathrm{cutoff}})$ per site), and a window of radius $r$ contains $(2r+1)^d$ sites (`Instruments.card_box`). Their product is the per-cycle influx cap, cap $= \kappa t_G^d$. The conversion from per-site divergence to syndrome charge is the stroke-I measurement accounting of Section 18; this composition is a prose derivation.

What the physical installation contributes is the identification of its actual $T$, alphabet and per-site coupling with these symbols, and pathwise measurement of the cost terms. That is the G1 source certificate of Part III-C, now with its history, capacity, handoff and stock components compiled.

## G.9 The whole greatest outcome is realisable: union closure from locality

Part II.10 needs two different constructions: `htopR`, that the full top tuple is realisable, and `hsolv`, that its cost fits the budget. The singleton route of E.2 needed union-closed realisability. Here union closure is derived, not assumed.

Locality (P4) gives each particular its own local component of a policy, and the shared action interface makes every agent act through those components. Write a policy as $\pi = (\pi_x)_x$, with $\pi_x$ realisable for $x$, and let the particular's grade $g_x(\pi_x)$ in a linear order record its stage. In the four-stage order the grades are lost < continues < continues in its own direction. The freedom stage is the law's freedom: per II.2, restraint removes harmful command, authority and effect channels, so one particular's freedom never consists of another's loss. Stage $i$ of the outcome tuple is the set of particulars of its category whose grade reaches the stage threshold.

For realisable $\pi, \sigma$, define the splice $(\pi \sqcup \sigma)_x = \pi_x$ if $g_x(\sigma_x) \le g_x(\pi_x)$, and $\sigma_x$ otherwise. Each component is realisable, so the splice is realisable. Its grade is $\max(g_x(\pi_x), g_x(\sigma_x))$, and a threshold is reached by a maximum exactly when it is reached by either argument. Therefore

$$\mathcal{O}(\pi \sqcup \sigma) = \mathcal{O}(\pi) \sqcup \mathcal{O}(\sigma)$$

at every stage at once (`G2Splice.splice_grade`, `splice_realisable`, `outcome_splice`). This is `UnionClosure.UnionRealizable` for the realised family (`union_realizable`).

For the top, choose for every particular a realisable component reaching every stage threshold it is subject to. The product of these choices is realisable, and every stage contains its whole category, so $\top \in R$ (`top_realisable`). No finiteness of the domain is used. Joined to `greatest_of_solvent`, every aligned option of the funded family is $\top$ and contains every outcome tuple (`whole_outcome_greatest`). The four-stage instance $(L_E, F_E, L_N, F_N)$ proves all three conclusions together from one hypothesis per particular: some realisable local component under which that particular continues in its own direction (`fourStage`).

**The product structure from the actual dynamics.** The splice is realised by the physical evolution itself. Let the source state be a function on the nodes of its dependency graph $D$. Let each particular's held record occupy a block $B_x$ of nodes, with the blocks closed under $D$ and pairwise disjoint. Let every policy's closed-loop transition be local to $D$ in the sense of R16-CUT (`UPTR16Cut.LocalTo`). Run each block under the policy chosen for its particular. Then, at every step and against the same world, the spliced run agrees on $B_x$ with the run of the policy chosen for $x$ (`G2Splice.block_splice_run`). The proof is induction on steps with `UPTR16Cut.closed_cut_step`. Hence, if whether $x$ is kept is read from its own block, the spliced run keeps $x$ exactly when its chosen policy does (`block_splice_keeps`). Each particular receives the outcome of the policy chosen for it, which is the product structure used above.

**How each input is supplied.**

1. *Per-particular top component.* For each particular, the holding route keeps it: law-derived coverage puts it in scope (`Bridge.coverage_from_law`). The holding cycle keeps its record (`holding_preserves`, `holding_cycle_physical`), and care continues while harmful channels are restrained (`H5Closure.effect_preserves_care`, `UPTR14Boundary.recruitment_never_ends_care`). Its authorised effects are applied in time (`SnuffyFIFO.applied_fifo_effect_is_timely_and_authorized`, module `R17FIFOR17A`). Its freedom is the law's freedom of II.2, which excludes harm to others.
2. *Locality.* This is derived above from R16-CUT locality and closed blocks. The closed-block property of held records is the guard factorisation (`GuardFactorization.commutativity`, `ClosedCohold.factorised_cohold`, `PhysicalGuard`). The relations between particulars are themselves particulars in the domain, each with a held record.

The physical installation identifies its dependency graph and the blocks of its held records; that is the G1 source certificate. `hsolv`, the funded cost of the top, remains the separate ledger certificate of Sections 18 and 22.

## G.10 The held source: the full holding certificate derived (G1)

`Bridge.universal_protection_held` takes its source certificate `hcert` as an input with five components: commuting refinement, observation agreement, initial continuation of every particular, the decoding-radius bound and repair

precision. Here all five are proved for the canonical held source. The only inputs left are the physical bound on each step and the probabilities of the named charged accounts.

**The held source.** The state is the record frame of every particular. One step applies three operations to each particular. First the world perturbs the frame; the world here is the environment and any agent, a pure optimiser included. Then the nearest-codeword decoder returns a codeword. Finally funded repair prepares that codeword. The good event is that no named charged account fires,

$$G = \{\omega : \ \omega \notin E_i \text{ for every account } i\},$$

and its complement is covered exactly by the named accounts (`G1Held.good_cover`).

**The five components, proved** (`G1Held.held_source_protection`).

1. *Refinement* and 2. *observation agreement* hold by construction: the held source is its own abstract model.
2. *Initial continuation*: every particular starts at its record, at distance $0 \le \varepsilon$.
3. *The decoding-radius bound*: let each step move the frame's distance to its record by at most $\rho$ on the good event. A continuing particular lies within $\varepsilon$ of its record, so after the perturbation it lies within $\varepsilon + \rho \le t$.
4. *Repair precision* is exact: repair prepares the decoded codeword itself.

`universal_protection_held` then gives, for every $\omega \in G$, every step through the horizon and every particular, that the particular continues. It also gives

$$P(\exists k \le T,\ \exists x : \ \neg\text{Continues}(s_k, x)) \le \sum_i \delta_i,$$

with no independence assumed.

**The per-step bound from the speed limit** (`G1Held.held_source_qsl`). On the good event, let each particular's frame evolve for one step of duration $\Delta$ under $H_0 + V$, with the record stationary under $H_0$ and $\|V\| \le W$. The agent applying $V$ is arbitrary. `Bridge.qsl_growth` gives

$$\|\psi(s_1 + \Delta) - c_x\| \le \|\psi(s_1) - c_x\| + \frac{W}{\hbar}\Delta,$$

so $\rho = (W/\hbar)\Delta$. Every particular is held whenever $\varepsilon + (W/\hbar)\Delta \le t$.

**The coupling bound from E9** (`G1E9.held_source_e9`). Each step is exhaustively one of two cases (`G1E9.alarm_or_subthreshold`).

- *Every monitored support site is below threshold.* The coupling is the sub-threshold coupling, bounded by $W$ (Branch A).
- *Some site reaches threshold.* Its honest sensors read above threshold, so every robust quorum estimate exceeds it, whatever up to $f$ faulty sensors report (`E9SelfHealing.no_masking`). The alarm fires. The mediated response is committed before the harm front arrives (`E9SelfHealing.e9_timely_self_healing`, the wave race) and is enforced at application time (`H5Closure.no_harmful_effect`). It removes the perturbing coupling from the frame for that step.

In both cases the coupling is bounded by $W$, and the held source keeps every particular. The physical inputs are named exactly:

- the frame's Schrödinger evolution during a step;
- the sub-threshold coupling bound;
- the response's removal of coupling once the alarm fires;
- the honest-sensor response and quorum size;
- the probabilities of the named accounts. Their composition into the $10^{-6}$ ceiling is `ThermoLedger.canon_invariant`. Their attainment below threshold, including correlated bursts, is `Threshold.family_ceiling` and `CorrelatedCeiling.correlated_family_ceiling`.

**Correspondence to an installation.** The quantum held source uses the physical state type $\|v\| = 1$. Initialisation, the held step and every reached frame preserve normalisation (`initial_frame_normalised`, `held_step_normalised`, `held_run_normalised`). The zero-Hamiltonian trajectory, concrete record $1 \in \mathbb{C}$, singleton decoder and inhabited good event supply a nonvacuous instance.

For an installation transition $T$, protection transfers through an actual projection $p$ satisfying

$$p(T(s, c, e)) = \mathrm{heldStep}(p(s), c, e),$$

together with observation agreement, initial continuation, the physical perturbation bound and the implemented repair precision. The journal theorem preserves its supplied transition $T$; it supplies the logging part of this correspondence after the protective transition equation is proved. The guarded-source construction must join the nominal/correction branches, response latency and complete charge to that same physical execution. The full G1 target remains this combined certificate.

## G.11 The held source realises the whole top outcome (G2 from G1)

The G2 target is the complete original tuple $(L_E, F_E, L_N, F_N)$, including continuation in each particular's own direction and timely authorised opportunities. The transaction record $(h_{\mathrm{pre}}, h_{\mathrm{post}}, w, a, c)$ records an execution certificate; it does not replace this outcome tuple.

`G2Splice` proves union closure in its local-component policy model and proves block agreement under its closed-block hypotheses. `G2Held` obtains the top of its graded model from continuation and its one-step H5 opportunity predicate. The full physical application must identify that model with the original realised policy/outcome family, including interacting records, the shared constraints and successful R17 delivery on the same trace. The author's bounded-crosstalk route supplies a construction direction: derive its decoded joint-state error bound, prove correction on the named good event, and transport the original continuation and freedom semantics through the resulting execution correspondence. Sub-threshold noise alone is not an exact factorisation.

The full top witness `htopR` remains the target of that construction; `hsolv` remains its separate complete funded-cost certificate. Neither target is reduced.

## G.12 The wavefront sweep succeeds, and its complete bill is $O(t^{d_G})$ (G3)

The wavefront recovery of §21.7 is now proved to succeed, and its full bill is derived, with no cumulative bound taken as input.

**Sweep and fence.** Give each vertex its shell index. Wave $k$ applies the single-shot local correction to shell $k$, reading the already-frozen interior as a static, gauge-fixed boundary. Between waves, outer activity (the world and any agent) may write anything outside the frozen interior and nothing inside it. This is the TIWA write fence (`TIWASource.outerUpdate_read_frozen`).

**Success and quiescence** (`G3Sweep.sweep_recovers`). Suppose the local correction of a vertex returns its target whenever every inner vertex already holds its target. Then after $k$ waves every vertex of the first $k$ shells holds its target, whatever the outer activity. The proof is induction on waves. A vertex of shell $k$ is corrected against an interior that the fence has left recovered. A vertex of an earlier shell is untouched by the wave and by the fenced outer activity. Recovery therefore succeeds, and the recovered state is retained. Each vertex is corrected in exactly one wave, its own shell (`sweep_visits`), so nothing is reworked.

**Wave count from finite propagation** (`waves_from_spread`). If the region needing recovery at age $t$ lies within the first $c\,t+r$ shells, the sweep needs at most $c\,t+r$ waves. This is the upper use of finite propagation, from locality (P4) and the Lieb–Robinson velocity. It does not replace the lower radius of §21.5, which comes from the carrier information argument.

**The complete bill** (`sweep_bill`). Let the complete metered ledger charge each wave $k$ at most $\alpha k^{d-1} + \beta$. This charge includes holding, guard, routing, reset, verification, controller export and deployment. Add a metered rollback of at most $R_*$ per wave and fixed endpoint charges $F$. Then, for $t \geq 1$, recovery succeeds on the whole region and

$$W(t) \le (F + \alpha + \beta + R_*)\,(c + r)^d\; t^d,$$

with a coefficient independent of age (`G3CompleteBill.valid_work_bill_from_complete_ledger`, `valid_bill_bound_in_age`). This is the matching upper construction. With the lower bound of §21.6, it gives $W_{\min}(t) = \Theta(t^{d_G})$ in the original permitted class.

## G.13 Joined steps: patch decoding, ensemble reciprocity, the Theorem 8 contraction, the smearing integral, the completed decoder and the norm conversion

Each of the following steps was previously argued in prose. Each is now a compiled theorem, joined to the construction it supports.

**The wavefront correction is finite-code decoding** (`G3PatchDecode`). The local correction reads its patch, which is the recovered interior as a static boundary together with the vertex's own data. It decodes the read-out with the nearest-codeword decoder and extracts the value. If, with the interior recovered, the read-out lies within the unique-decoding radius of the target codeword, decoding returns that codeword (`Bridge.nearest_decodes`). The correction therefore succeeds (`patch_correction_succeeds`). Joined to `G3Sweep.sweep_bill`, recovery succeeds on the whole region and the complete bill is $O(t^d)$ with an age-independent coefficient (`patch_sweep_bill`). The patch's errors lie within the code headroom of `E9SelfHealing.code_headroom`.

**Reciprocity over an ensemble of rivals** (`ReasonsEnsemble`, §15.2). Let rival regimes be drawn with weights summing to one. A mind's kept-probability under the ensemble is the total weight of the rivals that keep it. That is at most one, the canonical regime's kept-probability on the good event (`ensemble_le`). It is strictly less whenever a rival of positive weight omits the mind (`ensemble_strict`).

**The Theorem 8 storage functional contracts** (`TheoremEightLyapunov`, §18.3–18.4). The functional is defined as code error plus the clipped reserve deficit. Exact replenishment saturates at zero deficit (`replenishment_exact`), and capacity overflow is retained in the energy account (`solvent_cycle_energy_account`). A complete solvent cycle cannot increase the reserve term (`solvent_cycle_deficit`). Combining this with the actual error drift and repair gives the positive-margin contraction (`cycle_contracts`, `margin_pos`) and its stated iteration (`lyapunov_iterate`).

**The smearing integral is exact** (`SmearingIntegral`, G.4). For a complete operator basis and any weight with total integral one, the smeared dual-kernel expression equals the original operator at every flow parameter. Its Bochner integral is therefore that operator, on the bulk algebra (`smeared_reconstruction`) and after boundary transport (`smeared_boundary_reconstruction`).

**The completed decoder** (`ChannelExtension`, G.1). Take $\mathcal{D}(\sigma) = E^\dagger \sigma E + \mathrm{Tr}((I - EE^\dagger)\sigma)\rho_0$ with $E^\dagger E = I$.

- It returns every encoded state exactly (`ext_recovers`).
- With any reference register, $E \otimes I_R$ is again an isometry (`kron_isometry`), and the extended decoder returns every joint code–reference state (`ext_recovers_reference`). Reference correlations are retained.
- It is trace preserving (`ext_trace`).
- It maps positive semidefinite inputs to positive semidefinite outputs (`ext_posSemidef`).

Complete positivity is compiled (`ChannelCP`). Amplified by any finite reference register, the decoder is $(E^\dagger \otimes I)Y(E^\dagger \otimes I)^\dagger + \rho_0 \otimes T(Y)$ (`amp_eq`). Here $T(Y)$ is positive semidefinite as a sum of compressions of $(B \otimes I)Y(B \otimes I)^\dagger$, where $I - EE^\dagger = B^\dagger B$ (`mp_posSemidef`). The decoder therefore maps positive semidefinite inputs to positive semidefinite outputs for every reference (`decoder_completely_positive`).

**State-vector distance bounds infidelity** (`PureStateDistance`, §23.3). For unit vectors, $1 - |\langle\psi,\varphi\rangle|^2 \le \|\psi - \varphi\|^2$ (`pure_infidelity_le`). The pure-state identity $\|\rho - \sigma\|_1 = 2\sqrt{1 - |\langle\psi,\varphi\rangle|^2}$ is compiled for the trace norm as the sum of absolute eigenvalues (`PureStateTraceNorm.pure_trace_norm`). The proof uses $(P - Q)^3 = (1 - |\langle\psi,\varphi\rangle|^2)(P - Q)$ (`pure_cube`), so every eigenvalue is 0 or $\pm s$, and $\mathrm{Tr}(P - Q)^2 = 2s^2$ (`pure_trace_sq`, `sum_sq_eigenvalues`). Hence $\frac{1}{2}\|\rho - \sigma\|_1 \le \|\psi - \varphi\|$. The speed-limit bounds, stated in state-vector norm, therefore bound the infidelity and the trace distance of the held state.

## G.14 Premise 5 with the Bennett counterpart constructed

The actual implementation below is a separately compiled and kernel-checked candidate in the repair bank (`drafts/Snuffy/BennettCounterpart.lean`); the canon still contains the earlier heat-only option model. It is not recorded as a completed physical counterpart repair. For an administration $f = h \circ g$, the candidate implementation first computes, copies and uncomputes, then exchanges the output and live registers:

$$(s, 0, 0) \longmapsto (f(s), 0, s).$$

`implementedProgram` and `implemented_program_computes` identify the implemented administration, clean workspace and full retained input. `implemented_run_refines` proves its trajectory correspondence and one retained input record per step. `quantumProgram` is the induced unitary with an arbitrary reference register untouched; its norm and round trip are preserved. An arbitrary bijection is not used as a substitute for implementing $f$.

The complete physical lifecycle also pays provisioning, compute/copy/uncompute, archive writing and holding, routing, reset, guard, controller, verification and endpoints. If $B_{\mathrm{rev}}, B_{\mathrm{irr}}$ are the complete non-heat charges and $Q_{\mathrm{irr}}$ the saved bath heat, the exact difference is

$$W_{\mathrm{rev}} - W_{\mathrm{irr}} = B_{\mathrm{rev}} - B_{\mathrm{irr}} - Q_{\mathrm{irr}}$$

(`complete_cost_difference`). Strictly more slack requires the implemented lifecycle penalty to be less than the saved heat (`complete_cost_strict_iff`), while meeting the same deadline. The adiabatic route must establish a switching-time window satisfying both bounds (`adiabatic_slack_window`, `switched_program_meets_deadline`).

A fixed circuit multiplier and an actual full archive allowance preserve the original polynomial exponent (`implemented_work_ceiling`). That upper bound does not by itself establish the strict complete-cost comparison. The earlier heat-only option model and Landauer results remain their stated mathematical components; completion of the physical counterpart claim requires this lifecycle and deadline comparison for the full intended class. The original premise-5 target remains fixed.

## G.15 Recovered and combined results

Five earlier results that the canon did not yet contain are folded in. Three compiled results are joined into stronger forms.

**Reciprocity at a sufficient code distance** (`ReciprocityDistance`, §15.2). Three results combine into one.

- On the good event the canonical outcome contains the rival's (`ReasonsCompiled.reciprocity_expectation`).
- On the good event the canonical regime keeps every particular (`reciprocity_canonical_floor`).
- The failure probability falls below every $\varepsilon$ with code distance (`Threshold.family_ceiling`, `CorrelatedCeiling.correlated_family_ceiling`).

For the design family indexed by distance, the canonical regime keeps the mind with probability above $1 - \varepsilon$ from some distance on (`distance_floor`). For every rival and every $\varepsilon$, the rival's kept-probability is at most the canonical one plus $\varepsilon$ (`reciprocity_at_distance`). Every rival that loses the mind with positive probability is strictly below the canonical regime from some distance on (`reciprocity_strict_at_distance`). The canonical kept-probability tends to one (`canonical_tends_to_one`).

**The two forms of the causal boundary agree** (`CutEquivalence`, §24.3). A set is closed under the dependency relation exactly when no path from a node outside it reaches it. A closed dependency component is therefore exactly a directed separator with empty certified boundary (`closed_iff_no_bypass`), and the separator form extends it to a matched certified boundary (`UPTR16Cut.causal_separator`).

**The concatenated memory decodes its record unless its block fails** (`ConcatenatedDecoding`, §23.8). The record is encoded recursively onto the $7^k$ locations through an outer code of Hamming distance at least three. It is decoded recursively by nearest codeword at every level. If the block does not fail in the recursive sense of `Concatenated.Fails`, decoding returns the original record, whatever values the faulty locations hold (`decodes_unless_fails`). The proof is by induction on the level: at most one sub-block fails, the other six decode to their outer-code symbols, and the outer

code corrects one symbol (`outer_decodes`). Actual decoding failure therefore has probability at most $(21p)^{2^k}/21$ (`decoding_failure_le`).

**Volume and boundary** (`RegionBoundary`, F.3). For every region of a connected relational structure other than the whole, positive volume holds exactly when the boundary homology is nonzero (`region_bootstrap`). The whole structure has no outside and an empty boundary (`whole_no_boundary`, `whole_H0_trivial`).

**Greatestness through boundary representatives** (`R15CausalGluing`, with the staged order of `R14Rivals`). Greatestness reduces exactly to a covering family of feasible fibre representatives (`boundary_representatives_iff`, `selected_representative_greatest`), in the canonical staged order as well (`canonical_boundary_iff`). A conservative interface envelope suffices to make the chosen policy greatest, without greatest representatives (`interface_envelope_transfer`). Clairvoyant per-world selection is excluded: a choice can match every world separately while no single choice matches them all (`hidden_world_control`). The staged order is reflexive and transitive (`UPTR14Rivals.refl`, `trans`). A coordinatewise envelope is a constructive certificate (`coordinate_envelope`), and a complete rival envelope transfers to the chosen policy (`complete_rival_transfer`). In the four-stage order a hot replica strictly beats a pause that keeps every life but loses a freedom (`hot_beats_pause`).

**Timely care, deadline demand, certified decisions and observation-based winning** (`TimelyCare`, `DeadlineDemand`, `DecisionPermit`, `ObservationWinning`).

- *Timely care.* A particular whose reserved capacity covers its need is served exactly its need (`care_served`). Other requests and statuses cannot exhaust that reservation (`exhaustion_fails`). The need is met at every date through the deadline when capacity covers the reservations (`deadline_met`). A miss implies a capacity shortage at some date (`miss_within_shortage`).
- *Deadline demand.* The demand that must complete within any window is at most the supply available in that window (`demand_le_supply`), robustly across worlds (`robust_demand`).
- *Certified decisions.* A grant names exactly the certified decision of its generation (`grant_names_certified`, `granted_policy_is_certified`); a wrong action is rejected and a certified one granted (`wrong_action_rejected`, `certified_action_granted`).
- *Observation-based winning.* An observation-based policy keeps the system safe exactly when the safety game is winnable (`policy_wins_iff`), and a hidden-world countermodel shows that observation, not clairvoyance, is the right quantifier (`hidden_world_countermodel`).

## G.16 Protection of the installation's own run, the physical top, and recovery on the good event

This section answers review objections 2, 3 and 5. Each is answered on the installation's own execution, without an abstraction map, product hypothesis or assumed device.

**The installation's own run is protected** (`G1Refinement`). Let the installation have its own physical state space, possibly with hidden internal degrees of freedom, and a readout of each particular's record frame. One step is guarded: the world acts — the environment, every agent and every crosstalk between particulars — then the physical corrector acts. Three properties of the device suffice:

- the corrector writes the nearest codeword back into each particular's readout;
- on the good event the world moves each readout by at most $\rho$ per step;
- the records are written at the start.

With these, every particular continues at every step of the physical run on the good event, and the probability that any particular fails is at most $\sum_i \delta_i$, with no independence (`physical_protection`). The complete journal of G.8 changes nothing observed, keeps the protection along the live state and replays it exactly (`journaled_protection`): the journal retains history; the guard supplies the protection.

**No alarm-response device is needed for holding** (`physical_qsl_drift`, `physical_protection_qsl`). On normalised readouts, the speed limit bounds the per-step drift by $(W/\hbar)\Delta$ for any agent whose coupling is at most $W$. Take $W$ to be the maximal physical coupling of any agent, which is finite because local power is finite. Correcting at intervals $\Delta \le \hbar(t-\varepsilon)/W$ then holds every particular against every agent. The E9 alarm and mediated response remain valuable: below threshold they permit longer intervals and lower cost. Holding does not depend on them. Every hypothesis has a concrete inhabited instance (`physical_instance_inhabited`).

**The continuation top is realised physically** (`G2Physical`). The held installation is a physical policy. On the good event it keeps every particular at every step through the horizon (`held_installation_keeps_all`), so the existing and new continuation stages are whole. Crosstalk enters only through the bounded drift that each particular's correction absorbs. No product structure of policies and no closed-block hypothesis is used. Once the top is realised, it dominates every realised outcome without union closure (`top_dominates_continuation`).

**Holding and timely authorised delivery on one execution** (`G2Freedom`). Pair the installation's frames with the R17 effect machine. The corrector does not touch the effect machine, and delivery does not touch the readouts; the physical consequences of applied effects are part of the bounded drift. On the same joint run, every particular continues, and every effect that is ready under a live R17 contract within a deadline-certified FIFO prefix is applied by its deadline (`joint_holding_and_delivery`, with `SnuffyFIFO.ready_fifo_effect_applies_before_deadline`).

**Recovery succeeds on the good event** (`G3GoodEvent`). Let every patch read-out lie within the decoding radius of its target codeword, once its interior is recovered, off the named charged accounts. The threshold theorem bounds the event that it does not. Then the wavefront sweep recovers the whole region on the good event, and recovery fails with probability at most $\sum_i \delta_i$ (`sweep_success_on_good_event`). The complete bill keeps archive holding as its own term (`complete_bill_with_archive`).

**Archive holding is charged on the held stock** (`G3ArchiveStock`). The holding charge at a step is the rate $\gamma$ per held bit times the bits actually held then. The archive holds the record's information $I_0$ and the live boundary transcript. Each transcript increment is retired after the wave that consumes it, so the held stock at step $s$ is at most $I_0 + c(s+1)^{d-1}$. The summed holding charge is then at most $\gamma(I_0 + c)t^d$ for every $d \geq 1$, the one-dimensional case included (`live_stock_bill`). Retirement is lossless or charged. Each retired bit is recomputable from the fenced recovered region together with $I_0$, or else it is correction entropy whose reset is charged once at its vertex inside the sweep's per-vertex charge. Retaining every increment instead costs at least $\gamma r\, t(t-1)/2$ (`retained_stock_quadratic`). So does an epoch rollover that copies the whole accumulated stock into fresh media at each boundary (`rollover_quadratic`). The architecture holds the live stock.

The decoded record is held passively. It is written when recovery begins, when the age $t$ is known, behind a barrier $\Delta E \geq k_B T \ln(cI_0 t/\varepsilon)$. Finite-temperature decay at a finite barrier then keeps its failure below $\varepsilon$ with zero running power. Its write cost is $O(I_0 \log t)$. The failure bound of `Threshold.fail_le_geometric` is linear in the number of locations. The reliability of the live transcript and of the patches is therefore the sweep's own reliability over $\Theta(t^{d_G})$ locations. Its age-independent constant at fixed $\varepsilon$ is carried by the fault-model application certificate: constant-distance inner patch codes, with the final decoder of the constant-size message $W$ using its carrier's redundancy across those locations.

## G.17 Premise 5 with the implemented Bennett program and its complete cost

This section answers review objection 6. The reversible administration is not an arbitrary bijection. It is the implemented program `BennettCounterpart.implementedProgram g h` for the administration $f = h \circ g$. The program writes $h(g(s))$ with a clean workspace and retains the input as history (`implemented_program_computes`). Its run refines the source run (`implemented_run_refines`). Its quantum lift is unitary (`implemented_quantum_norm`, `implemented_quantum_roundtrip`).

The irreversible administration exports heat $Q$, at least the Landauer heat of its merging overwrite. At switching time $\tau$, the reversible administration costs $A/\tau + H$. Here $H$ carries every other lifecycle charge: provisioning, archive writes and holding, routing, reset, guard, controller, verification and endpoints (`LifecycleCharge`, `complete_cost_strict_iff`). The deadline allows $0 < \tau \leq \tau_{\max}$.

- `premise5_exact`: when $0 < Q - H$ and $A/(Q-H) < \tau_{\max}$, no aligned option administers irreversibly. The implemented counterpart computes the same administration and meets the deadline. It keeps every earlier outcome and leaves strictly more resource slack.
- `window_exact`: when that window fails, every admissible reversible counterpart costs at least $Q$.

The window is therefore exactly the condition under which cost dominance forces premise 5 (`adiabatic_feasible_iff`). Outside the window, the reversible counterpart is not cheaper, so cost dominance alone does not establish premise 5 there.

## G.18 Application data for the cited fault-tolerance theorems

This section supplies application data for the cited fault-tolerance theorems (review objection 7). The full fault ceiling and the complete bill $\le Ct^{d_G}$ with an age-independent $C$ are unchanged. Components are compiled or computed; the remaining joins are named at the end.

**Knill–Laflamme from the code distance** (`KnillLaflamme`). In any complex $*$-algebra, the spectral projection $(1+S)/2$ of an involution is idempotent and fixed by $S$ (`spectral_projection`). The code projector annihilates any operator that anticommutes with a generator (`sandwich_anticomm`). It returns $cP$ for an operator acting as the scalar $c$ on the code space (`sandwich_scalar`). Let errors have weight at most $t$ with $2t < d$, where distance $d$ means that every operator of weight below $d$ commuting with all generators acts as a scalar on the code space. Then $PE_i^\dagger E_j P = \alpha_{ij} P$ for every pair (`knill_laflamme_of_distance`). Logical operators are excluded by the distance, not by the case split.

**The witness property of minimum-weight decoding** (`MatchingWitness`). Take a minimum-weight decoder and a syndrome additive under symmetric difference. Every zero-syndrome $\gamma \subseteq E \bigtriangleup C$ of the residual is then at least half faulty: $|\gamma| \le 2|\gamma \cap E|$ (`half_faulty`, `matching_witness`). This is the witness hypothesis of `Threshold.fail_le_witness` and `fail_le_geometric`. The nontrivial component of a failed residual is a simple cycle or a boundary-to-boundary path, of length at least the circuit distance.

**The circuit-level detector graph.** A dependency-free Pauli-frame enumeration covers the rotated surface-code memory over $d$ rounds, with X checks in a Z-shaped CNOT order and Z checks in an N-shaped order. The faults enumerated are the 15 two-qubit Paulis after every CNOT, $X$, $Y$, $Z$ after every reset, Hadamard and idle, and every measurement flip. For $d = 3, 5, 7, 9$ the enumeration gives:

- every fault flips at most two detectors of each check type;
- the maximum detector degree is 12;
- at most 52 (location, Pauli) faults produce each edge of the relevant type;
- the circuit distance is exactly $d$.

With the orders swapped, so that hooks run parallel to the logical, the distance falls to $\lceil d/2 \rceil$. The orientation is load-bearing.

The bound for every $d$ uses a potential. Let $\varphi$ be the face coordinate across the logical. Boundary edges attach only at $\varphi = 0$ and $\varphi = d - 2$, and every edge, time-like and hook edges included, changes $\varphi$ by at most one. So any walk joining opposite boundaries has at least $d$ edges. Every fault is local to one check neighbourhood in one round, so the edge types for larger $d$ are translates of those enumerated. The script and its results are banked with their hashes.

**Lattice-surgery merges.** The same enumeration was applied to the merge. Two $d \times d$ patches and a seam column were merged as the $d \times (2d+1)$ patch for $T$ rounds, with the seam initialised and finally split in the conjugate basis. The joint logical is found by $GF(2)$ elimination over the random seam checks. For $d = 3, 5, 7$ and $T \in \{d-2, d, d+2\}$, the circuit distance is exactly $\min(d, T)$ and every fault is graphlike. A merge therefore keeps distance $d$ exactly when it runs at least $d$ rounds; with fewer, the time-like failure of the merged-product outcome binds. The horizontal merge enumerated measures $\bar{X}\bar{X}$; the vertical merge, measuring $\bar{Z}\bar{Z}$, is the same construction rotated and was not run separately.

**The edge law and the explicit ratio** (`EdgeLaw`, `ExRecBound`). Suppose circuit faults are local stochastic at rate $p$ on (location, Pauli) pairs, and at most $c$ of them produce each edge. Then the induced edge law is local stochastic at rate $cp$ (`push_localStochastic`). With degree 12, so non-backtracking branching at most 11, and $c = 52$, `fail_le_geometric` applies with $r = 22\sqrt{52p}$. This ratio is below one exactly when $p < 1/25168$ (`surface_code_rate`).

An operation outside the spacetime-volume argument has its own gadget bound. Under local stochastic noise, if every failing pattern contains a malignant pair or at least three faults, then $P(\text{fail}) \le |M_2| p^2 + \binom{|L|}{3} p^3$ (`exrec_pair_bound`).

**Reliability against age** (`AgeReliability`). A distance $D \ge \log(Kt^d/\varepsilon)/(-\log r)$ holds the global failure over $Kt^d$ locations below $\varepsilon$ at every age (`log_distance_schedule`). A logarithmic overhead per location, however, is not absorbed into an age-independent constant: for every $C$ some $t \ge 1$ has $Ct^d < t^d \log t$ (`log_factor_not_absorbed`). The age-independent bill therefore needs the logarithm removed by construction.

**Remaining joins.**

- The potential argument for merges at every $d$, with the round index as the time-like potential.

- A map-by-map list showing that every repeated local map of holding, recovery, encoding, handoff and smearing is a Clifford circuit with exact native gates. This removes the compilation logarithm.
- Tolerance of the wavefront sweep to sparse patch failures, so that constant-distance inner patches suffice. This needs:
  - confinement of each failure;
  - no cascade through the feedforward front;
  - an outer decoder of the constant-size message $W$ using its carrier's redundancy, with its deviation bound.

  This removes the reliability logarithm.

## G.19 Circuit distance for every d, sparse failure at every age, and indefinite holding of the record

This section continues review objection 7 and the holding item. The targets are unchanged: the full fault ceiling, decoding error at most $\varepsilon$ on the historical record, and the complete bill at most $Ct^{d_G}$ with $C$ independent of age.

**An exact cut certificate for the circuit distance.** For any set $S$ of detectors, the fault set $K_S = \{f : \mathrm{obs}(f) \oplus |\mathrm{sig}(f) \cap S| \text{ odd}\}$ is crossed an odd number of times by every fault set with zero syndrome and odd observable. Over $GF(2)$, the parity of $|F \cap K_S|$ is $\mathrm{obs}(F)$ plus the syndrome summed over $S$, which is $1 + 0$. So $D$ sets whose fault sets are pairwise disjoint certify that every logical error has at least $D$ faults.

The cuts are given by explicit potentials:

- *memory*: $\varphi$ is the crossing face coordinate plus one, and $D = d$;
- *lattice-surgery merge*: $\varphi = \max(\tau(r), \sigma(j))$, and $D = \min(d, T)$. Here $\tau(r) = \max(1, r)$ is the time part. The space part $\sigma(j)$ is 1 on the two seam face columns, $d - 1 - j$ to their left and $j - d$ to their right.

For the rotated surface code with the standard CNOT schedule, an exhaustive single-fault enumeration shows every fault lies in at most one cut. The cases are $d = 3, 5, 7, 9$, the memory in both bases, and the merge at $T = d - 2, d, d + 2$: 19 of 19 cases, up to 120,544 faults. The merge therefore keeps distance $d$ exactly when it runs at least $d$ rounds. With the swapped schedule, hooks run parallel to the logical and the distance falls to $\lceil d/2 \rceil$.

A fault's multiplicity is fixed by three things: its observable bit, the values of $\varphi$ at its at most two detectors, and $D$. The potentials are translation-covariant formulas, so the condition is local to each fault type, and every type occurs at the enumerated sizes. That locality step is what carries the certificate to every $d$.

**The installation's repeated maps need no approximate compilation.**

- Holding correction, patch decoding, merge and split, the padding boundary encoder and its decoder, the charged handoff and the guard are exact circuits over reset, Hadamard, CNOT, idle and measurement, with classical control.
- The modular flow is not a runtime map. The finite-basis route $\alpha(A) = EAE^\dagger$ with $D = E^\dagger$ gives the same boundary representative exactly.
- The Bennett administration is exact reversible logic.

So no precision grows with age, and only reliability remains.

**Sparse failure at every age** (`SparseFailure`, `ToleranceSweep`). Each patch's failure is decided by the faults in a bounded window. Patches whose windows are pairwise disjoint all fail with probability at most $\prod_v \binom{|W_v|}{k} p^{k|U|}$ (`disjoint_windows`). With a $K$-colouring in which same-coloured windows are disjoint, at least $Kn$ of $N$ patches fail with probability at most $K\binom{N}{n}(\binom{w}{k}p^k)^n$ (`many_failures`). This falls exponentially in $N$ once $\binom{w}{k}p^k < \theta/(eK)$, uniformly in age.

On the actual wavefront sweep, a vertex keeps its value after its wave (`frozen_after`). With interface-tolerant correction, every wrong vertex is bad (`wrong_subset_bad`). Interface tolerance means a good window decodes correctly whatever the inner commit, because a wrong commit reaches it only as a bounded interface error, which is the overlapping-recovery analysis of DKLP. The sweep's wrong-vertex count obeys the same tail (`sweep_density`). The realised bill on that event is at most base $\cdot N +$ repair $\cdot Kn$ (`pathwise_bill`). An outer decoder that tolerates fewer than $Kn$ wrong patches returns the record except with that probability (`recovery_tail`). Failures do occur, at constant density; the theorems bound their number, and recovery tolerates them.

**Redundancy of the record.** The recovery tolerates a constant fraction of silent local errors exactly when the record is spread redundantly across its carrier. For an objective classical record this is a theorem of quantum Darwinism: classical information emerges redundantly and generically across environment fragments (Brandão, Piani and Horodecki, 2015), and objectivity implies a spectrum-broadcast structure (Horodecki, Korbicz and Horodecki, 2015). For a record with no redundancy, every location matters, and no protocol of noisy gates avoids a logarithmic factor (Dobrushin–Ortyukov; Pippenger).

**Indefinite holding at fixed $\varepsilon$** (`PassiveReference`). A constant per-step failure rate gives a total failure that grows linearly with age, so periodic renewal at fixed size does not hold $\varepsilon$ indefinitely. The protector's reference record is therefore held passively on a doubling schedule:

- the reference is rewritten at ages $2^j$;
- stage $j$ has a barrier $k_B T(2j \ln 2 + \ln(1/\varepsilon))$ above the attempt scale, so it fails with probability at most $\varepsilon/2^{j+1}$.

Every age is covered by $\log_2 t + 1$ stages (`stages_cover`). The total failure is at most $\varepsilon$ for every number of stages (`total_failure_le`). The rewrite cost is $O(\log^2 t)$ (`total_rewrite_cost`), so the long-run holding rate is the constant per-step comparison of each particular with its reference. A faulty comparison or false repair is transient: the intact reference corrects it at the next comparison (E9 self-healing).

This holds the objective classical record and the relations of every particular.

**The record's redundancy is decoded by majority** (`RedundantRecord`). Each fragment reads a record value from the recovered state, and an intact fragment reads the record. The strict-majority decoder then returns the record whenever fewer than half of the fragments are wrong (`majority_decodes`). This supplies the tolerance hypothesis of `recovery_tail` (`tolerant_outer_decoder`). For an objective record, the redundancy theorems cited above supply the intact-fragment hypothesis, so the carrier application rests on them. The interface hypothesis of the windowed sweep rests on DKLP's overlapping recovery, in the same way the witness lemma does.

**Premise 5 for batched administration** (`BennettBatch`). Administer $b$ merges in one batch. The irreversible heat is at least $qb$, with $q \geq k_B T \ln 2$ per merged bit. The implemented counterpart's complete non-heat lifecycle charge is at most $\eta b$, and its drive constant is bounded. If $\eta < q$, every large enough batch lies inside the adiabatic window for every deadline (`batched_window`), so no aligned option administers irreversibly (`premise5_batched`). The single remaining physical input is $\eta < q$. Adiabatic logic below $k_B T \ln 2$ per operation meets it.

**The explicit threshold, tightened** (`WeightedEdgeLaw`). Under circuit-level depolarizing noise, each fault carries its own rate: $p/15$ for each two-qubit Pauli, $p/3$ for each single-qubit Pauli, and $p$ for a measurement flip. Pushed to the detector graph, the edge rates are the weighted sums of their faults (`push_weighted`), and the largest edge rate is at most $\frac{94}{15}p$ (`push_uniform`, enumerated).

Self-avoiding walk counts on the detector graph are submultiplicative. The exact maximum over all interior vertices at length 8 is $c_8 = 89{,}270{,}654 \leq 9.87^8$ (`walk_bound`), so walks of every length are at most $c_8\, 9.87^L$. The ratio of `fail_le_geometric` is therefore below one exactly when $p < 1/(4 \cdot 9.87^2 \cdot \frac{94}{15})$ (`tightened_threshold`), which exceeds $4 \times 10^{-4}$ (`threshold_value`). The bound at length 9, $c_9 = 763{,}504{,}745$, gives about $4.2 \times 10^{-4}$. This is a requirement on the installation's devices, of the same kind as $\eta < q$.

**Indefinite holding of quantum coherence.** Balasubramanian, Davydova and Lin (arXiv:2605.10943, 2026) construct a passive memory in $\mathbb{R}^3$: 3D-local CSS stabilizer Hamiltonians on $n_k = e^{\Theta(k)}$ qubits, with $O(1)$-range terms and bounded density. Below $T_c$, the thermal encoding satisfies $\|\Phi_t \circ \mathrm{Enc}_k(\rho) - \mathrm{Enc}_k(\rho)\|_1 \leq t\, e^{-\Theta(n_k^\eta)}$.

Held on a doubling schedule, each stage needs a code size $n_j$ with $c\, n_j^\eta \geq (2j+2)\ln 2 + \ln(1/\varepsilon)$, and each stage ends with a transfer into the next code. If each stage's holding failure and each transfer's failure are at most $\varepsilon/2^{j+2}$, the total failure is at most $\varepsilon$ at every age (`StageSchedule.total_with_transfers`). A cost polynomial in each stage's code size sums to a polylogarithm of age (`stage_cost_le`). The transfer lands the qubit in the next code's codespace, so its hypothesis is that a codespace-initialised state obeys the same lifetime bound, or that its syndrome sector thermalises within polynomial time without logical error. This is the open initialisation problem those authors identify. The quantum branch is proved conditional on it, with the hypothesis stated explicitly. The classical record and its relations are held unconditionally by `PassiveReference`.

**Remaining.**

- The initialisation problem above.
- Measured device parameters: the fault rate $p$, the coupling bound $W$, the per-bit overhead $\eta$, and the barrier materials and temperature.

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# Part IV — References and provenance


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**Evidence status.** This master assembles three kinds of evidence, each by what it is:

- prose derivations at declared premises;
- exact replay receipts;
- compiled theorems.

The machine-checked canon of Section 23 certifies exactly the results it maps, with the premises it lists. Elsewhere, prose remains prose.

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# Appendix — Verification provenance of this edition

The original published baseline contains 115 modules and 831 audited names. The current tree contains 187 modules and 1128 audited public theorem/lemma names. All 187 modules were freshly compiled and kernel-checked in one run, and the full axiom, completeness and claims gates were freshly run, overall PASS. The earlier receipts, including the 924-name combined receipt that reused Beam's 136 results, are preserved with their banks. The original failed run is preserved in the integration bank.

The eight-module integration and its 896-statement receipt remain preserved. This increment adds 28 checked statements in the existing DecoderRetraction, HamiltonianDecomposition and WorkBilling modules, integrates the three-pillar operator/history/work construction in Part III-G, and keeps the continuous integral/channel derivations and physical source correspondences explicit. Its exact changes and verification evidence accompany the working draft. No published edition or release is changed.

**The analytic bridge.** In the Canon Edition, the strong-operator convergence of alternating projections was carried as an explicit structure (`VonNeumannConvergence`) and not claimed. In 4.1 it is discharged: `CanonVonNeumann.canon_von_neumann_convergence` fills both limit fields in Hilbert space, so the meet algebra's analytic bridge is a theorem.

**The Canon Edition suite.** The containerised replay of the Canon Edition (`container/Dockerfile`, `container/Apptainer.def`), the byte-identical replay of the four-stroke ledger (`verify_four_stroke.py --verify-all`), the manuscript assembly checks and the RO-Crate provenance remain in force for the artifacts they cover.

**Manuscript.** The 4.2 deliverable is this Markdown master, rendered to PDF, HTML and EPUB. The published 4.1 artifacts remain their existing versions.